Ajan Ahmed, PhD

Contact

- **Email:** aahmed@clarkson.edu
- **Location:** Edmonton, Alberta, Canada
- **LinkedIn:** linkedin.com/in/ajan-ahmed
- **GitHub:** github.com/ahmedajan
- **Portfolio / Website:** ahmedajan.github.io/website
- **Google Scholar:** scholar.google.com/citations?user=BID713QAAAAJ
- **ORCID:** 0009-0002-7000-7809

- **Work authorization:** Open Work Permit (Canada), no sponsorship required; Permanent Residence application in progress

Education

Ph.D., Electrical and Computer Engineering — Clarkson University, Potsdam, NY, USA

- **Dates:** January 2024 – 2026
- **Department / Lab:** Department of Electrical and Computer Engineering; AI Vision, Health, Biometrics, and Applied Computing (AVHBAC) Lab
- **Advisor(s):** Dr. Masudul Imtiaz, Associate Professor, Electrical and Computer Engineering
- **Committee members:**
  - Dr. Masudul Imtiaz (Chair / Advisor), Associate Professor, Electrical and Computer Engineering, Clarkson University
  - Dr. Stephanie Schuckers, Bank of America Distinguished Professor in Computing & Informatics, Computer Science, University of North Carolina at Charlotte (also affiliated with ECE, Clarkson University)
  - Dr. Faraz Hussain, Associate Professor, Electrical and Computer Engineering & Computer Science, Clarkson University
  - Dr. Mahesh Banavar, Academic Dean and Professor, Electrical and Computer Engineering, Clarkson University
  - Dr. Shafique Ahmad Chaudhry, Associate Professor, Reh School of Business (courtesy appointment, Computer Science), Clarkson University
- **Thesis / Dissertation title:** *Standardization of Voice Biometrics towards Improved and Reliable Speaker Recognition*
- **Thesis summary:** ISO/IEC standardizes per-sample biometric quality metrics for fingerprint, face, and iris (e.g. NFIQ 2, OFIQ), but not for voice, and traditional speech-quality metrics do not reliably predict speaker verification performance. This dissertation introduces the Voice Quality Index (VQI), an ISO/IEC-aligned, utility-based, per-sample quality metric that predicts how useful a voice recording will be for speaker recognition. VQI produces two explainable scores (signal quality, VQI-S, and voice distinctiveness, VQI-V), is trained on actual verification outcomes from ~2.6 M utterances, and generalizes across held-out datasets and verification systems. The work also contributes two real-world voice datasets, an open-source desktop application, and a longitudinal analysis of speaker verification in children, including age-normalization methods that partially recover performance lost to vocal development.
- **Research questions:**
  1. Do traditional speech-quality metrics predict speaker verification outcomes?
  2. Can a speech quality score trained on actual recognition performance predict verification outcomes?

3. Does it generalize across systems, datasets, and over time?

- **What I worked on:**
  - **Data collection:** designed and ran two human-subject voice data collections:
    - *VPQAD* (Voice Pre-Processing and Quality Assessment Dataset): 50 adults, dual-microphone (Audio-Technica AT2020 + Shure SM58, simultaneous), 44.1 kHz/16-bit, 2 real-world environments (cafeteria, engineering lab), text-dependent and text-independent speech. Published on IEEE DataPort and in *IEEE Data Descriptions*.
    - *VSEA-DC* (Voice Sample Enrollment Analysis Data Collection): 100 adults, 6 real-world environments (quiet lab, dining area, noisy HVAC lab, inside running vehicle, telephone calls from lab and vehicle), 2 platforms (studio recorder at 48 kHz and iPhone 16 Pro call recording), 600 recordings, ~40–50 hours.
  - **Speech-quality metric study:** analyzed whether families of speech-quality metrics (SNR-based, perceptual, voice quality) predict speaker verification performance (EER, AUC-ROC, d′, minDCF) across three self-supervised speaker embedding models (WavLM, Wav2Vec2, HuBERT); found no family reliably predicts performance, motivating a utility-based metric. Published in *Signals* (2026).
  - **VQI pipeline:** built an end-to-end pipeline from raw audio to a 0–100 per-sample quality score:
    - Data: 11 datasets (~2.6 M utterances), incl. VoxCeleb1/2, Common Voice (EN), LibriSpeech, VOiCES, VCTK, CN-Celeb, VPQAD, VSEA-DC; augmentation with MUSAN noise and simulated room impulse responses.
    - Preprocessing: 16 kHz mono conversion, voice activity detection, loudness normalization.
    - Utility-based labelling: consensus of three speaker verification systems (ECAPA-TDNN, ResNet34-SE, ECAPA2) over 1.2 M mated and 28 M non-mated pairs, producing a balanced 20,288-sample training set plus a 50,000-sample held-out validation set.
    - Features: 705 hand-crafted, explainable acoustic features (544 signal-quality, 161 voice-distinctiveness), reduced to 430 + 133 via a 3-stage selection (zero-variance removal, Spearman correlation filtering, Random Forest importance pruning); avoided using recognition-model embeddings so the score works with any verification system.
    - Modelling: compared 8 model families (Random Forest, Ridge, SVM/SVR, MLP, LightGBM, XGBoost, TabNet, Logistic Regression) as classifiers and regressors with hyperparameter search under 5-fold cross-validation; compared dimensionality reduction methods (PCA, Factor Analysis, ICA).
    - Deployment: selected Ridge Regression for VQI-S and XGBoost for VQI-V, balancing AUC-ROC, Error-vs-Reject performance, and inference time.
    - Evaluation: 5 datasets × 5 speaker verification systems (2 architectures held out from training: x-vector, WavLM-SV) plus a commercial system (Neurotechnology VeriSpeak) as external reference; Error-vs-Reject Curves (ISO/IEC utility measure), ranked DET curves, and dual-score quadrant analysis.
  - **Open-source software:** released a VQI desktop application and command-line tool on GitHub (record or upload audio, dual score gauges, per-feature explanation of each score with guidance on how to improve it, waveform/spectrogram views, exportable reports; all 28 trained models selectable).
  - **Longitudinal children study:** analyzed speaker verification in 292 children (ages 4–13 at first enrollment, up to 20 collections over ~9 years) with four systems (VeriSpeak, ECAPA-TDNN, ResNet, ECAPA2) and 10 hand-crafted-feature i-vector/PLDA systems (12,096 mated cross-age pairs); used linear mixed-effects regression (LMER) to identify drivers of degradation, causal feature ablation to locate responsible feature dimensions, and compared z-score, linear, and polynomial age-normalization methods.
  - **Emotion analysis:** evaluated VQI on the Emotional Speech Dataset (ESD; 20 speakers, 5 emotions, 35,000 utterances).
- **Key contributions / results:**
  - Showed that no family of traditional speech-quality metrics reliably predicts speaker verification performance.
  - Developed VQI, a dual-score (signal quality + voice distinctiveness), explainable, ISO/IEC-aligned voice quality metric; trained on ~2.6 M utterances from 11 datasets.
  - Achieved AUC-ROC of 0.88 (VQI-S, Ridge) and 0.91 (VQI-V, XGBoost) on held-out validation data, with inference in 0.007 ms and 0.04 ms respectively; best model family (SVM) reached up to 0.94–0.95 AUC-ROC.

- Rejecting the lowest-quality 20% of samples reduced speaker verification error (FNMR) by 7.66% (VQI-S) and 6.12% (VQI-V); error dropped on every verification system and dataset tested, including systems never used in training.
- Showed the two scores capture different failure sources, enabling actionable feedback (e.g. poor signal → re-record; low distinctiveness → speaker limitation).
- Found VQI-V varies over six times more than VQI-S across emotions (median 82 Neutral vs 43 Sad), indicating emotion affects voice distinctiveness rather than signal quality.
- In children, predicted FNMR (at 0.1% FMR) rises from ~3% to over 20% as voices develop; identified time gap, pitch change, formant spacing (vocal-tract length), enrollment age, and puberty as main drivers.
- Causal ablation showed restoring just two formant dimensions (F4 frequency and bandwidth) reduces FNMR by 10–15 percentage points.
- Polynomial age normalization outperformed linear and z-score methods; age-normalizing LPCC coefficients cut FNMR by up to ~35% on held-out speakers.
- Contributed 2 new datasets (VPQAD, VSEA-DC) and 1 open-source software tool.

- **Highlights:**
  - Developed the Voice Quality Index (VQI), the first ISO/IEC-aligned, utility-based voice quality metric that predicts speaker verification performance, trained on ~2.6 M utterances from 11 datasets; published in *IEEE Access* (2026).
  - Designed and ran 2 human-subject voice data collections (VPQAD: 50 participants, dual microphone; VSEA-DC: 100 participants, 6 environments, 2 devices, ~40–50 h) and released VPQAD publicly.
  - Benchmarked 8 ML model families (Random Forest, Ridge, SVM, MLP, LightGBM, XGBoost, TabNet, Logistic Regression) with 5-fold cross-validation and hyperparameter search, reaching up to 0.95 AUC-ROC.
  - Reduced speaker verification error (FNMR) by rejecting low-quality samples, with gains on every dataset and on verification systems never used in training.
  - Showed that traditional speech-quality metrics fail to reliably predict verification performance across 3 self-supervised models (*Signals*, 2026).
  - Quantified voice-aging effects in 292 children over ~9 years (FNMR rising from ~3% to over 20%) using mixed-effects models and causal ablation, and developed age normalization that cuts FNMR by up to ~35%.
  - Released an open-source VQI desktop app and CLI with explainable per-feature feedback.
- **Limitations & future work:** longitudinal normalization fit to one dataset; English-only training; no deepfake detection yet; consensus labels rather than human judgments; not real-time. Next steps: multilingual retraining (Common Voice, MLS, VoxLingua), deepfake flagging, on-device real-time scoring, and pushing VQI toward an ISO/IEC voice-quality reference alongside NFIQ 2 and OFIQ.
- **Day-to-day research duties (PhD, 2024–2026):**
  - Framed research questions, reviewed literature, and wrote study designs and experiment plans before each project, including patent and prior-art searches (Google Patents, Espacenet) during the literature reviews for VQI and the deepfake and denoising pipelines, and before choosing open-source licences for the released software
  - Managed lab GPU workstations and HPC jobs (SLURM): environment setup, CUDA / driver updates, job scheduling, and storage quotas
  - Kept experiments reproducible: version control with Git / GitHub, pull requests and code review, conda / pip environments, Docker containers, fixed random seeds, and config files
  - Tracked experiments and model runs (Weights & Biases / TensorBoard, spreadsheets) and kept lab notebooks and README documentation
  - Automated repetitive work with Python / Bash scripts and CI checks (GitHub Actions, pytest, Black / Ruff)
  - Built data pipelines and data stores for large audio and metadata collections (pandas, Parquet / HDF5, SQLite / PostgreSQL), with data validation and de-duplication
  - Orchestrated the data and training pipelines as scheduled workflows (Apache Airflow DAGs, later Prefect flows; Kubeflow Pipelines tried on the k3s cluster): ingestion, cleaning, feature extraction, labelling, training, and

evaluation steps with retries, logging, and versioned outputs tracked in MLflow; made each step idempotent (content-hashed outputs, skip-if-done checks) so reruns were safe, ran backfills over the full corpus when a feature extractor or labeller changed, and handled schema evolution of the Parquet feature tables (added and renamed columns with versioned schemas)

- Handled sensitive human-subjects data: IRB protocols and amendments, consent tracking, de-identification, access control, and secure backups
- Set recording and data-quality standards for the voice data collections (VPQAD, VSEA-DC): calibrated microphone gain and recording levels against a reference tone and a sound level meter at the start of each session, logged device, room, and background-noise metadata, and ran per-file quality checks (clipping, SNR, sample rate, channel count, duration) before recordings entered the pipeline
- Wrote dataset documentation (datasheets, metadata dictionaries, licences) for public releases on IEEE DataPort and GitHub
- Integrated LLM tools (Claude Code, GitHub Copilot, ChatGPT / GPT-4o, Claude API) into coding, text-data generation, literature review, and writing, verifying all outputs
- Built a conversational retrieval-augmented generation (RAG) assistant (multi-turn chat with context tracking and step-by-step reasoning over retrieved sources) for semantic search and summarization of lab papers, reports, and study protocols (LangChain / LlamaIndex, FAISS / Chroma vector stores, OpenAI and Claude APIs); tested answer quality with RAGAS, LangSmith traces, and LLM-as-judge scoring
- Improved retrieval for the assistant: hybrid search (BM25 keyword plus dense embeddings), cross-encoder re-ranking, query rewriting, and chunking experiments, measured with recall at k, MRR, and NDCG on a labelled question set; served the keyword index on Elasticsearch / OpenSearch, and tried Vespa for combined keyword, vector, and ranking-expression search in one engine
- Chose the context strategy per question type: preloaded small, stable context (lab procedures, study metadata, schemas) into long-context prompts, used retrieval or tool calls for large or changing data, and compared preload, RAG, and hybrid approaches on quality, latency, and cost with the regression suite
- Used the assistant on regulatory and contractual documents: answered compliance questions from IRB protocols, consent forms, sponsor reports, and data-use and licence agreements, citing the governing clause, with answers checked against the source text
- Added paper and dataset recommendations to the lab assistant for literature reviews: content-based recommendation from embedding similarity and each member's reading history, re-ranked with their feedback (saved, dismissed, and cited items), and checked with simple ranking metrics (precision at k, NDCG); added collaborative signals from implicit feedback (saves, opens, citations) with matrix factorization (LightFM) alongside the content embeddings, used an LLM to write short summaries and topic tags for each paper and dataset as extra item features so new (cold-start) items could be recommended, and compared an LLM re-ranker with the cross-encoder and gradient-boosting (LambdaRank) re-rankers on NDCG, latency, and cost
- Improved grounding and memory in the assistant: answers cite retrieved passages and are checked against them, conversation memory and context windows are managed across long sessions, and per-user preferences personalize answers; tested for hallucination and faithfulness
- Added agent guardrails and failure recovery: tool-call validation, retries and fallbacks when a step failed, output schema checks, and full trace logging (LangSmith) to debug a single interaction end to end; versioned prompts and agent workflows in Git and LangSmith, and watched latency, token cost, and errors on tracing dashboards to tune cost and speed; used chain-of-thought and ReAct-style prompting for multi-step agent reasoning, few-shot example selection with calibration checks, structured output contracts (JSON schemas), and prompt (context) caching to cut cost and latency
- Added intent checks before every agent tool call: compared the user's request and the agent's stated purpose with the actual tool name and payload using embedding similarity and zero-shot (NLI) classification, produced a confidence score, and allowed, blocked, or sent the call for human approval; wrote the allow / deny rules as policy-as-code (Open Policy Agent, with AWS Cedar in a trial) fed by these scores
- Ran the intent and prompt-injection checks inline in the assistant's API request path (FastAPI middleware) with small ONNX classifiers and cached embeddings to keep added latency low, and measured false blocks and missed attacks on a labelled set of benign and adversarial requests

- Red-teamed the lab's LLM assistant and agents before use: prompt-injection and jailbreak tests (including injected instructions inside retrieved documents), tool-call abuse, data-leakage checks, and an automated security test harness (promptfoo, garak, PyRIT) that ran on every change; hardened the system with input and output filters, tool allow-lists, least-privilege data access, and human approval for risky actions
- Mapped the lab assistant's governance, security, and privacy controls to AI standards and guidance: ISO/IEC 42001 (AI management system: AI risk and impact assessment, roles, human oversight, monitoring, and records), the NIST AI Risk Management Framework, the OWASP Top 10 for LLM Applications, and the UK AI Cyber Security Code of Practice, alongside the privacy rules for human-subjects data (IRB, de-identification, least-privilege access)
- Mined usage logs of the lab LLM tools for failure cases and weak labels, and fed them back into prompts, retrieval, and the regression suite
- Built feedback loops for the lab assistant: active feedback (thumbs up / down, corrections) and passive signals (re-asked questions, abandoned sessions, copied answers), sent as events through a Kafka topic into the monitoring store, with KPIs for each user journey (answer accepted, time to answer, escalation rate) on dashboards; consumed the event stream with a small real-time pipeline (Kafka consumer groups with windowed aggregations) that updated the KPIs and the model router's bandit statistics within seconds
- Streamed row-level changes from the lab's PostgreSQL study database with change data capture (Debezium into Kafka), so dashboards and the assistant's index updated in near real time; ran statistical and ML anomaly detectors on the event stream (rolling z-scores and seasonal baselines, Isolation Forest) to separate noise from real problems such as stalled uploads, missing sessions, or sudden drops in recording quality, and used an LLM step to correlate each alert with recent data, code, and config changes and return likely root causes with confidence scores and links to the evidence (logs, diffs, records)
- Took over code and data pipelines written by earlier students (longitudinal children dataset, CITeR deepfake project) and by partner-university collaborators (West Virginia University), and refactored them into tested, documented modules that fit the lab's shared pipeline
- Split the lab assistant and model services into microservices (retrieval, embedding, classifier, router, and monitoring services) that talk over REST and gRPC; used asyncio and thread pools for concurrent I/O and request batching, cached embeddings and features in Redis, and load-tested the request path with Locust to keep tail latency (p95 / p99) low under many concurrent users
- Used the assistant as an agent-assist tool for the lab's help requests (data access, equipment, scheduling, IRB questions): classified each incoming request by intent, retrieved the relevant procedure, and drafted a reply for a lab member to review and send; tracked time to resolution, share of drafts sent with few edits, and requester satisfaction; served the assistant to lab members as a chatbot in a web chat page and in the lab's Microsoft Teams channel (Bot Framework connector)
- Analyzed participant questionnaires from the data collections (Qualtrics / Google Forms): cleaned demographic and self-report responses, coded open-ended answers with NLP (topic modelling with BERTopic, sentiment and theme classification with LLMs checked by hand), and reported summaries to the team
- Trained a small judge model (LoRA fine-tuned open model) on human-graded examples to score assistant answers cheaply, and checked its agreement with human review before relying on it
- Made evaluation cheaper and faster: stratified sampling of test items, cached model outputs, and cheaper judges for easy items, checking that results matched the full run
- Reviewed LLM outputs and data by hand for many hours (generated scripts, extracted fields, assistant answers), and wrote review guidelines so human labels stayed consistent
- Built an evaluation harness for the LLM tools: a fixed regression suite of questions and records with reference answers, LLM-as-judge graders checked against human review (including spot checks for when the judge was wrong or biased), and a run on every prompt, retrieval, or model change before it shipped; set acceptance thresholds (release gates) for each metric, showed uncertainty with bootstrap confidence intervals, and ran significance tests before calling one version better
- Used LLMs for information extraction and classification: pulled structured fields from session notes, consent logs, and collection records into SQL tables, with prompt tuning and output checks; extracted structured fields

from unstructured PDF documents (sponsor reports, study protocols, papers) with layout-aware parsing (PyMuPDF, Unstructured, Tesseract OCR) and LLM structured outputs, feeding the RAG index
- Turned recorded conversations into searchable knowledge: transcribed consented lab meetings and sponsor review calls with Whisper (faster-whisper), attributed speakers with diarization and speaker embeddings, extracted decisions, action items, and entities into structured records with LLM structured outputs, and added the transcripts to the assistant so answers cite speaker and timestamp; all processing ran on lab servers with self-hosted models, and nothing was sent to outside services
- Evaluated that pipeline stage by stage: word error rate for transcription, diarization error rate and attribution accuracy for speakers, precision and recall of extracted fields against hand labels, and faithfulness of summaries checked against the source transcript
- Built a knowledge graph of the lab's research assets (papers, datasets, data collections, models, experiments, and the decisions and entities extracted from meetings) in Neo4j (Cypher), filled from the LLM extraction pipeline, and combined graph lookups with vector search in the assistant (GraphRAG-style retrieval) to answer multi-hop questions such as which models were trained or tested on which datasets
- Fine-tuned BERT / DistilBERT text classifiers (Hugging Face) to label free-text session notes and compared them with prompted LLMs on accuracy, cost, and speed
- Distilled the prompted-LLM note classifier into a DistilBERT student trained on the LLM's labels and soft scores, cutting latency and cost while keeping accuracy close to the teacher on a held-out gold set
- Calibrated the probabilities of the note classifier and the assistant's intent router (temperature scaling for the DistilBERT models, Platt and isotonic scaling for the gradient-boosting and SVM models), checked them with reliability diagrams and expected calibration error, and used the calibrated scores to set routing and abstain thresholds
- Built silver labels with LLMs, drew a stratified sample with quotas per class and source for human adjudication into a gold set, and measured inter-annotator agreement (Cohen's kappa) before using the labels
- Applied NLP across the lab's text data: named-entity recognition and rule-based extraction (spaCy) of dates, devices, environments, and consent fields from free-text session notes; sentence-embedding semantic search and de-duplication (sentence-transformers); topic modelling of the research literature (BERTopic) to guide literature reviews; text normalization, tokenization, and grapheme-to-phoneme conversion (phonemizer, g2p) for speech corpora
- Used agentic LLM tools (Claude Code, LangChain agents) to automate multi-step coding and data tasks
- Built multi-step agents with LangGraph and Google Agent Development Kit (ADK) that call search, SQL, and document tools, served with self-hosted models (Ollama, vLLM) for private data; exposed the lab's SQL database and document store to the assistant and agents through Model Context Protocol (MCP) tool servers, and rebuilt the literature-review and data-collection-planning workflows with CrewAI and AutoGen to compare them with the LangGraph version on quality, cost, and ease of control; designed the LangGraph workflows as stateful graphs that combine deterministic steps (validation, SQL lookups, rules) with LLM reasoning, tool calling, conditional routing, checkpointed memory, and human-in-the-loop approval steps
- Designed multi-agent workflows (planner, retriever, and checker agents) for multi-step research tasks such as literature reviews and data-collection planning, with step-by-step plans, tool use, and self-checks under latency, cost, and privacy limits; rebuilt the same workflow with the OpenAI Agents SDK (handoffs, guardrails, tracing) to compare it with the LangGraph version
- Evolved the assistant from read-only answers to taking actions on users' behalf (booking participant sessions in the scheduling tables, filing data-access requests, updating collection records) with confirm-first prompts, tool allow-lists, dry runs, and rollback for failure containment
- Built a deep-search (deep research) mode for literature reviews: an agent that plans sub-questions, iteratively searches the web, scholarly APIs (Semantic Scholar, arXiv, Crossref), and the lab's own index, reads and reflects on what it found, decides when to search again, and writes a cited report; evaluated it against hand-written review summaries for coverage and citation accuracy
- Added a voice interface to the lab assistant: streaming speech recognition (faster-whisper) with voice activity detection, the LLM answer, and text-to-speech (XTTS) streamed back to the browser over WebSockets and

WebRTC, with barge-in (the user can interrupt), and latency measured for each stage (speech end to first audio)
- Integrated the assistant with the lab's existing systems the way enterprise software is integrated: single sign-on (OAuth 2.0 / OpenID Connect) so retrieval and tools respected each user's data permissions, REST APIs and webhooks to the scheduling and data-request tools, encryption in transit and at rest, and audit logs of every tool call for data-security review
- Stored datasets and ran GPU training in the cloud as well as on HPC (Azure Blob Storage / Data Lake, AWS S3 and EC2, Google Cloud / Colab); ran large Spark feature-extraction jobs on Databricks (notebooks, scheduled jobs, Delta Lake tables) with runs tracked in MLflow; on Azure, tracked runs and registered models in Azure Machine Learning workspaces and deployed a scoring model as an Azure ML managed online endpoint in a cloud trial, with Azure OpenAI used for LLM calls
- Deployed the RAG assistant API and small model services on Google Cloud Run, called frontier models (Gemini, Claude) through Vertex AI and direct APIs, and queried experiment and usage logs in SQL (PostgreSQL, BigQuery) to size problems and check fixes; grounded the assistant's answers on structured study and usage data in SQL warehouses (PostgreSQL, BigQuery, and Snowflake in a trial)
- Set up CI/CD for research code and the VQI software with GitHub Actions: automated tests, builds, and versioned releases
- Added security checks to the CI/CD pipeline: dependency vulnerability scanning (Dependabot, pip-audit), static code analysis (CodeQL, Bandit), and secret scanning, and fixed the reported issues before releases
- Deployed model services and the RAG assistant in Docker containers orchestrated with Kubernetes (k3s) on lab servers, with model management and experiment tracking in MLflow; in a cloud trial, ran the same containers on Amazon EKS with models hosted on SageMaker endpoints, and queried the Parquet feature and log tables in S3 with Amazon Athena; ran SageMaker training jobs on spot instances, right-sized GPU instance types, and scaled idle endpoints to zero to keep GPU and cloud costs down; catalogued the S3 Parquet tables in the AWS Glue Data Catalog (Glue crawlers) for Athena, ran the feature-extraction ETL as AWS Glue PySpark jobs with Glue Data Quality rules (completeness, ranges, uniqueness), chained the steps with AWS Step Functions, and used AWS Lambda functions for event-driven triggers when new recordings landed in S3
- Monitored deployed models and pipelines (performance tracking, data drift, latency, and errors) with logging and dashboards, and resolved issues; set retraining triggers (drift or a performance drop past a threshold) that re-ran the training pipeline on fresh data, and promoted the retrained model only after it passed the evaluation gates
- Applied AIOps practices to the lab's model services and RAG assistant: drift detection on inputs and scores (PSI and Kolmogorov–Smirnov tests in Evidently), Prometheus / Grafana dashboards with automated alerts, anomaly detection on service logs and metrics, runbooks, and rollback to the last good model or prompt version; designed the services as cloud-native components (stateless containers with health checks, horizontal autoscaling on Kubernetes, managed cloud storage and model endpoints)
- Led responsible-AI checks before release: bias and fairness testing with per-group error rates and disparity measures across sex, age group, language, accent, and recording device (Fairlearn, per-group DET curves), explanations of model behaviour (SHAP, per-feature explanations), and model cards documenting intended use, limitations, and model risk for sponsor and IRB review; validated model behaviour against ISO/IEC standards and sponsor requirements
- Documented data lineage (raw audio to features to labels to models, versioned with DVC), assumptions, and model cards, keeping records audit-ready for sponsors and IRB review
- Wrote solution designs, data models (entity-relationship diagrams and table specs for the study database and feature store), and integration specifications for the assistant's tool APIs before building them, and broke larger initiatives into work plans and milestones
- Built data visualizations and dashboards (Matplotlib, seaborn, Plotly Dash, Power BI) to report results and data-collection progress to advisors and sponsors
- Defined lab coding standards, reusable pipeline templates, and documentation practices, and trained new members on them

- Set lab practice for using AI coding assistants: generated code is tested and reviewed like a colleague's pull request before it is merged; reviewed generated code for correctness, fit with the existing architecture, integration risk, edge cases, and secure design
- Prepared figures, tables, slides, posters, progress reports, and technical reports for advisors, collaborators, and external stakeholders
- Explained methods, results, uncertainty, and trade-offs to non-technical and senior audiences: sponsor review boards (FBI, DHS, CITeR members), IRB reviews, and department leadership
- Coordinated with collaborators at other universities and with commercial SDK vendors (e.g. Neurotechnology) on licensing, data sharing, and joint experiments
- Assessed third-party and out-of-the-box models before adopting them: benchmarked the commercial Neurotechnology VeriSpeak SDK against open-source systems, and cloud LLM APIs (OpenAI, Claude, Gemini) against self-hosted open models on the lab regression suite, documenting capabilities, limitations, cost, latency, and data-governance implications
- Onboarded new lab members: codebase walkthroughs, data-access setup, lab SOPs, and study-protocol training
- Scheduled participants, rooms, and devices for data collections with a simple optimization model (integer programming in PuLP / OR-Tools), and forecast recruitment and collection progress with time-series models to plan resources; tracked equipment and consumables inventory (microphones, batteries, SD cards, participant gift cards) and planned reorders from the forecast collection schedule (demand forecasting and replenishment); ran Monte Carlo simulations of no-shows, session lengths, and device failures to test each schedule and give completion dates with confidence ranges, and used an LLM with a JSON schema to turn plain-language scheduling rules from staff ("no sessions before 9 on Mondays", "keep the vehicle sessions together") into structured constraints for the optimizer, validated before use
- Measured the effectiveness of participant-recruitment campaigns for the data collections (email lists, flyers, class announcements, social media posts): tracked response, sign-up, and show-up rates per channel, computed cost per enrolled participant and the return on the gift-card and printing budget, and shifted outreach and budget toward the channels that worked best
- Supported lab operations: equipment and software procurement, recording-studio setup, and scheduling participants and lab spaces

- **Methods, tools & technologies:**
  - *Speaker recognition:* ECAPA-TDNN, ResNet34-SE, ECAPA2, x-vector, WavLM-SV, i-vector/PLDA, Neurotechnology VeriSpeak (commercial SDK), S-Norm score normalization
  - *Self-supervised speech models:* WavLM, Wav2Vec2, HuBERT
  - *Machine learning:* Random Forest, Ridge, Logistic Regression, SVM/SVR, MLP, LightGBM, XGBoost, TabNet; hyperparameter search, k-fold cross-validation; PCA, Factor Analysis, ICA
  - *Signal processing / acoustic features:* voice activity detection, SNR, HNR, clipping, reverberation (C50), bandwidth, F0/pitch, formants, MFCC, LPCC; data augmentation (MUSAN, room impulse responses)
  - *Statistics:* linear mixed-effects regression, causal ablation, Spearman correlation
  - *Biometric evaluation:* ISO/IEC quality framework, EER, FNMR/FMR, minDCF, d′, AUC-ROC, DET curves, Error-vs-Reject Curves
  - *Research operations:* human-subject data collection design, multi-environment and multi-device recording
  - *Programming languages / frameworks:* Python, R, MATLAB; PyTorch, scikit-learn, XGBoost, SpeechBrain, Hugging Face (see Technical Skills)
- **Scholarship:** $6,750 Merit Scholarship (50% tuition), ECE Department, Clarkson University (Spring 2024); see Awards
- **Research sponsors:** CITeR (Center for Identification Technology Research) and NSF grants 1650503, 2413228, 2501916
- **Publications from this degree:** VPQAD dataset descriptor (IEEE Data Descriptions, 2024); speech-quality metrics study (Signals, 2026); speech enhancement comparison (Big Data and Cognitive Computing, 2026); ELAD-SVDSR

dataset descriptor (IEEE Data Descriptions, 2026); VQI (IEEE Access, 2026); children longitudinal study (IEEE TBIOM, under review; revision submitted). Full citations in Publications.

- **PhD coursework completed:**
  - EE 523 Introduction to Biometrics (A)
  - EE 622 Advanced Biometrics (A)
  - CS 570 Deep Learning (A)
  - EE 513 Intelligent System Design (A)
  - IA 651 Applied Machine Learning (B+)
  - IA 503 Introduction to Programming (Pass)
  - EE 693 Directed Study: Voice Quality Assessment for Speaker Recognition (A); EE 694 Directed Study: Deepfake Audio Synthesis and Detection (A)
  - CS 572 Image Understanding
  - EE 507 Computer Networks
  - EE 551 Digital Control
  - EE 566 Computer Architecture
  - EE 597 Instrumentation & IoT for Engineers
  - EE 603 Statistical Methods: Reliability and Life Testing (statistical analysis in SAS and R)
  - EE 691 Special Topics: Mechanosensitive Ion Channel Research (Piezo1 electrophysiology and imaging)
  - EE 999 Special Topics (×3): Neuroengineering; MEMS and Micro-Engineering; Machine Learning
  - EE 610 ECE Graduate Seminar (6 semesters)
- **Funding / Scholarships / Assistantships:**
  - Graduate Teaching Assistantship (TA), Department of ECE, Clarkson University: January 2024 – June 2026
  - Graduate Research Assistantship (RA), AVHBAC Lab, Clarkson University: January 2024 – present, covering all PhD research (see Research Experience)
- **Honors / Awards:** 1st Prize, 3-Minute Thesis (RAPS, Spring 2024); Best Graduate Oral Presentation, CS & Computational Modeling (RAPS, April 2024); $6,750 ECE Merit Scholarship (Spring 2024). Best Presenter Award (Poster), MCCNNY (April 2024). See Awards.
- **Other activities during degree:** (clubs, lab management, collaborations; Awards and activities sections later)

M.Eng., Electrical and Computer Engineering — Southern Illinois University Carbondale, Carbondale, IL, USA

- **Dates:** August 2022 – December 2023 (degree awarded December 16, 2023)
- **Department:** School of Electrical, Computer, and Biomedical Engineering
- **Concentration:** Biomedical Engineering (Neuroengineering)
- **Advisor:** Dr. Chilman Bae, Assistant Professor, School of Electrical, Computer, and Biomedical Engineering (research: bioelectrical engineering, neuroscience, mechanobiology / Piezo channels)
- **Research:** Mechanosensitive Piezo1 ion channels in neuroinflammatory astrocytes; publication in *Frontiers in Cellular Neuroscience* (2023). Details in Research Experience.
- **Funding:** Graduate Research Assistantship, Dr. Chilman Bae's lab (see Research Experience); Lab Instructor, BME 338L (see Teaching Experience)
- **Coursework (from SIUC transcript):**
  - *Biomedical engineering:* BME 417 Neuroengineering (A); BME 438 Medical Instrumentation (A); BME 532 Biomedical Imaging (A); BME 338 Biomedical Measurements (A); BME 338L Biomedical Measurements Lab (A)
  - *Computing & embedded systems:* ECE 517 Edge Computing (A); ECE 536 Embedded Systems (A); ECE 568 Machine Learning (A−); ECE 419 Computer Systems Reliability (A−)
  - *Microelectronics & hardware:* ECE 459 MEMS and Micro-Engineering (A−); ECE 522 VLSI Circuit Testing (B)
  - *Power:* ECE 589 Electric Power Distribution (C+)

- *Research:* ECE 592 Special Investigations (A): independent research on Piezo1 mechanosensitive channels in neuroinflammation (Dr. Bae's lab)

- **Honors / Awards:** (Awards section, later)

B.Sc., Electrical and Electronic Engineering (EEE) — North South University, Dhaka, Bangladesh

- **Dates:** January 2018 – August 2021
- **Department:** Department of Electrical and Computer Engineering (ECE)
- **Concentration:** High Voltage and Power Engineering
- **Thesis:** *Development of IoT-Based Real-Time Patients Vital Physiological Parameters Monitoring System Using Smart Wearable Sensors* (embedded systems). Team: Ajan Ahmed (lead), Shefat Ara Moon, A. S. M. Sabbir Hassan, Nusrat Jahan Fatima. Journal version: see Publications J9.
- **Thesis advisor:** Dr. Mohammad Monirujjaman Khan, Professor, Department of Electrical and Computer Engineering, North South University (research: wearable antennas, body-area networks, mm-wave / 5G antennas, IoT and wireless healthcare, power electronics, renewable energy)
- **Thesis summary:** An affordable, easy-to-use IoT system for remote monitoring of patients' vital signs, aimed at improving healthcare access in Bangladesh and other developing countries. Wearable sensors connected to an Arduino microcontroller measure physiological parameters; the microcontroller conditions the analog signals and sends the data over Bluetooth to a smartphone app that both patients and physicians can use.
- **Thesis work:**
  - Led a 4-member thesis team from concept to final report
  - Chose and interfaced wearable vital-sign sensors (e.g. heart rate / pulse, SpO_2, body temperature, ECG, blood pressure) to Arduino analog inputs
  - Designed signal conditioning, amplification, and filtering of analog sensor data in firmware
  - Programmed the Arduino firmware (C / C++) for real-time acquisition and processing
  - Built Bluetooth communication to a smartphone app for real-time display to patients and doctors
  - Designed the circuit, built the prototype, integrated the hardware, and tested and validated readings against reference devices; delivered a working low-cost wearable prototype
  - Kept a bill of materials with unit cost, chose components for low cost and availability, designed the enclosure in SolidWorks, and built and tested several units against reference devices
  - Wrote the thesis report and defended it
- **Highlights:**
  - Led a 4-person team to design and build a low-cost wearable IoT system for real-time remote monitoring of patients' vital signs (heart rate, SpO_2, temperature, ECG, blood pressure).
  - Interfaced physiological sensors with an Arduino, implemented analog signal conditioning and real-time firmware in C / C++, and streamed data over Bluetooth to a smartphone app for patients and doctors.
  - Prototyped, integrated, and validated the hardware against reference devices; wrote and defended the thesis.
- **Directed research:** Fuzzy-based energy-efficient control of a three-phase induction motor (EEE 498, Spring 2021, Dr. Riasat Khan). Details in Research Experience.
- **Coursework (NSU BS EEE curriculum):**
  - *EEE core:* EEE 141 / 241 Electrical Circuits I & II (+ labs); EEE 111 / 311 Analog Electronics I & II (+ labs); EEE 211 Digital Logic Design (+ lab); EEE 221 Signals and Systems; EEE 361 Electromagnetic Fields & Waves; EEE 312 Power Electronics (+ lab); EEE 321 Introduction to Communication Systems (+ lab); EEE 342 Control Engineering (+ lab); EEE 362 Power Systems (+ lab); EEE 363 Electrical Machines (+ lab)
  - *Power System Engineering concentration (High Voltage):* EEE 465 High Voltage Engineering; EEE 462 Switchgear and Protection; EEE 461 Power System Operations & Reliability
  - *Capstone and design:* EEE 299 Junior Design Project; EEE 499 / 499B Senior Design Project I & II (thesis); EEE 498 Directed Research

- *Mathematics:* MAT 120 / 130 / 250 Calculus I–III; MAT 350 Engineering Mathematics; MAT 125 Linear Algebra; MAT 361 Probability and Statistics
- *Science and computing:* PHY 107 / 108 Physics I & II (+ labs); CHE 101 Chemistry I (+ lab); BIO 103 Biology (+ lab); CSE 115 Programming Language I (+ lab)
- *Engineering practice:* EEE 452 Engineering Economics; CEE 110 / EEE 154 Engineering Drawing
- *University core:* ENG 102 / 103 Composition; ENG 111 Public Speaking; PHI 104 Introduction to Ethics; economics, political science, sociology, and history of Bangladesh

- **Honors / Awards:** (Awards section, later)

Research Experience

Graduate Research Assistant — AVHBAC Lab, Clarkson University, Potsdam, NY, USA

- **Dates:** January 2024 – present (the GRA role IS the PhD research; one continuous role; never split it into 'PhD Researcher' + 'GRA' on resumes)
- **Supervisor:** Dr. Masudul Imtiaz
- **Context:** the lab's funded voice-biometrics, face-biometrics, and child-biometrics research (NSF ERI #2501916, CITeR, FBI) throughout the PhD and after the dissertation defense, including the CITeR face image quality project (PR25, Fall 2026); all PhD research projects (PR1–PR16) were done in this role; project details are under Projects (PR1–PR8) and Data Collections (DC1).
- **Day-to-day research duties:** worked independently on frontier research goals in speech and acoustic AI: algorithm design (VQI dual-score quality index, age normalization, emotion-aware GAN voice conversion), literature review and tracking of the latest research, data collection and analysis; contributed ideas in weekly lab brainstorming sessions and sponsor project-planning meetings; collaborated in cross-functional teams with senior researchers, engineers, and product managers from CITeR industry and government member organizations; analyzed and interpreted findings and identified where they could be applied in products (voice enrollment screening, deepfake detection, forensic audio triage); presented findings to internal (lab, department) and external (CITeR sponsors, FBI, international workshops) stakeholders, explaining complex technical concepts clearly; turned research into robust, scalable, real-world tools (open-source VQI software, public datasets)
- **Highlights:**
  - Drove the lab's voice-biometrics research program after the dissertation defense, taking VQI to publication in *IEEE Access* (June 2026) and releasing the open-source VQI software.
  - Authored a standards-based evaluation study ("Beyond ROC-AUC", arXiv 2026) benchmarking 7 biometric matchers across face, voice, iris, and fingerprint with bootstrap confidence intervals.
  - Took the children's longitudinal speaker-verification study (292 children, 9 years) through first-round review at IEEE TBIOM and submitted the revision, and co-authored an ECG emotion-recognition benchmark accepted at IEEE UEMCON 2026.
  - Mentored 5 graduate and undergraduate lab members on research design, code review, presentations, and paper writing.

Graduate Research Assistant — Dr. Chilman Bae's Lab, School of Electrical, Computer, and Biomedical Engineering, Southern Illinois University Carbondale, Carbondale, IL, USA

- **Dates:** August 2022 – December 2023
- **Supervisor:** Dr. Chilman Bae
- **Context:** neuroengineering and mechanobiology lab studying mechanosensitive ion channels (Piezo1) in the nervous system
- **Research question:** Does neuroinflammation regulate the Piezo1 mechanosensitive ion channel in mouse cerebellum astrocytes (C8-S cell line)?

- **What I worked on:**
  - Experimental design and data analysis for patch-clamp electrophysiology (cell-attached and outside-out) with high-speed pressure clamp stimulation
  - LPS-induced neuroinflammation model; pharmacology (Yoda1 agonist, GsMTx4 antagonist); Piezo1 gene silencing (siRNA)
  - Calcium imaging (Cal-520 AM, confocal microscopy) and wound-healing cell-migration assays
  - Extensive hands-on microscopy: brightfield, phase contrast, DIC, epifluorescence, confocal, live-cell imaging
  - Spectral imaging on the Olympus FV3000 (TruSpectral detectors): acquired lambda-scan (x, y, λ) image cubes, a hyperspectral-style data format, and separated overlapping fluorophore emissions (Cal-520, Fluo-4, DAPI / Hoechst, GFP-tagged constructs) and cell autofluorescence with linear spectral unmixing from reference spectra; used non-negative matrix factorization (NMF) and PCA in Python for blind unmixing when reference spectra were missing
  - Ratiometric calcium imaging with Fura-2 (340 / 380 nm excitation): calibrated fluorescence ratios to intracellular Ca^{2+} concentration (in-situ Rmin / Rmax and Kd, Grynkiewicz equation), linking the measured optical signal to a chemical quantity
  - Imaging-sensor calibration and data quality: dark-frame and flat-field correction, background subtraction, photobleaching correction, fixed laser power, detector gain, and pinhole settings, checks with fluorescent reference slides and beads across sessions and microscopes, and SOPs so images from different days and setups stayed comparable; saved acquisition metadata with every file for traceability
  - Automated image analysis in Python (tifffile, scikit-image, NumPy): stack loading, drift registration, cell segmentation, ROI $\Delta F/F_0$ traces, and per-image quality metrics (saturation, focus, signal-to-background) that flagged bad acquisitions for re-imaging
  - Statistical analysis (Mann-Whitney, one-way ANOVA), figures, and manuscript editing
- **Key results:**
  - Neuroinflammation (LPS) significantly increased mechanosensitive channel currents in astrocytes ($p = 0.0042$) and lowered the half-activation pressure from 39.0 to 30.3 mmHg ($p < 0.01$), i.e. sensitized Piezo1
  - Silencing Piezo1 normalized the currents, calcium influx, and cell migration, identifying Piezo1 as the driver
  - Co-authored a paper in *Frontiers in Cellular Neuroscience* (2023), cited 36 times
- **Highlights:**
  - Contributed to experimental design and data analysis for a study showing that neuroinflammation sensitizes Piezo1 mechanosensitive channels in astrocytes, published in *Frontiers in Cellular Neuroscience* (2023; 36 citations).
  - Performed and analyzed patch-clamp electrophysiology (cell-attached and outside-out, high-speed pressure clamp), confocal calcium imaging, and wound-healing migration assays.
  - Maintained astrocyte cultures and ran LPS inflammation models, pharmacology (Yoda1, GsMTx4), and siRNA gene-silencing experiments.
  - Operated brightfield, phase-contrast, DIC, epifluorescence, confocal, and live-cell microscopes, and quantified images in ImageJ / Fiji.
  - Ran statistical analyses (Mann-Whitney, one-way ANOVA) in GraphPad Prism and Origin, showing a left shift in half-activation pressure from 39.0 to 30.3 mmHg ($p < 0.01$); produced figures and edited the manuscript.
  - Maintained lab records, SOPs, reagents, and equipment under biosafety practice.
  - Co-wrote the 2-page summary and 10-page proposal for an NSF Engineering Research Initiation award ($200K over 24 months, 2023) on a novel protective agent for noise-induced hearing loss acting on Piezo mechanosensitive channels.

Research Assistant — Dr. Mohammad Monirujjaman Khan's Group, Department of Electrical and Computer Engineering, North South University, Dhaka, Bangladesh

- **Dates:** September 2021 – July 2022

- **Supervisor:** Dr. Mohammad Monirujjaman Khan, Professor
- **Context:** research group working on wearable and mm-wave antennas, IoT, remote healthcare, and applied software systems; ran several projects in parallel
- **Projects:**
  - **Smart helmet with rear view and accident detection** (IEEE CCWC 2022, first author, 14 citations): designed an embedded safety helmet with a rear-view camera, accelerometer / gyroscope-based crash detection, and GPS / GSM emergency alerts
  - **Compact wideband V-band / W-band antenna for healthcare** (IEEE CCWC 2022, first author): designed and simulated a compact mm-wave antenna for body-centric healthcare communication; analyzed return loss, bandwidth, gain, radiation patterns, and efficiency
  - **60 GHz transparent antenna for body-centric communication** (ICAEEE 2022, co-author): evaluated a transparent mm-wave antenna for wearable healthcare communication
  - **Smart telemedicine system** (IEEE CCWC 2022, first author): built a telemedicine platform connecting patients and doctors for remote consultation, with vital-sign data
  - **Info Hospital: web / mobile healthcare system** (ICCMC 2022, first author): developed a web and mobile app for hospital information, doctor search, appointments, and health records
  - **Online mental well-being mobile app for COVID-19** (ICCMC 2022, first author): developed a mobile app with mental-health resources, self-assessment, and support features during the pandemic
  - **Web-based online financial aggregator service** (ICCMC 2022, first author): developed a web platform that brings financial products and services together for comparison
- **Highlights:**
  - Ran 7 parallel research projects in embedded systems, mm-wave antennas, and healthcare / fintech software, producing 7 IEEE conference papers in one year (6 as first author) at CCWC, ICCMC, and ICAEEE.
  - Designed a smart motorcycle helmet with a rear-view camera, heart-rate-based crash and drowsiness detection, and GPS / GSM emergency alerts (14 citations).
  - Designed and simulated compact mm-wave antennas for body-centric healthcare communication at 60, 94, and 110 GHz, including a transparent 60 GHz glass antenna (7.42 dB gain, 65.77% efficiency, 5.95 GHz bandwidth).
  - Built 4 full-stack web and mobile applications: a telemedicine platform (video calls, messaging, payments), a hospital-information system, a mental-health consultation app (Java, Jitsi video), and a fintech product aggregator with custom search algorithms.
  - Wrote and presented papers at international IEEE conferences.

Undergraduate Researcher (Directed Research) — Department of Electrical and Computer Engineering, North South University, Dhaka, Bangladesh

- **Dates:** January 2021 – April 2021 (Spring 2021, EEE 498)
- **Supervisor:** Dr. Riasat Khan (now Associate Professor, ECE, North South University)
- **Project:** *Fuzzy-Based Energy-Efficient Control of a Three-Phase Induction Motor*
- **What I worked on:**
  - Developed a fuzzy logic controller that adjusts motor excitation (flux / voltage) to the load, minimizing losses and improving the energy efficiency of a three-phase induction motor, especially at light loads
  - Modelled the induction motor, inverter, and drive (V/f and vector control) in MATLAB / Simulink
  - Designed the fuzzy inference system: membership functions, rule base, and defuzzification
  - Compared against conventional constant V/f and PI control on efficiency, power loss, speed and torque response, and stability
  - Analyzed copper and iron losses, power factor, and efficiency under varying load and speed; wrote the research report and presented the findings
- **Highlights:**

- Designed a fuzzy logic controller for energy-efficient control of a three-phase induction motor that adjusts flux / excitation to the load to minimize copper and iron losses, especially at light loads.
- Modelled the motor, inverter, and drive (V/f and vector control) in MATLAB / Simulink, and built the fuzzy inference system (membership functions, rule base, defuzzification) with the Fuzzy Logic Toolbox.
- Benchmarked the controller against conventional constant-V/f and PI control on efficiency, power loss, power factor, and speed / torque response across load conditions; reported and presented the results.

Work Experience

Antenna Engineering Intern — BlackBerry Limited (formerly Research In Motion, RIM), Waterloo, ON, Canada

- **Dates:** May 2016 – August 2017 (16-month internship)
- **Context:** antenna design and RF engineering for mobile / wireless devices at Canada's flagship smartphone and secure-communications company.
- **Duties:**
  - Designed, simulated, and optimized antennas for mobile and wireless devices (cellular LTE / 3G, Wi-Fi 2.4 / 5 GHz, Bluetooth, GPS / GNSS, NFC)
  - Built 3D electromagnetic models of antennas in device housings (PCB ground planes, batteries, displays, metal frames) and ran full-wave simulations
  - Designed impedance-matching networks using Smith charts and lumped-element tuning to maximize bandwidth and efficiency
  - Built and tuned antenna prototypes (copper-tape and flex antennas, feed and matching components) and supported PCB layout reviews
  - Measured S-parameters (return loss, VSWR, input impedance) with vector network analyzers, and radiation patterns, gain, and efficiency in anechoic chambers
  - Ran over-the-air (OTA) performance tests (TRP, TIS) and supported SAR / hearing-aid-compatibility and regulatory pre-compliance testing (FCC, ISED Canada, PTCRB / carrier requirements)
  - Studied user-interaction effects (hand and head phantoms, detuning) and antenna isolation / coexistence between radios
  - Automated measurement data collection and post-processing with MATLAB / Python scripts; built test reports and design documentation
  - Worked with mechanical, PCB, RF, and test engineering teams in design reviews and build cycles
  - Supported antenna designs through the phone build stages (EVT, DVT, and PVT) into volume production: tuned antennas on each build, tracked open issues, and signed off antenna performance for production release
  - Worked with antenna suppliers and contract manufacturers on part tolerances, assembly variation, and yield: reviewed flex, stamped-metal, and LDS antenna parts for manufacturability (DFM / DFA) and diagnosed build failures such as detuning from assembly gaps
  - Helped set production-line antenna test limits and test fixtures, and analyzed failures and rework data from builds
  - Suggested lower-cost matching components and simpler antenna parts that kept radio performance, and imported mechanical CAD (SolidWorks / Creo) into simulations
- **Tech stack:** CST Studio Suite, Ansys HFSS, Keysight ADS, FEKO, MATLAB, Python; Keysight / Rohde & Schwarz vector network analyzers, spectrum analyzers, anechoic / OTA chambers (Satimo / ETS-Lindgren), SAR measurement systems, hand / head phantoms, soldering and prototyping tools; Altium / Cadence (layout review)
- **Highlights:**

- Designed, simulated, and optimized mobile-device antennas (LTE, Wi-Fi, Bluetooth, GNSS, NFC) during a 16-month antenna engineering internship at BlackBerry (RIM) in Waterloo.
- Built and tuned antenna prototypes and matching networks, validating them with VNA S-parameter measurements and anechoic-chamber radiation-pattern, efficiency, and OTA (TRP / TIS) testing.
- Supported SAR and regulatory pre-compliance testing (FCC, ISED, carrier requirements) and automated measurement analysis with MATLAB / Python.
- Worked with mechanical, PCB, and RF test teams through device build cycles and design reviews.

Machine Learning Engineer — ImransLab, Dhaka, Bangladesh

- **Dates:** April 2018 – July 2019 (surgical-item detection, sepsis detection); August 2020 – February 2021 (COVID-19 detection GUI)
- **Context:** paid client product (not research) delivered through ImransLab: a GUI application for doctors that detects COVID-19 from lung images (chest X-ray and CT) with machine-learning models, built by the teacher (Ajan) with a student team.
- **Duties:**
  - Gathered requirements with the client, agreed on scope, deliverables, and timelines, and gave regular progress updates
  - Worked in a cross-functional team with the client's product manager, engineers, and doctors to turn requirements into product features
  - Collected and curated public medical-imaging datasets (COVID-19 Radiography Database, COVIDx, SARS-CoV-2 CT-scan data) and organized labels for COVID-19, pneumonia, and normal cases
  - Built the preprocessing pipeline: resizing, intensity normalization, contrast enhancement (CLAHE), data augmentation, and handling of DICOM and PNG / JPEG files
  - Trained and compared convolutional neural networks with transfer learning (VGG16, ResNet50, DenseNet121, InceptionV3, EfficientNet) in TensorFlow / Keras and PyTorch
  - Handled class imbalance with class weights and oversampling; tuned hyperparameters with k-fold cross-validation
  - Evaluated models with accuracy, sensitivity, specificity, precision, F1, ROC-AUC, and confusion matrices, keeping patient-level splits to avoid data leakage
  - Added explainability with Grad-CAM heatmaps so clinicians could see which lung regions drove each prediction
  - Experimented with lung segmentation (U-Net) before classification
  - Designed and built a GUI application for doctors: upload chest X-ray / CT images, get a COVID-19 / pneumonia / normal prediction with confidence scores and Grad-CAM heatmaps, and export a patient report
  - Gathered feedback from doctors on usability and workflow, and improved the interface over several release iterations
  - Built the desktop GUI (PyQt / Tkinter) and a web version with a REST API (Flask), packaged the app for installation on clinic computers (PyInstaller, Docker), and handled version releases
  - Wrote a user manual, trained doctors and staff to use the app, and delivered code, trained models, and documentation to the client
  - Protected patient data: removed identifying DICOM metadata, stored records locally with access control, and followed the client's privacy requirements
  - Tested the app end to end (unit tests with pytest, test images, and doctor acceptance testing) and fixed bugs before each release
  - Made inference fast on ordinary clinic computers by exporting models to ONNX / TensorFlow Lite and running on CPU
  - Hosted the web version on a cloud server (AWS EC2 / Heroku) for remote access, with logging to track usage and errors, and monitored model performance after release

    - Versioned models and datasets, kept a changelog, and maintained the app after hand-over with updates and support
    - Led and mentored the student team: split tasks, reviewed code, and managed the project with Git / GitHub
- **Tech stack:** Python, TensorFlow / Keras, PyTorch, torchvision, OpenCV, scikit-image, pydicom, scikit-learn, NumPy, pandas, Matplotlib, Grad-CAM; PyQt / Tkinter (desktop GUI), Flask, HTML / CSS / JavaScript (web GUI), REST API, SQLite (patient records); PyInstaller, Docker; ONNX / ONNX Runtime, TensorFlow Lite; pytest; AWS EC2 / Heroku; Git / GitHub; Google Colab / Kaggle GPUs
- **Highlights:**
    - Delivered a healthcare AI product: a GUI application that lets doctors detect COVID-19 from chest X-ray and CT images with deep-learning models, from requirements to release.
    - Built image preprocessing and augmentation pipelines and trained transfer-learning CNNs (ResNet50, DenseNet121, EfficientNet) in TensorFlow / Keras and PyTorch.
    - Evaluated models with sensitivity, specificity, and ROC-AUC using patient-level splits, and added Grad-CAM explanations for clinicians.
    - Built desktop (PyQt) and web (Flask, REST API) interfaces with Grad-CAM heatmaps and patient reports, improved them from doctors' feedback, and trained clinical users.
    - Led a development team through development, testing, and release.
- **Earlier ImransLab projects (April 2018 – July 2019; tools limited to what existed then):**
    - **Retained surgical item detection (gauze / sponge left inside the body) from ultrasound, CT, and X-ray** (AI for surgical-safety review of post-operative scans; retained gauze is also called gossypiboma)
        - Framed the clinical problem with surgeons and radiologists, and defined annotation criteria and quality standards for retained gauze, sponges, and look-alike structures
        - Collected and curated de-identified ultrasound videos and frames, CT volumes, and X-rays (DICOM), and organized labelling with radiologists in 3D Slicer, ITK-SNAP, and LabelImg, with inter-annotator agreement checks and adjudication
        - Built preprocessing for each modality: CT windowing and resampling to isotropic voxels, intensity normalization, ultrasound speckle reduction and cropping, and 2D / 3D patch extraction
        - For ultrasound specifically: worked from B-mode frames and cine loops, handled speckle noise (median, anisotropic diffusion, and wavelet filtering), depth-dependent gain and shadowing artifacts, probe and machine variation, and normalized images across devices before training; segmented targets across cine-loop frames with temporal smoothing and optical-flow-guided mask propagation (video segmentation)
        - Trained and compared segmentation models (U-Net, attention U-Net, 3D U-Net, V-Net, FCN) and detection models (YOLOv3, Faster R-CNN, RetinaNet, Mask R-CNN) in TensorFlow / Keras and PyTorch
        - Used transfer learning from ImageNet and self-supervised pre-training on unlabelled scans (convolutional autoencoders and pretext tasks such as rotation prediction and context restoration) to cope with few labelled cases
        - Handled class imbalance and hard negatives (surgical clips, drains, bowel gas, calcifications) with focal / Dice loss, hard-negative mining, and heavy augmentation
        - Evaluated with Dice, IoU, sensitivity, specificity, mAP, and FROC with patient-level splits, analyzed failure modes case by case with clinicians, and added Grad-CAM and saliency maps for review
        - Compressed the models (knowledge distillation, pruning, quantization) and served them through a review tool that flags suspicious regions for the radiologist
        - Built the processing and visualization path for the review tool: batch inference over scan series, overlays of predicted masks and boxes, and per-case reports; documented models, training settings, and dataset procedures for engineers
        - Added geometric analysis of each 3D segmentation: object volume, centroid, and extent from the voxel spacing, surface meshes with marching cubes (scikit-image, VTK) for 3D review, and surface-distance checks of mask quality (Hausdorff distance, average surface distance) alongside Dice
    - **Sepsis (septicemia) early detection from clinical data**

- Built early-warning models that predict sepsis hours before onset from vital signs, laboratory results (e.g. lactate, white blood cell count, creatinine), demographics, and clinical notes in hospital records
- Prepared longitudinal patient time series: irregular sampling, missing-value imputation, forward filling, time windows, and patient-level train / test splits to avoid leakage (MIMIC-III)
- Defined sepsis cohorts and labels from billing diagnosis and procedure codes (ICD-9, Angus criteria; claims-style admission records) together with clinical criteria (Sepsis-3); used the admission and health-insurance fields (payer type: Medicare, Medicaid, private, self-pay) to describe cohorts and check model performance across payer groups
- Trained and compared gradient boosting (XGBoost, LightGBM), logistic regression, LSTM / GRU, and 1D convolutional models, with SHAP explanations of the main risk drivers
- Fused structured data (vitals, labs) with clinical-note text features (TF-IDF, word2vec / fastText embeddings) and chest X-ray CNN features where available (multimodal fusion)
- Built the clinical-note NLP pipeline: text cleaning, sentence splitting, abbreviation expansion, negation handling (NegEx-style rules), medical concept extraction (MetaMap / cTAKES and spaCy rules), and TF-IDF and word2vec features for symptom and infection mentions
- Evaluated with AUROC, AUPRC, sensitivity at fixed alert rates, calibration, and lead time before onset, and reviewed false alarms with clinicians
- Wrote model documentation and validation reports, and presented results to clinical partners
- Adjusted alert thresholds with clinicians to balance sensitivity against false-alarm burden

Network and IT Administrator — McGill University, Montreal, QC, Canada

- **Dates:** September 2015 – April 2016
- **Context:** IT and network administration for a research lab environment serving ~65 students, with lab servers, workstations, instruments, and multi-terabyte research data.
- **Duties:**
  - Designed and implemented the lab network, giving 65 students secure access to computers and shared storage across building labs
  - Provisioned and imaged workstations for new students; created user accounts and access permissions on the secure lab network
  - Procured lab computers, instruments, and peripherals: gathered requirements, compared vendors and quotes, and handled purchasing and asset tracking
  - Administered and monitored multiple lab servers for uptime, performance, patching, and security; reduced downtime through proactive monitoring
  - Designed and ran backups of several terabytes of research data replicated across geographically separate locations (disaster recovery)
  - Managed user accounts, file-share permissions, and access control; enforced security practices (updates, antivirus, firewall rules, least-privilege access)
  - Diagnosed and resolved hardware, software, network, and printer problems for students and staff; kept an issue log and documentation
  - Installed and licensed research and productivity software; maintained lab computers and instruments
  - Wrote how-to guides for students and onboarding documentation for the lab network
- **Tech stack:** Windows 10 / Server, Active Directory, Group Policy, Linux (Ubuntu / CentOS) servers, TCP/IP, DHCP, DNS, VLANs, switches and routers, Wi-Fi, VPN, firewalls, NAS / SAN storage, RAID, SMB / NFS file shares, backup software (Veeam / rsync / robocopy / cloud backup), remote desktop (RDP / SSH), monitoring tools (Nagios / Zabbix / PRTG), imaging tools (Clonezilla / WDS), PowerShell and Bash scripting, ticketing / issue logs
- **Highlights:**
  - Designed and implemented a secure lab network giving 65 students access to shared computers and storage across multiple building labs at McGill University.

- Built a multi-site backup strategy protecting several terabytes of research data across geographically separate locations.
- Administered and monitored lab servers to cut downtime and harden security (patching, access control, firewall rules).
- Procured, configured, and supported lab computers and instruments; onboarded new students with workstation setup and secure network access.

Accounting Clerk — McGill University, Financial Services, Montreal, QC, Canada

- **Dates:** September 2015 – April 2016 (held alongside the Network and IT Administrator role)
- **Context:** university central finance office handling payments, invoices, and financial records for McGill departments.
- **Duties:**
  - Processed accounts payable and receivable: checked, coded, and entered vendor invoices, payment requests, and receipts
  - Reconciled accounts, statements, and ledgers; investigated and fixed discrepancies
  - Prepared and processed expense reports, reimbursements, and journal entries using the university's chart of accounts (fund / organization / account / program codes)
  - Kept accurate financial records and supporting documentation for audits
  - Answered departmental, vendor, and staff enquiries about payments and account status
  - Prepared spreadsheets and summary reports for supervisors; supported month-end close
  - Followed internal controls, approval workflows, and confidentiality requirements for financial data
- **Tech stack:** university ERP / financial systems (e.g. Banner Finance), Microsoft Excel (reconciliations, pivot tables, lookups), Microsoft Office, document-management and scanning systems
- **Highlights:**
  - Processed accounts payable and receivable, expense reimbursements, and journal entries for McGill University departments with high accuracy under internal controls.
  - Reconciled accounts and ledgers, resolved discrepancies, and kept audit-ready financial records.
  - Answered payment enquiries from departments, vendors, and staff, and prepared Excel reports to support month-end close.

Employee Relations — Volt Workforce Solutions, Montreal, QC, Canada

- **Dates:** September 2014 – April 2016
- **Context:** Volt is an international staffing and workforce-solutions company; employee-relations role supporting assigned contract and temporary personnel.
- **Duties:**
  - Evaluated the performance of assigned employees and recommended hiring, promotion, termination, and disciplinary actions
  - Investigated employee claims and workplace concerns through confidential interviews, fact-finding, and documentation, working with HR on resolutions
  - Created and maintained employee files, logs, and HR records accurately and confidentially
  - Handled onboarding paperwork, scheduling, attendance and timesheet tracking, and communications with clients and staff
  - Applied company policies and employment standards consistently; escalated sensitive cases appropriately
  - Completed administrative reporting and correspondence
- **Tech stack:** Microsoft Office (Excel, Word, Outlook), HR information systems / applicant-tracking systems, timesheet and scheduling tools
- **Highlights:**

- Evaluated employee performance and recommended hiring, promotion, termination, and disciplinary actions for assigned personnel at an international staffing firm.
- Investigated employee claims through confidential interviews and worked with HR to resolve them fairly and in line with policy.
- Maintained accurate, confidential employee files and logs and handled HR administration for assigned staff.

Publication Data Mining Research Assistant — McGill University, Montreal, QC, Canada

- **Dates:** January 2015 – April 2015
- **Context:** bibliometric project cataloguing and analyzing the research publications of McGill professors.
- **Duties:**
  - Gathered all publications by McGill faculty from Web of Knowledge / Web of Science and exported the records as HTML
  - Wrote Java programs to parse the HTML and extract structured fields (authors, titles, journals, years, citations, affiliations) into CSV files
  - Cleaned, de-duplicated, and validated the extracted data, and loaded it into a relational database (MySQL) as an ETL pipeline for statistical analysis in SAS and Excel
  - Analyzed publication statistics and built charts and dashboards in Excel (pivot tables, trend and comparison charts)
  - Presented findings to the supervising professor
- **Tech stack:** Java (HTML parsing: jsoup / regular expressions, file I/O), Web of Science / Web of Knowledge, CSV, Microsoft Excel (pivot tables, charts)
- **Highlights:**
  - Built a Java data-extraction pipeline that parsed Web of Science records into structured CSV datasets of McGill faculty publications.
  - Analyzed and visualized publication statistics in Excel and presented the insights to the supervising professor.

Computer Programmer and Technician — Dogma Diagnosis and Hospital Ltd., Dhaka, Bangladesh

- **Dates:** May 2013 – August 2013
- **Duties:**
  - Resolved hardware, software, and network problems for 20–25 doctors and clinical staff
  - Set up, calibrated, and maintained an eye-tracking system and other specialized research / diagnostic computers
  - Supported clinical workstations running hospital-information and patient-record systems, and kept clinical data backed up and secure
  - Backed up and maintained 8 lab hard drives; managed data integrity and storage
  - Installed and updated software, configured peripherals and printers, and wrote small scripts and programs to automate routine tasks
  - Documented configurations and fixes
- **Tech stack:** Windows, Microsoft Office, eye-tracking hardware and software, backup utilities, basic scripting (batch / Python / Java), networking basics
- **Highlights:**
  - Delivered IT support to 20–25 doctors, and set up and maintained specialized diagnostic equipment, including an eye-tracking system.
  - Backed up and maintained 8 lab drives, protecting clinical and research data.

Customer Support Consultant — HAF General Hospital and Diagnostic Centre Ltd., Dhaka, Bangladesh

- **Dates:** May 2013 – August 2013
- **Duties:**
  - Provided technical support to hospital staff, affiliates, and doctors for VPN, telephony, hardware / software, wireless, and other IT issues
  - Triaged, assigned, and escalated tickets to the right IT teams as the link between IT departments and end users
  - Followed up with users until issues were resolved and communicated status clearly to non-technical staff
  - Kept records of recurring issues and resolutions
- **Tech stack:** VPN clients, IP / PBX telephony, Wi-Fi, Windows, Microsoft Office, help-desk ticketing
- **Highlights:**
  - Supported hospital staff and doctors with VPN, telephony, wireless, and hardware / software issues, triaging and escalating tickets across IT teams as the main point of contact.

Research Funding & Grant Proposals

Funded Projects

- **NSF ERI (Award #2501916): "ERI: Advancing Biometric Identification for Younger Children through Age-Adaptive Technology Innovations"**. National Science Foundation, Engineering Research Initiation program; July 2025 – June 2027; **total $200,000**. PI: Dr. Masudul H. Imtiaz (Clarkson University). **Role:** co-wrote the proposal; graduate researcher on the funded project. **Scope:** age-adaptive biometric identification for infants and toddlers (ages 0–4) across face, fingerprint, iris (and voice): quality-aware capture devices, longitudinal studies of how biometric traits change in early childhood, and privacy-compliant protocols with healthcare and education partners. **Outputs I contributed to:** children's longitudinal speaker verification study (IEEE TBIOM, under review); synthetic fingerprints for children under four (IEEE UEMCON 2026); VQI (*IEEE Access*, 2026) acknowledges this grant.
- **FBI CJIS (via West Virginia University): "Quality Checking and Enrolling High-Quality Voice Samples to Ensure Accessibility and Usability of Data"**. FBI Criminal Justice Information Services Division, subaward through WVU; Spring 2024 – Spring 2025; **total $108,000**. PI: Dr. Masudul Imtiaz (Clarkson). **Role:** co-wrote the proposal; graduate researcher. **Outputs:** voice quality assessment research that led to the Voice Quality Index (VQI; *IEEE Access*, 2026) and the speech-quality metrics study (*Signals*, 2026).
- **NSF CITeR 25F-04W: "Open-Source Facial Image Quality Toolbox Based on ISO/IEC 29794-5"**. Fall 2026; multi-university (Clarkson University and West Virginia University); **total $45,000**. Team: **A. Ahmed**, A. Storey, Dr. M. Imtiaz (Clarkson), Dr. P. Gyawali (WVU). **Role:** co-wrote the proposal; lead researcher; built the FUQI face-quality models, labels, and evaluation; wrote and presented the final report at the Fall 2026 CITeR Program Review. Details in PR25.
- **NSF CITeR 25S-04C: "Longitudinal Progression of Biometric Recognition in Fingerprint, Face, Iris, and Voice"**. Team: Dr. S. Schuckers, Dr. M. Imtiaz, M. R. Sumi, N. Venkataswamy, **A. Ahmed**, L. Igene, A. Hossain (Clarkson); progress poster Fall 2025, final report Spring 2026 (ORNL, Oak Ridge, TN). Center for Identification Technology Research (NSF IUCRC); Fall 2025 – Spring 2026; **total $45,000**. Co-PI: Dr. Masudul Imtiaz. **Role:** co-wrote the proposal; graduate researcher; data collector for all modalities; led the voice data analysis. **Outputs:** children's longitudinal speaker verification study (292 children, 9 years; IEEE TBIOM, under review).
- **NSF CITeR 24F-02C: "Development of Extended-Length Audio Dataset for Advanced Deepfake Synthesis and Detection"**. Spring 2025 – Fall 2025; **total $45,000**. PI: Dr. M. Imtiaz; team: R. Vijaykumar, **A. Ahmed** (Clarkson). **Role:** co-wrote the proposal; student co-investigator; led data curation; presented the progress-report poster to sponsors (CITeR Spring 2025 Program Review, Idiap, Switzerland). **Sponsor reporting:** 1-page proposal and new-proposal presentation (CITeR Fall 2024, Niagara Falls, NY, Oct 2024); progress poster (Spring 2025, Idiap,

Switzerland); final report (Fall 2025, JHU Applied Physics Laboratory, Laurel, MD, Nov 2025). **Outputs:** ELAD-SVDSR dataset (IEEE DataPort) and *IEEE Data Descriptions* paper (2026).

- **NSF CITeR 23F-03C: "Evaluating and Progressing the State of the Art in Deepfake Audio Generation and Detection"**. Funding agency: U.S. Department of Homeland Security (DHS), through CITeR; Spring 2024 – Fall 2024; **total $45,000**. PI: Dr. M. Imtiaz; team: Y. Sukhdeve, **A. Ahmed**, R. Vijaykumar (Clarkson). **Role:** co-wrote the proposal; student researcher; invited speaker at the Fall 2025 CITeR Webinar. **Sponsor reporting:** final report presentation (CITeR Fall 2024 Phase II, Niagara Falls, NY, Oct 29, 2024). **Outputs:** deepfake-ready multi-microphone voice data, deepfake generation and evaluation pipeline, SPEAD dataset.
- **NSF CITeR 23S-07C: "Modified EKF-ResNet Architecture for Automatic Speaker Recognition from Noisy Environment"**. NSF CITeR; **total $45,000**. PI: Dr. M. Imtiaz; team: **A. Ahmed** (lead student researcher), S. Purnapatra (Clarkson). **Role:** co-wrote the proposal; led the project; invited talk at the International Martigny Biometrics Workshop 2025 (EAB / CITeR / Idiap), Switzerland. **Sponsor reporting:** presented the final report (CITeR Spring 2024 Program Review, WVU, Morgantown, WV, May 16, 2024). **Outputs:** noisy-environment adult voice data (VPQAD), denoising pipeline, groundwork for VQI.
- **NSF CITeR 25S-01CW-J: "A Multi-Layered Framework for Robust Deepfake Audio Detection & Tamper Analysis"** (funded from Spring 2025); **total $45,000**. Multi-university project: Dr. M. Imtiaz and **A. Ahmed** (Clarkson), Dr. N. Karimian and Dr. S. Tehranipoor (West Virginia University). **Role:** co-wrote the proposal and presented it to CITeR industry and government members at the Spring 2025 Program Review, Idiap Research Institute, Martigny, Switzerland.
- **FBI (FY25 Programs Research and Standards Unit): "Evaluating the Effectiveness of Synthesized and Enhanced Speech for Speaker Recognition Models"**, through West Virginia University; awarded December 2024; project period Spring 2025 – Spring 2026; **total $120,000**. Co-PI: Dr. Masudul Imtiaz (Clarkson). **Role:** co-wrote the 1-page initial proposal and, after initial acceptance, the 2-page detailed proposal.
- **FBI Human Subjects Research: "Voice Sample Enrollment Analysis"**. FBI-funded human-subjects study; awarded Fall 2024; **total $50,000**. PI: Dr. Masudul Imtiaz (Clarkson). **Role:** wrote the detailed Institutional Review Board (IRB) human-subjects research proposal; led the resulting data collection. **Outputs:** VSEA-DC dataset (100 participants, 6 environments, 2 devices, ~40–50 h; see Projects), used as a real-world test set for VQI.
- **NSF ERI CIVIC: "Longitudinal Study of Child Biometrics"**. National Science Foundation, Engineering Research Initiation (CIVIC); awarded 2025; **total $200,000**; separate from award #2501916. PI: Dr. Masudul Imtiaz. **Role:** co-wrote the 10-page detailed proposal.
- **NSF ERI: "The Effect of Novel Protective Agent of Noise Induced Hearing Loss on Piezo Mechanosensitive Channel"**. National Science Foundation, Engineering Research Initiation; awarded 2023; **total $200,000 over 24 months**. PI: Dr. Chilman Bae (Southern Illinois University Carbondale). **Role:** co-wrote the 2-page proposal summary and the 10-page detailed proposal during the M.Eng. **Related outputs:** Piezo1 research in *Frontiers in Cellular Neuroscience* (2023) and *Biophysical Journal* abstract (2023).

Research Support

- Dissertation research supported by CITeR and NSF grants 1650503, 2413228, and 2501916.
- $6,750 ECE Department Merit Scholarship (Spring 2024); Graduate Teaching Assistantship (2024–2026); Graduate Research Assistantship (2024–present). See Awards and Education.

Resume bullets (grant writing)

- Co-wrote 12 funded research proposals with faculty totaling over $1.1M: 3 NSF Engineering Research Initiation awards ($200K each: child biometrics and a longitudinal child-biometrics ERI CIVIC award at Clarkson, and noise-induced hearing loss / Piezo channels at SIUC), 3 FBI-funded projects ($120K synthesized and enhanced speech for speaker recognition, $108K voice-sample quality checking, $50K voice enrollment human-subjects study), and 6 CITeR (NSF IUCRC) projects ($45K each, including the 25F-04W face image quality toolbox).
- Presented proposals and progress reports to industry and government sponsors, including a funded multi-university (Clarkson–West Virginia University) proposal at the CITeR Spring 2025 Program Review in Switzerland.

- Turned funded projects into peer-reviewed outputs: 3 public datasets, an *IEEE Data Descriptions* paper, and the groundwork for the Voice Quality Index (*IEEE Access*).

Projects

PhD Projects — AVHBAC Lab, Clarkson University (2024–2026)

PR1. Voice Quality Index (VQI): Utility-Based Voice Quality Assessment for Speaker Recognition

- **Dates:** 2024 – 2026 · **Role:** lead researcher and developer (dissertation core) · **Supervisor:** Dr. Masudul Imtiaz; collaborator Dr. Stephanie Schuckers
- **Funding:** FBI CJIS "Quality Checking and Enrolling High-Quality Voice Samples" ($108K), CITeR, NSF #2501916
- **Overview:** ISO/IEC has standardized per-sample quality metrics for fingerprint, face, and iris, but not for voice. VQI is the first utility-based, explainable voice quality metric: it predicts how useful a recording will be for speaker verification and returns two 0–100 scores, VQI-S (signal quality) and VQI-V (voice distinctiveness).
- **Duties:**
  - Defined the problem against the ISO/IEC 29794-1 "quality as utility" framework, and designed dual-score outputs that give actionable feedback (re-record vs. speaker limitation)
  - Built a large-scale data pipeline over 11 datasets (~2.6 M utterances; VoxCeleb1/2, Common Voice, LibriSpeech, VOiCES, VCTK, CN-Celeb, VPQAD, VSEA-DC) with MUSAN / RIR augmentation, 16 kHz resampling, VAD, and loudness normalization; parallelized processing of millions of files with Dask / PySpark and stored features in Parquet tables for SQL queries
  - Designed consensus labelling from 3 speaker-verification systems (ECAPA-TDNN, ResNet34-SE, ECAPA2) with s-norm over 1.2 M mated and 28 M non-mated pairs, producing a balanced 20,288-sample training set
  - Tried custom asymmetric objectives in XGBoost and LightGBM (custom gradient and Hessian) that penalize over-rating poor-quality samples more than under-rating good ones, since accepting a bad recording costs more
  - Ran large-scale embedding inference for labelling across multiple GPUs (PyTorch DistributedDataParallel, torch.multiprocessing) with batched, mixed-precision (FP16) inference, and tuned batch size and data loading to raise throughput
  - Wrote GPU code for pairwise scoring of millions of trial pairs (CUDA kernels via CuPy / Numba, batched matrix operations) and profiled bottlenecks with the PyTorch Profiler, NVIDIA Nsight, and py-spy
  - Engineered 705 explainable acoustic features (SNR, HNR, clipping, C50, bandwidth, F0, formants, harmonic richness, etc.) and a 3-stage selection pipeline (430 + 133 kept)
  - Benchmarked 8 model families (RF, Ridge, SVM / SVR, MLP, LightGBM, XGBoost, TabNet, LR) as classifiers and regressors with 5-fold CV and hyperparameter search; compared 7 dimensionality-reduction settings (PCA, FA, ICA)
  - Trained, evaluated, and versioned hundreds of model configurations concurrently (model families × targets × feature sets × folds) as SLURM job arrays and parallel workers, logged every run's metrics and artifacts to MLflow, and kept all 28 released models loadable side by side in the app with a shared inference and monitoring path
  - Clustered recordings by their acoustic features (k-means, DBSCAN, PCA / UMAP views) to find quality groups, and flagged anomalous recordings with Isolation Forest and robust z-scores before labelling
  - Evaluated with ISO/IEC Error-vs-Reject Curves, ranked DET, and dual-score quadrant analysis across 5 datasets and 5 providers, including held-out x-vector and WavLM-SV; analyzed emotion effects on the ESD corpus
  - Framed quality-based rejection as selective prediction (a reject option that flags inputs the system should not be trusted on): compared rejecting samples by VQI with rejecting by the verification system's own match-score confidence on Error-vs-Reject curves, and checked how well the scores were calibrated

    - Built and open-sourced a desktop app and CLI (record / upload, dual gauges, waveform and spectrogram, per-feature explanations, report export)
    - Generated plain-language feedback text for each quality issue with templates and LLM drafting, and tested readability with users
    - Added opt-in, anonymous usage logging to the VQI software (feature use, session length, GUI vs CLI and batch use, errors, and which quality feedback users saw), and combined it with GitHub release downloads and issue reports
    - Analyzed the software usage patterns (SQL, pandas, dashboards) to find where users got stuck and which quality problems were most common, and used the findings to prioritize improvements such as clearer feedback text, faster batch scoring, and a simpler install
    - Built real-time streaming audio in the app: full-duplex capture and playback with PortAudio / sounddevice callbacks, ring buffers, chunked VAD, and live quality scoring on short frames
    - Kept all audio processing and scoring on the user's own device (no cloud upload) for privacy, with local per-user enrollment so feedback adapts to each speaker (privacy-first on-device personalization)
    - Compressed the models for on-device inference: knowledge distillation of the ensemble into a compact student model, feature and tree pruning, and INT8 post-training quantization for fast CPU inference on ordinary laptops
    - Optimized end-to-end latency and throughput of the audio pipeline: vectorized feature extraction (NumPy, Numba), C++ extensions for hot loops (pybind11), a Rust prototype of the feature extractor (PyO3), multiprocessing for batch scoring, and model export to ONNX Runtime and TensorRT
    - Served the exported ONNX / TensorRT models and the speaker-embedding models on the lab GPU server with NVIDIA Triton Inference Server (dynamic batching, concurrent model instances, model ensembles for preprocessing plus scoring, gRPC and HTTP endpoints), and load-tested throughput and latency with perf_analyzer
    - Used LLMs (Claude Code, GitHub Copilot, ChatGPT / GPT-4o) for code generation, refactoring, debugging, unit tests, and documentation across the data pipeline, feature extractors, model-training scripts, and the desktop app / CLI
    - Used LLMs to draft plain-language, per-feature feedback messages for the app (what each quality issue means and how to fix it), then reviewed and edited them
    - Used LLM assistants for literature search and summarization, manuscript editing, and drafting responses to reviewers
    - Wrote the *IEEE Access* paper as corresponding author and handled peer review; presented at the ECE Graduate Seminar
- **Tech stack:** Python, NumPy, pandas, SciPy, scikit-learn, XGBoost, LightGBM, pytorch-tabnet, PyTorch, SpeechBrain, Hugging Face, librosa, Praat / Parselmouth, torchaudio, Matplotlib; PyQt / Tkinter, PyInstaller; HPC / GPU; Git / GitHub; LLMs: Claude (Claude Code), ChatGPT / GPT-4o, GitHub Copilot
- **Highlights:**
    - Developed the Voice Quality Index, the first utility-based voice quality metric that predicts speaker-verification performance, cutting false non-matches by up to 38% when the lowest-quality 20% of samples are rejected (*IEEE Access*, 2026).
    - Built an end-to-end ML pipeline over 2.6 M utterances and 11 datasets, from consensus labelling with 3 speaker-verification systems to 705 engineered features and 8 benchmarked model families.
    - Validated generalization on 5 datasets and 5 verification systems, including 2 never used in training, with ISO/IEC Error-vs-Reject and DET analysis.
    - Shipped an open-source desktop app and CLI with explainable, per-feature quality feedback (github.com/ahmedajan/VQI-v1).
- **Outputs:** J1 (*IEEE Access*), T2 seminar, VQI software

PR2. Speaker Verification in Children: A 9-Year Longitudinal Study

- **Dates:** 2025 – 2026 · **Role:** lead researcher (first and corresponding author) · **Team:** F. Ahmad, Y. Sukhdeve (mentees), Dr. S. Schuckers, Dr. M. Imtiaz
- **Funding:** NSF CITeR "Longitudinal Progression of Biometric Recognition in Fingerprint, Face, Iris, and Voice" ($45K); NSF ERI #2501916
- **Status:** under review at IEEE TBIOM (first-round comments addressed; revision submitted)
- **Overview:** How well does speaker verification hold up as children's voices develop? This study analyzes 292 children (ages 4–18) from an IRB-approved Clarkson longitudinal biometrics study, recorded over 16 semiannual collections spanning 9 years, to our knowledge the largest and longest such dataset.
- **Duties:**
  - Curated and cleaned 9 years of longitudinal child voice data (subject selection, age estimation, mated cross-age pairs by time gap)
  - Ran a commercial system (VeriSpeak), 3 deep-learning backends (ECAPA-TDNN, ResNet, ECAPA2), and 10 interpretable i-vector / PLDA pipelines built on hand-crafted feature families (MFCC, LPCC, formants, pitch, etc.)
  - Reported FNMR at FMR = 0.1% with bootstrapped confidence intervals per age and time gap; built heatmaps by enrollment age × time gap
  - Modeled drivers with nested LMER and GLMM mixed-effects models (time gap, enrollment age, pitch, formant drift, duration, puberty)
  - Treated the collection as an observational study: separated the effect of time gap from confounders (enrollment age, puberty, recording session and device) with inverse-probability weighting and double machine learning (DoubleML / EconML, gradient-boosting nuisance models), and estimated heterogeneous effects by age group with causal forests
  - Designed an intervention-based causal ablation and age-normalization method (z-score, linear, polynomial growth models) and re-scored verification
  - Used LLMs (Claude, ChatGPT) to write and debug R / Python analysis code for the LMER / GLMM models and plots, summarize related literature, and draft the reviewer response
  - Wrote the manuscript, led the response to reviewers, and supervised 2 student co-authors
- **Tech stack:** Python, R (lme4, lmerTest), statsmodels, NumPy / SciPy, scikit-learn, Kaldi-style i-vector / PLDA, SpeechBrain, Neurotechnology VeriSpeak SDK, Matplotlib / seaborn; LLMs: Claude, ChatGPT
- **Highlights:**
  - Led a 9-year longitudinal study of speaker verification in 292 children (ages 4–18), showing FNMR climbs to ~33% after 5 years and ~60% after 7 years at FMR = 0.1%.
  - Identified the drivers of degradation (time gap, pitch, and formant drift) with LMER / GLMM mixed-effects models and causal feature ablation across 14 verification systems.
  - Developed age normalization that recovers up to 35% of lost performance on i-vector systems and 9–12% on ECAPA-TDNN / ResNet (IEEE TBIOM, under review).
- **Outputs:** R1 (TBIOM, under review)

PR3. VSEA: Voice Sample Enrollment Analysis (FBI-funded; NDA)

- **Dates:** FY24 (Fall 2024 – 2025) · **Role:** project lead; wrote the IRB proposal, ran the data collection, did the analysis, and wrote the final technical report
- **Funding:** FBI human-subjects research award, "Voice Sample Enrollment Analysis" ($50K)
- **Status:** completed; final technical report delivered to the sponsor (results under a non-disclosure agreement)
- **Overview:** Investigated whether traditional audio quality metrics can predict speaker-recognition performance, a key operational question for investigative and forensic work where analysts must decide which recordings are usable. Preliminary analyses on existing datasets showed little correlation, so a controlled dataset (VSEA-DC) was designed, collected, and systematically degraded to test the finding across commercial, research-grade, and lightweight speaker-recognition systems.

- **Duties:**
  - Wrote the IRB human-subjects protocol, consent materials, and participant scripts
  - Designed and ran VSEA-DC: 100 adult participants, 6 real-world environments (quiet lab, dining area, noisy HVAC lab, running vehicle, 2 telephone channels), 2 platforms (Inni Oasis R1 studio recorder; iPhone 16 Pro call recording); measured ambient noise levels per environment; 598 high-quality baseline recordings (~40–50 h)
  - Designed a denoising protocol that systematically changes quality across 5 processing levels, and an enrollment–probe verification framework
  - Computed audio quality metrics (SNR, PESQ, STOI, jitter, shimmer) and scored verification with 3 systems: Neurotechnology VeriSpeak (commercial), SpeechBrain ECAPA-TDNN (research-grade), and Resemblyzer (lightweight GE2E)
  - Ran correlation and variance-explained analysis across 7,645 comparisons (2,965 VeriSpeak, 2,340 SpeechBrain, 2,340 Resemblyzer), with distribution and scatter-matrix visualization
  - Wrote the final technical report for the FBI, including operational recommendations, risk and limitations analysis, and future directions
- **Tech stack:** Python, NumPy, pandas, SciPy (Pearson / Spearman), statsmodels, Matplotlib / seaborn, pesq, pystoi, Parselmouth (jitter / shimmer), noisereduce / denoising tools, SpeechBrain, Resemblyzer, Neurotechnology VeriSpeak SDK, LaTeX; Inni Oasis R1, iPhone 16 Pro, Audacity
- **Highlights:**
  - Led an FBI-funded ($50K) speaker-recognition study end to end: wrote the IRB protocol, collected 598 recordings from 100 participants across 6 real-world environments and 2 devices, and delivered the final technical report.
  - Turned the finding into the design basis for the Voice Quality Index, a utility-based quality metric later published in *IEEE Access*.

PR4. Synthesized and Enhanced Speech for Speaker Recognition (FBI-funded)

- **Dates:** 2025 – 2026 · **Role:** graduate researcher; co-wrote the 1-page initial and 2-page detailed proposals
- **Funding:** FBI FY25 Programs Research and Standards Unit, via West Virginia University ($120K)
- **Overview:** Evaluated how speech enhancement (denoising) and synthetic speech affect speaker-recognition models, for forensic and law-enforcement use.
- **Duties:**
  - Benchmarked deep-learning speech enhancement (Wave-U-Net, CMGAN, U-Net) on unseen real-world corpora (SpEAR, VPQAD, Clarkson) with SNR, PESQ, and VeriSpeak speaker-verification scores
  - Ran enhancement models in streaming mode on overlapping audio chunks (overlap-add), measuring real-time factor, latency, and GPU memory, and applied FP16 and TorchScript / torch.compile to speed up inference
  - Evaluated synthetic and cloned speech against speaker-recognition systems, and helped build interpretable deepfake-detection baselines on the Fake-or-Real dataset
  - Analyzed trade-offs between noise suppression, perceptual quality, and speaker-identity preservation; wrote reports for the sponsor
- **Tech stack:** Python, PyTorch, Wave-U-Net, CMGAN, U-Net, pesq, pystoi, librosa, scikit-learn, Neurotechnology VeriSpeak SDK, NVIDIA RTX 4080
- **Highlights:**
  - Evaluated 3 deep-learning speech-enhancement models on unseen real-world data for an FBI-funded ($120K) project, showing a trade-off: U-Net gave up to +235% SNR, CMGAN the best PESQ, and Wave-U-Net the best speaker-identity preservation (up to +30% VeriSpeak).
  - Co-wrote the funded proposals and contributed evaluation analysis to a peer-reviewed benchmark (*Big Data and Cognitive Computing*, 2026).
- **Outputs:** J4 (BDCC), C2 (MDTS)

PR5. Deepfake Audio Generation, Detection & Extended-Length Voice Datasets (CITeR program)

- **Dates:** Spring 2024 – 2026 · **Role:** student researcher and co-writer of 3 funded CITeR proposals; curated the ELAD collection
- **Funding:** NSF CITeR 23F-03C (DHS-funded, $45K), 24F-02C ($45K), 25S-01CW-J with West Virginia University ($45K)
- **Overview:** A multi-phase program to create deepfake-ready voice datasets recorded in real conditions, generate high-quality voice clones, and evaluate and detect them, including a multi-layered deepfake detection and tamper-analysis framework.
- **Duties:**
  - Generated voice deepfakes with 6+ TTS / voice-cloning systems (Tortoise TTS, ElevenLabs, YourTTS, XTTS, MeloTTS, MetaVoice) and tuned hyperparameters
  - Sped up large batch generation with autoregressive transformer TTS models (XTTS, Tortoise): KV caching, FP16, batching, and serving with vLLM / TensorRT-LLM-style inference engines on multiple GPUs
  - Profiled the TTS decoding loop and attention (PyTorch Profiler, Nsight), tried FlashAttention / scaled-dot-product attention kernels and torch.compile, and measured real-time factor and quality after each change
  - Piloted a 5-microphone synchronized collection (8 subjects), then curated ELAD: 36 participants × 45 minutes on 5 devices (AT2020, SM58, TOZO earbuds, Inni Oasis R1, spy pen)
  - Annotated and cleaned audio (Audacity), removed silences and fillers, and applied speaker diarization and VAD
  - Evaluated speech recognition and diarization on the recordings: word error rate (WER) with Whisper transcripts against reference scripts, and diarization error rate (DER) with pyannote.audio
  - Attributed speakers in the multi-speaker and long recordings by combining pyannote diarization with speaker-recognition embeddings (ECAPA-TDNN) matched against each participant's enrollment, and checked attribution errors by hand
  - Compared 4 signal-processing denoisers (spectral gating, VAD-adaptive, spectral statistics, multi-band) against Meta's deep-learning Denoiser
  - Tested deepfakes against VeriSpeak (FAR-calibrated thresholds) and WeSpeaker cosine similarity
  - Threat-modelled attacks on voice biometric systems (replay, text-to-speech and voice-conversion deepfakes, audio tampering and splicing, channel manipulation) and mapped each to detection and evaluation methods, following ISO/IEC 30107 presentation attack detection concepts; this fed the multi-layered deepfake detection and tamper-analysis framework proposal
  - Used LLMs (ChatGPT / GPT-4o, Claude) to generate reading scripts and phoneme-rich prompts (narrations, dialogues, tongue twisters) for SPEAD and the extended-length recordings, and to write batch scripts for TTS / voice-cloning generation; tuned prompts and used LLM-as-judge scoring to check text naturalness and phoneme coverage
  - Processed the recorded speech into text with ASR (Whisper), aligned transcripts to the scripts, and analyzed the text (word error rate, phoneme and word frequencies, readability) to check script coverage and reading errors
  - Served open-source LLMs locally with vLLM and SGLang for high-throughput batch text generation, tuning batching and GPU memory use
  - Benchmarked and tuned LLM inference for batch text generation: throughput and latency versus batch size, paged KV cache and prefix caching, continuous batching, 4-bit and 8-bit weight quantization (AWQ, GPTQ) and FP8 / INT8, and speculative decoding with a small draft model; compared a dense model with a mixture-of-experts model (Mixtral) for quality, speed, and GPU memory, and checked that speed-ups did not reduce output quality with LLM-as-judge scoring
  - Ran small language models (SLMs: Llama 3.2 1B / 3B, Phi-3-mini, Qwen2.5, Gemma 2B) locally on laptop CPUs and GPUs with 4-bit and 8-bit quantization (llama.cpp / GGUF, Ollama, bitsandbytes, AWQ / GPTQ) so participant-related text never left the device, and compared their quality, latency, and memory use with cloud LLM APIs

- Built a simple local–cloud arbitration step that routed each generation request to an on-device SLM or a cloud API (GPT-4o, Claude) by task difficulty, latency, cost, and privacy, and did the same for speech generation (local XTTS / Tortoise vs. the cloud ElevenLabs API)
- Changed models behind the router safely: weighted routing between models, staged rollout of new models and prompts on a small share of requests, and attribution of quality changes to each change through the regression suite; ran A/B comparisons between prompt and model versions on live lab requests and tested the differences for statistical significance; used LiteLLM as a single gateway to OpenAI, Claude (Anthropic API and AWS Bedrock in a cloud trial), Gemini, and local models, with per-route logging and cost tracking
- Tuned the router's weights with a contextual multi-armed bandit (epsilon-greedy, then Thompson sampling) that learned from judge quality scores, latency, and cost per request, so traffic shifted toward the best model for each kind of request
- Tested new routing policies offline on historical request logs before rollout: replayed logged requests with their recorded quality, latency, and cost, estimated each policy's value with off-policy evaluation (inverse propensity scoring and doubly robust estimates using the logged routing probabilities), and promoted a policy only when it beat the current one with confidence intervals that excluded zero
- Fine-tuned a small language model with LoRA / QLoRA to classify generated prompts by intent and style (narration, dialogue, tongue twister, question) and filter off-topic text before recording
- Post-trained a small open language model for script generation: supervised fine-tuning on accepted scripts, then preference tuning with DPO (Hugging Face TRL) on pairs ranked by LLM-as-judge scores and human spot checks, and compared quality, phoneme coverage, and cost with the base and cloud models
- Compared DPO with a small reinforcement-learning-from-feedback run: trained a reward model on the same preference pairs and fine-tuned the policy with PPO (Hugging Face TRL), and compared quality, stability, and cost
- Wrote the ELAD manuscript, co-created the SPEAD dataset, and presented to CITeR sponsors (invited Fall 2025 webinar)

- **Tech stack:** Python, PyTorch, Coqui TTS, Tortoise TTS, ElevenLabs API, noisereduce, facebookresearch/denoiser, pyannote.audio, webrtcvad / Silero VAD, WeSpeaker, VeriSpeak SDK, Audacity, IEEE DataPort; LLMs: ChatGPT / GPT-4o, Claude
- **Highlights:**
  - Co-wrote 3 funded CITeR proposals ($135K total, including DHS funding) on deepfake audio synthesis and detection with Clarkson and West Virginia University.
  - Generated voice deepfakes with 6+ TTS / voice-cloning systems that fooled a commercial speaker-recognition engine (VeriSpeak match scores up to 91).
  - Curated ELAD-SVDSR (36 speakers × 45 min, 5 microphones, deepfakes included) and co-created SPEAD (100 speakers, phoneme-rich speech), both public on IEEE DataPort.
- **Outputs:** J5 (ELAD-SVDSR), SPEAD dataset, T3 talk

PR6. CUWavestarGAN: Emotion-Aware GAN for Cross-Domain Voice Deepfake Detection

- **Dates:** Summer 2025 · **Role:** lead researcher (first and corresponding author) · **Team:** Y. Sukhdeve (mentee), Dr. F. Hussain, Dr. M. Imtiaz
- **Status:** completed research project with a full manuscript (not being published)
- **Overview:** Voice deepfake detectors overfit to narrow prosodic and synthesis-specific artefacts and fail on unseen emotional styles. CUWavestarGAN is a multi-discriminator GAN framework that performs emotion-conditioned many-to-many voice conversion to create emotion-augmented spoofs, which train detectors to learn domain-invariant cues instead of superficial emotional signatures.
- **Duties:**
  - Designed a composite multi-discriminator GAN with emotion-conditioned many-to-many voice conversion (StarGAN-VC family), hierarchical prosody disentanglement, and hybrid multi-objective losses (adversarial, cycle-consistency, speaker-identity, emotion-transfer, and a differentiable PESQ-proxy loss)

- Built a zero-shot deepfake speech generator: linguistic encoder, hierarchical prosody encoder, dual-attention (Bahdanau) decoder producing mel-spectrograms, speaker-embedding conditioning, and a HiFi-GAN vocoder
- Built the audio preprocessing pipeline: FFmpeg resampling to 24 kHz mono, multi-band spectral gating, dynamic-range compression, fixed-length cropping and padding
- Generated emotion-augmented training data and retrained anti-spoofing detectors (RawNet2, AASIST) on ASVspoof 2019 LA plus emotional speech (ESD)
- Ran cross-emotional, cross-synthesizer, and cross-dataset evaluations with paired t-tests, plus ablations of loss weights and discriminator criteria
- Used LLM coding assistants (Claude Code, GitHub Copilot) to implement and debug the GAN, TTS, and evaluation code and to organize ablation runs
- Fine-tuned transformer front-ends (Wav2Vec2, WavLM) for spoof detection with parameter-efficient fine-tuning (LoRA / QLoRA via Hugging Face PEFT)
- Trained sequence models (transformer encoders, attention decoders, HiFi-GAN vocoder) with multi-GPU data and model parallelism (PyTorch DDP / FSDP) and mixed precision
- Evaluated speaker similarity with Neurotechnology VeriSpeak; wrote the manuscript; guided a graduate collaborator

- **Key results:** 89.5% emotion-transfer accuracy while keeping speaker similarity (VeriSpeak 0.70 after emotion conversion vs. 0.72 for neutral deepfakes); emotion-enriched spoofs cut detection EER by 53% (RawNet2) and 56% (AASIST) under cross-emotional evaluation ($p < 0.01$), with robust cross-synthesizer and cross-dataset generalization
- **Tech stack:** Python, PyTorch, torchaudio, librosa, HiFi-GAN, StarGAN-VC-style voice conversion, RawNet2, AASIST, ASVspoof 2019 LA, ESD, FFmpeg, Neurotechnology VeriSpeak SDK, NVIDIA GPUs
- **Highlights:**
  - Designed CUWavestarGAN, a multi-discriminator GAN for emotion-conditioned voice conversion, reaching 89.5% emotion-transfer accuracy while keeping speaker identity (VeriSpeak similarity 0.70).
  - Cut voice-deepfake detection EER by 53–56% (RawNet2, AASIST) on unseen emotional spoofs by training detectors on GAN-generated, emotion-augmented data ($p < 0.01$).
  - Built the full pipeline: zero-shot TTS with dual-attention decoding and a HiFi-GAN vocoder, hybrid losses (speaker, emotion, PESQ-proxy), and cross-dataset and ablation evaluation.

PR7. Speech Quality Metrics vs. Speaker Recognition Under Degradation

- **Dates:** 2025 · **Role:** sole student author (all CRediT roles except supervision)
- **Overview:** Do standard speech-quality metrics predict how well self-supervised speaker-verification models perform on degraded audio?
- **Duties:** designed 21 degradation conditions (noise 0–20 dB SNR, reverberation RT60 0.3–1.0 s, telephone / MP3 / AAC codecs); ran WavLM, Wav2Vec2, and HuBERT verification on TIMIT, CHiME-6, and Common Voice; computed 8 quality metrics and 4 performance metrics; ran correlation and multiple-regression analysis; wrote the paper
- **Tech stack:** Python, Hugging Face Transformers, PyTorch, FFmpeg, pesq, pystoi, MUSAN, RIR, SciPy, statsmodels
- **Highlights:**
  - Quantified how noise, reverberation, and codecs affect 3 self-supervised speaker-verification models, finding quality metrics explain up to 80% of minDCF variance for HuBERT but only 35% for Wav2Vec2 (*Signals*, 2026).
- **Outputs:** J2 (*Signals*)

PR8. Beyond ROC-AUC: Standards-Based Biometric Performance Reporting

- **Dates:** 2026 · **Role:** lead researcher (first author)
- **Overview:** Showed that the widely reported full ROC-AUC can hide or even reverse system rankings at the low false-match rates used in deployment, and argued for ISO/IEC 19795-1 operating-point reporting.
- **Duties:** benchmarked 7 pretrained matchers across face (ArcFace, FaceNet), voice (ECAPA-TDNN, x-vector), iris (open-iris, log-Gabor), and fingerprint (SourceAFIS) on LFW, VoxCeleb1, CASIA-Iris-Thousand, and FVC2002;

implemented TMR / FNMR at fixed FMR, DET, EER, and minDCF with bootstrap CIs and paired tests; used LLM assistants (Claude Code, ChatGPT) for evaluation code, bootstrap / significance-test scripts, and manuscript editing; wrote the paper

- **Tech stack:** Python, PyTorch, facenet-pytorch, InsightFace, SpeechBrain, open-iris, SourceAFIS, NumPy / SciPy, Matplotlib
- **Highlights:**
  - Benchmarked 7 biometric matchers across 4 modalities and showed ROC-AUC can reverse system rankings at FMR = 10^{-3}, making the case for DET / FNMR@FMR as the primary metric (arXiv, 2026).
- **Outputs:** P1 preprint

PR9. VPQAD: Voice Pre-Processing and Quality Assessment Dataset

- **Dates:** 2024 · **Role:** lead (first author)
- **Overview:** A public dual-microphone voice dataset recorded in real noisy environments, for developing and validating preprocessing and speech-enhancement methods.
- **Duties:** designed the protocol and IRB materials; recruited and recorded 50 participants (ages 18–40) in a cafeteria and an engineering lab on AT2020 and SM58 simultaneously; collected text-dependent and text-independent speech at 44.1 kHz; curated and anonymized the data; ran SNR / SegSNR quality profiling; released the data and MATLAB / Python scripts; wrote the paper
- **Tech stack:** Audio-Technica AT2020, Shure SM58, Python data-collection software, MATLAB, Python, IEEE DataPort, GitHub
- **Highlights:**
  - Built and released VPQAD, a public dual-microphone voice dataset of 50 speakers recorded in real noisy environments, now used in follow-up speech-enhancement and deepfake studies (*IEEE Data Descriptions*, 2024).
- **Outputs:** J7, VPQAD on IEEE DataPort

PR10. CUADS: Multimodal Affective Dataset & Emotion Recognition Benchmark

- **Dates:** 2024 – 2026 · **Role:** co-designed the protocol; led participant recruitment and data collection; co-author
- **Overview:** A public dataset of ECG, PPG, and GSR recordings plus Big Five personality scores from 38 participants watching emotion-eliciting film clips, followed by a unified cross-dataset benchmark of deep-learning ECG emotion recognition.
- **Duties:** co-designed the data-collection protocol (Shimmer3 sensors, 20 film clips, ARDT mobile app); recruited participants and supervised all sessions; contributed CUADS to the combined benchmark with ASCERTAIN and DREAMER (119 participants, 3,216 trials); helped design 10-fold cross-validation and hyperparameter search for temporal CNN, PETSFCNN, and CNN-LSTM models
- **Tech stack:** Shimmer3 ECG / PPG / GSR, ARDT / ARRC toolkits (Python), NeuroKit2, PyTorch, scikit-learn
- **Highlights:**
  - Co-designed and led data collection for CUADS, a public multimodal physiological emotion dataset (38 participants, ECG / PPG / GSR), and contributed it to a cross-dataset deep-learning benchmark accepted at IEEE UEMCON 2026.
- **Outputs:** J6 (CUADS), C1 (UEMCON)

PR11. Emotion Detection from Gait and Balance with Wearable Sensors

- **Dates:** 2024 · **Role:** lead analyst (first author) · **Collaborators:** S. Crowell, A. Boolani
- **Overview:** Can anger be detected from how people walk and balance?
- **Duties:** processed APDM wearable IMU data from 107 participants (modified CTSIB balance test, 2-minute walk); engineered and selected features; trained and tuned ML classifiers; controlled for age, height, and weight with ANCOVA; wrote and presented the paper; reconstructed each participant's walking and balance trials as an animated 3D skeleton from the IMU orientation (quaternion) streams in Blender and Unity to review motion with

collaborators; trained a small conditional variational autoencoder (and tried a diffusion model) to generate realistic synthetic gait and sway sequences for data augmentation, checked with a real-vs-synthetic classifier and feature-distribution comparisons

- **Tech stack:** APDM Opal IMU sensors, Python, scikit-learn, pandas, SciPy / statsmodels
- **Highlights:**
  - Detected anger from wearable gait and balance data with 85.3% accuracy using a 5-feature support vector classifier on 107 participants (IEEE HI-POCT 2024).
- **Outputs:** C3

PR12. Speech Emotionality and Speaker Recognition Accuracy

- **Dates:** Spring 2024 · **Role:** sole author
- **Overview:** Studied how emotional state (anger, sadness, fear, joy, etc.) changes pitch, tone, speed, and volume and degrades speaker recognition, and proposed an emotion-robust pipeline (emotion-diverse data, MFCC / prosodic features, CNN / LSTM models).
- **Duties:** reviewed emotional-speech effects on ASR; designed the data, preprocessing, and modelling strategy; compared FAR / FRR across 8 emotions; presented the work at RAPS (3MT and oral)
- **Tech stack:** Python, librosa, openSMILE, PyTorch / Keras (CNN, LSTM), ESD / RAVDESS / CREMA-D / IEMOCAP
- **Highlights:**
  - Won 1st Prize (3-Minute Thesis) and Best Graduate Oral Presentation at Clarkson RAPS 2024 for research on how emotion affects speaker recognition, showing lower FAR / FRR across 8 emotional states with CNN / LSTM models.
- **Outputs:** RAPS awards (see Awards)

PR13. Deep-Learning Speech Preprocessing and Quality Metrics (MCCNNY Poster)

- **Dates:** Spring 2024 · **Role:** lead
- **Overview:** Applied CNN denoising to MS-SNSD noisy speech, measured SNR, MOS, and PESQ before and after, and correlated them with VeriSpeak recognition accuracy; the seed of the Signals paper and VQI.
- **Tech stack:** Python, PyTorch / TensorFlow, MS-SNSD, pesq, VeriSpeak SDK
- **Highlights:**
  - Won Best Presenter (Poster) at MCCNNY 2024 for correlating deep-learning speech enhancement and quality metrics with commercial speaker-recognition accuracy.

PR14. Speaker Recognition in Noisy Environments: Modified EKF-ResNet Architecture (CITeR 23S-07C)

- **Dates:** January 2024 – Fall 2024 (progress reported at the CITeR Spring 2024 Program Review; invited talk at the International Martigny Biometrics Workshop, May 2025)
- **Role:** lead student researcher; co-wrote the funded proposal
- **Team:** A. Ahmed, S. Purnapatra, Dr. M. Imtiaz (Clarkson University)
- **Funding:** NSF CITeR (IUCRC), $45K (see Research Funding)
- **Overview:** Noisy environments sharply reduce automatic speaker recognition accuracy, and most public datasets simulate noise artificially. This project built a voice preprocessing and denoising pipeline, assembled real-world noisy and longitudinal voice datasets, and measured how denoising affects commercial speaker verification (Neurotechnology VeriSpeak), comparing classical (Kalman filter) and deep-learning (Wave-U-Net) denoising. Findings fed directly into the VPQAD dataset, the speech-enhancement benchmark (BDCC 2026), and the Voice Quality Index.
- **Duties:**
  - Reviewed the literature on speaker recognition under noise, speech enhancement, and speech-quality metrics
  - Designed and ran real-world noisy adult voice collection (51 subjects in cafeterias and teaching labs with live background speech), including the IRB protocol, recording setup, and participant scheduling

- Curated evaluation data: the Clarkson longitudinal child voice dataset (14+ collections over 7+ years, ages 4–18) and the public SpEAR corpus (clean speech with controlled Gaussian noise)
- Built the preprocessing pipeline: resampling, normalization, high-pass Butterworth filtering, (extended) Kalman filtering, energy-based VAD segmentation, framing (1.5–46.5 ms, 50% overlap), MFCC extraction
- Measured the acoustics of the recording rooms (cafeteria, labs, HVAC lab, vehicle cabin): played exponential sine sweeps and short pulses through a loudspeaker and recorded them, recovered room impulse responses by deconvolution (matched filtering), picked direct-path and echo arrival times (time of flight), and estimated reverberation decay (RT60, C50) and background-noise spectra, then used them to check the simulated impulse responses and noise used for augmentation
- Implemented and adapted Wave-U-Net (1D CNN source separation trained on MUSDB18 / ccMixter) for speech denoising
- Built evaluation tooling: frame-wise FFT SNR, before / after quality metrics, and VeriSpeak 1:1, 1:N (ranked candidate lists for speaker identification), and N:N match scoring across subjects, collections, and ages
- Trained and tested a neural-network speaker-recognition baseline on SpEAR (train / validation / test splits)
- Analyzed why the Kalman filter was inconsistent across subjects and moved to deep-learning denoising; proposed a ResNet speaker model and a meta-learning (MAML) denoiser for future work
- Wrote progress reports and presented to CITeR industry and government members; managed code, data, and documentation

- **Key results:**
  - Kalman filtering raised SNR substantially on some samples (e.g. from 0.23 dB to 37.65 dB) but was inconsistent across subjects
  - Wave-U-Net increased child-voice SNR by 26.99–53.88 dB and consistently improved VeriSpeak match scores across collections
  - Speaker-recognition accuracy on the SpEAR test set: 30% unfiltered → 60% with the Kalman filter → **70% with Wave-U-Net**

- **Tech stack:**
  - Languages: Python, MATLAB
  - Signal processing: SciPy signal, NumPy (FFT), librosa, FilterPy / pykalman, MATLAB Signal Processing Toolbox (butter, filtfilt), webrtcvad / energy VAD, python_speech_features
  - Deep learning: PyTorch / TensorFlow (Wave-U-Net), ResNet speaker embeddings; MUSDB18, ccMixter training data
  - Biometrics: Neurotechnology VeriSpeak SDK (1:1, 1:N, N:N matching)
  - Data and tools: Audacity, Audio-Technica AT2020 and Shure SM58 microphones, Jupyter, Git / GitHub, Matplotlib

- **Highlights:**
  - Led a CITeR-funded ($45K) project on speaker recognition in noisy environments, building a denoising and preprocessing pipeline (Kalman filter, VAD, MFCC, Wave-U-Net) evaluated with a commercial speaker-recognition engine (Neurotechnology VeriSpeak).
  - Collected a real-world noisy voice dataset from 51 adults in cafeterias and labs, and combined it with a 7-year longitudinal child voice corpus (ages 4–18) and the SpEAR benchmark for validation.
  - Raised speaker-recognition accuracy from 30% to 70% on noisy test data and increased child-voice SNR by up to 54 dB using deep-learning (Wave-U-Net) denoising, outperforming classical Kalman filtering.
  - Presented results to industry and government sponsors (CITeR Program Review) and in an invited talk at the International Martigny Biometrics Workshop (Switzerland); the findings led to the VPQAD dataset and the Voice Quality Index.

PR15. FPGA Acceleration of Biometric Recognition (Review)

- **Dates:** 2025 – 2026 · **Role:** co-author (conceptualization, methodology, writing, visualization)

- **Overview:** A 33-page survey of 2021–2025 FPGA / FPGA-SoC implementations of face, fingerprint, iris, speaker, and finger-vein recognition for edge devices.
- **Duties:** co-designed the review scope and methodology; wrote the speaker-recognition and evaluation sections; built comparison tables of accuracy / EER, latency, resources, and power; identified design patterns (streaming dataflow, quantization, HLS, CPU–FPGA partitioning)
- **Highlights:**
  - Co-authored a 33-page review of FPGA-based biometric recognition across 5 modalities, synthesizing accuracy, latency, resource, and power trade-offs for edge deployment (*Electronics*, 2026).
- **Outputs:** J3

PR16. Mentored Projects (student-led, under my guidance)

- **Synthetic fingerprints for children under four** (F. Ahmad; IEEE UEMCON 2026): evaluation and selection of 32,000 generated pediatric fingerprints down to 1,985 using NFIQ 2, NBIS, and DINOv2 similarity
- **Interpretable deepfake-audio detection baselines** (F. Ahmad; IEEE MDTS 2026, oral): prosodic and spectral features with classical classifiers on Fake-or-Real, RBF-SVM ~93% accuracy / ~7% EER
- **AIDrawPen: AI-powered drawing pen for pediatric rehabilitation** (L. Akanji; IEEE MDTS 2025 poster): BLE (EFR32MG) + IMU pen with a deep-learning gesture model above 95% accuracy
- **SPEAD: Speech Phoneme-Enriched Audio Dataset** (Y. Sukhdeve; IEEE DataPort 2025): 100 speakers, phoneme-rich speech
- **Highlights:** Guided 4 student-led projects to 3 accepted IEEE conference papers / posters and a public dataset.

PR23. Vision-Enabled Robotic Prosthetic Arm for Children (lab project led by Dr. M. A. B. Sarker; contributor)

- **Dates:** 2024 – 2025 · **Role:** contributing team member in the AVHBAC Lab (not an author on the paper) · **Lead:** Md Abdul Baset Sarker (then PhD student, now postdoc at Howard University); PI Dr. Masudul Imtiaz
- **Overview:** a low-cost, lightweight AI vision-enabled prosthetic arm and hand for children aged 10–12 with upper-limb disabilities: a micro camera, Xilinx KV260 FPGA, force sensors, IMUs, and a time-of-flight distance sensor drive servo-actuated fingers for adaptive grasping without EMG training (paper: "A Vision-Enabled Prosthetic Hand for Children with Upper Limb Disabilities," Sarker et al., arXiv:2504.15654; IEEE Sensors in Spotlight 2025 best poster honourable mention). Project page: baset-sarker.github.io
- **Tech stack:** Python, PyTorch, YOLOv7-tiny, CNNs, Xilinx Kria KV260 FPGA (Vitis AI, DPU), quantization-aware training, OpenCV, MPU-6500 IMUs, force sensors, VL6180X time-of-flight sensor, SG90 servo motors, PLA / silicone 3D-printed parts

PR24. AI Face-Capture for First-Time Passport Photos of Young Children (lab project led by Dr. M. A. B. Sarker; contributor)

- **Dates:** 2024 · **Role:** contributing team member in the AVHBAC Lab (not an author on the preprint) · **Lead:** Md Abdul Baset Sarker; PI Dr. Masudul Imtiaz
- **Overview:** an AI pan-tilt-zoom (PTZ) camera system on an NVIDIA Jetson Nano that captures passport-compliant face images of young children by detecting the face, analyzing 68 facial landmarks, automatically adjusting pan, tilt, and zoom, and checking geometric and photographic quality against ISO/IEC 19794-5 rules (preprint: "AI Face-Capture for First-Time Passport Application," Sarker and Imtiaz, Preprints 2024, doi:10.20944/preprints202406.0392.v1; companion system: "An Open-Source Face-Aware Capture System," Electronics 2024). Project page: baset-sarker.github.io
- **Tech stack:** Python, OpenCV, Dlib (face detection, 68-point landmarks), NVIDIA Jetson Nano, IMX477 12 MP PTZ camera with Arducam PTZ controller, ISO/IEC 19794-5 face-image rules

PR25. FUQI: Face Utility Quality Index — Open-Source Facial Image Quality Toolbox Based on ISO/IEC 29794-5 (CITeR 25F-04W)

- **Dates:** Fall 2026 · **Role:** co-wrote the funded proposal; lead researcher; first author of the final report and presenter at the Fall 2026 CITeR Program Review · **Team:** Aaron Storey (Clarkson), Dr. Prashnna Gyawali (West Virginia University), Dr. Masudul Imtiaz (Clarkson)
- **Funding:** NSF CITeR project 25F-04W ($45,000; multi-university, Clarkson and WVU); co-wrote the proposal
- **Status:** final report presented at the Fall 2026 CITeR Program Review; open-source GitHub release (OFIQ + FUQI toolbox), written executive summary (Spring 2027), and a paper on partner-free utility labels and interpretable face quality planned
- **Overview:** Face recognition fails most often on poor photos (blur, pose, lighting, occlusion, low resolution), and operators need to know before enrolment or search whether a photo is good enough and what to fix. Existing tools give either many separate measurements or a single unexplained score. FUQI (Face Utility Quality Index) is one score that predicts whether a face photo will be recognized, learned from real face-recognition results and built on the ISO/IEC 29794-5 (OFIQ) quality components, so it also points to the cause. It is the face counterpart of the Voice Quality Index (PR1).
- **Duties:**
  - Framed the problem as utility-based face image quality (ISO/IEC 29794-1 "quality as utility") on top of the ISO/IEC 29794-5 components, carrying the VQI approach from voice to face
  - Set up OFIQ v1.0.3 (the open-source ISO/IEC 29794-5 reference implementation) and CR-FIQA, and selected and downloaded 7 public face datasets: CelebA, XQLFW, CPLFW, CALFW, AgeDB, TinyFace, and CFP (web photos, degraded copies, large pose and age gaps, wide age range, native low-resolution faces, frontal and profile photos)
  - Scored 432,549 images with all 28 OFIQ measures (native and 0–100 scales: lighting and exposure, sharpness, compression, colour, background, head pose, face size and framing, eyes, mouth and expression, occlusion, and the unified quality score UQS) and with CR-FIQA, and handled failures such as tiny faces where face detection fails
  - Built a person-disjoint 80/20 train / test split with 5-fold cross-validation, kept people who appear in several datasets together, held CFP out completely as an unseen dataset, and kept CFP people out of training
  - Aligned and embedded 279,121 faces with 4 label matchers (AdaFace IR-101 on WebFace12M, ArcFace ResNet-100 on Glint360K, LVFace-B transformer on Glint360K, EdgeFace-Base) and kept FaceNet (Inception-ResNet on VGGFace2) as an independent, held-out test matcher; verified each matcher (AUC, EER, TAR at FMR 1e-3)
  - Designed continuous consensus utility labels: cosine similarity of embeddings, each photo's "stranger level" (mean similarity to reference strangers), same-person scores corrected by the two photos' stranger levels, near-duplicates removed, a least-squares split of every same-person score into each photo's own share (removing the partner photo's effect), per-matcher percentiles, and the mean over the 4 matchers
  - Found and fixed two label problems: the first label followed the quality of a person's other photos (fixed with each photo's own share), and S-norm normalisation tracked quality for LVFace (fixed with mean-only normalisation)
  - Trained two FUQI variants, FUQI-C (27 OFIQ components only, fully explainable) and FUQI-CU (components plus UQS), with 4 tree models (LightGBM, XGBoost, Random Forest, CatBoost) and 3 dataset weightings under 5-fold participant-disjoint cross-validation
  - Evaluated with ISO/IEC 29794-1 Error-versus-Discard Characteristic (EDC / ERC) curves and pAUC-based gain on 624k same-person and 14M different-person comparisons at FMR 1e-3 across 5 matchers, DET curves by quality thirds, Spearman rank agreement, quality-bin staircases, and per-component analysis
  - Checked OFIQ components against CelebA attribute labels (hats, glasses, smiles, open mouths, blur) with Cliff's delta and one-sided Mann-Whitney tests, finding that OFIQ Sharpness misses blurry photos while UQS and CR-FIQA catch them
  - Compared FUQI with the two leading scores (OFIQ UQS and CR-FIQA), reported honestly where CR-FIQA stays ahead, and analyzed why its advantage does not carry over to an unseen matcher (training-data overlap with

the label matchers)
    - Kept the large pipeline reliable with checkpoint checks that caught OFIQ text-encoding and non-ASCII path issues, detector crashes on tiny faces, and GPU library mismatches; ran embedding extraction in batches on GPUs and stored per-image tables for analysis
    - Used LLM coding assistants (Claude Code, GitHub Copilot) to implement and debug the scoring, labelling, and evaluation code, reviewing every change
    - Wrote the final report and presented it to CITeR industry and government members at the Fall 2026 Program Review; prepared the deliverables (FUQI models, Python scoring and evaluation code using OFIQ v1.0.3, derived per-image quality tables and utility labels for the 7 datasets without redistributing images, evaluation workbook)
- **Tech stack:** Python, OFIQ v1.0.3 (C++ reference implementation of ISO/IEC 29794-5), CR-FIQA, AdaFace, ArcFace, LVFace, EdgeFace, FaceNet, InsightFace-style face alignment, PyTorch, ONNX Runtime, OpenCV, NumPy, pandas, SciPy, scikit-learn, LightGBM, XGBoost, CatBoost, Matplotlib; NVIDIA GPUs; LLM coding assistants (Claude Code, GitHub Copilot)
- **Highlights:**
    - Co-wrote the funded CITeR proposal ($45K) and led the project that built FUQI, an open, explainable face image quality score that predicts whether a photo will be recognized, on top of the ISO/IEC 29794-5 (OFIQ) quality components.
    - Built consensus utility labels from several deep face matchers (AdaFace, ArcFace, LVFace, EdgeFace), with a held-out matcher (FaceNet) and a held-out dataset for honest testing.
    - Trained explainable tree models (LightGBM, XGBoost, Random Forest, CatBoost) and evaluated them with ISO/IEC 29794-1 error-versus-discard and DET analysis against OFIQ and CR-FIQA.
    - Presented the final report to CITeR industry and government members at the Fall 2026 Program Review.
- **Outputs:** CITeR final report and presentation (T7); open-source toolbox and executive summary (Spring 2027); paper planned

Data Collections

Summary: 6 human-subjects data collections totaling 500+ participants, including a 9-year multimodal longitudinal child biometrics collection in schools; 4 of the resulting datasets are public on IEEE DataPort.

DC1. Clarkson Longitudinal Child Biometrics Collection (school-based, multimodal)

- **Dates:** ongoing since 2016 (CITeR / NSF study); my involvement 2024 – present (every semester), ~7 school visits per semester across 3 schools (elementary, middle, and high school); 35+ school visits so far (Spring 2024 – Fall 2026)
- **Role:** data collector on the collection team at every collection since 2024, capturing whichever modality (fingerprint, face, iris, or voice) was needed on a given day; later used the voice data for the 9-year longitudinal speaker verification study
- **Scale:** 379 enrolled children (ages 4–18) from local schools; 16 semiannual collections over ~9 years; multimodal biometrics (fingerprint, face, iris, voice); 292 children with repeat voice sessions used in the longitudinal analysis; one of the largest and longest child biometric datasets of its kind
- **Funding:** CITeR / NSF (incl. NSF ERI #2501916; CITeR "Longitudinal Progression of Biometric Recognition in Fingerprint, Face, Iris, and Voice")
- **Duties:**
    - Took part in ~7 on-site collection days per semester at 3 partner schools (elementary, middle, high school) since 2024; set up portable capture stations, and operated whichever capture device was needed that day (fingerprint, face, iris, or voice)
    - Worked with school administrators and teachers on scheduling, classroom logistics, and minimal disruption to instruction

- Followed IRB child-research requirements: parental consent, child assent, privacy, secure data handling, and age-appropriate instructions
- Guided young children (from age 4) through capture sessions patiently, keeping sessions short, engaging, and repeatable
- Ran on-site quality checks and recaptures; logged session metadata (subject ID, session, device, conditions)
- Transferred, organized, de-identified, and backed up session data for longitudinal analysis across collections

- **Highlights:**
  - Collected multimodal biometrics (fingerprint, face, iris, voice) from children aged 4–18 on ~7 school visits per semester across an elementary, a middle, and a high school (2024–present), as part of a 9-year, 16-collection longitudinal study of 379 enrolled children.
  - Coordinated on-site collection days with schools, following IRB child-research protocols (parental consent, child assent, privacy) and running quality checks and recaptures in the field.
  - Used the collected voice data for a 9-year longitudinal analysis of children's speaker verification (IEEE TBIOM, under review).

DC2. VSEA-DC: Voice Sample Enrollment Analysis Data Collection (FBI; NDA)

- 100 adults · 6 real-world environments · 2 platforms · 598 baseline recordings (~40–50 h) · project lead; wrote the IRB protocol. See PR3.

DC3. VPQAD: Voice Pre-Processing and Quality Assessment Dataset

- 50 adults (ages 18–40) · cafeteria and engineering lab · dual microphone (AT2020 + SM58) · text-dependent and text-independent · public on IEEE DataPort · lead. See PR9, J7.

DC4. ELAD-SVDSR: Extended-Length Audio Dataset for Synthetic Voice Detection and Speaker Recognition

- Pilot 8 subjects × 3 × 15 min, then 36 participants × 45 min · 5 synchronized microphones · deepfakes included · public on IEEE DataPort · curated the collection and wrote the paper. See PR5, J5.

DC5. CUADS: Clarkson University Affective Research Dataset

- 38 participants · ECG / PPG / GSR (Shimmer3) · 20 emotion-eliciting film clips · Big Five personality · public on IEEE DataPort · co-designed the protocol, recruited participants, supervised all sessions. See PR10, J6.

DC6. SPEAD: Speech Phoneme-Enriched Audio Dataset

- 100 speakers (+18 low-phoneme-density controls) · narrations, dialogues, tongue twisters · public on IEEE DataPort · co-creator (mentee-led). Reading material generated with LLMs (GPT-4o / Claude) and checked for phoneme coverage. See PR16.

Highlights: Designed, ran, or contributed to 6 IRB-approved human-subjects data collections (500+ participants), including a 9-year multimodal child biometrics study in schools, and released 4 public datasets on IEEE DataPort.

M.Eng. Project — Southern Illinois University Carbondale (2022–2023)

PR17. Piezo1 Mechanosensitive Channels in Neuroinflammation

- **Dates:** August 2022 – December 2023 · **Role:** graduate research assistant · **Supervisor:** Dr. Chilman Bae
- **Funding:** co-wrote an NSF ERI proposal awarded $200K (noise-induced hearing loss and Piezo channels)
- **Overview:** Investigated whether neuroinflammation sensitizes Piezo1 channels in astrocytes, and tested a novel Piezo1 activator with wound-healing assays.
- **Duties:** experimental design and data analysis for patch-clamp electrophysiology with high-speed pressure clamp; LPS inflammation models, pharmacology (Yoda1, GsMTx4), siRNA silencing; confocal calcium imaging and extensive microscopy; wound-healing migration assays; statistics in GraphPad Prism / Origin; manuscript editing; co-wrote the NSF ERI proposal

- **Tech stack:** Axon MultiClamp 700B, Digidata, pClamp, HSPC-2 pressure clamp, Olympus FV3000 confocal, ImageJ / Fiji, GraphPad Prism, Origin
- **Highlights:**
  - Contributed experimental design and analysis to a neuroscience study showing that neuroinflammation sensitizes Piezo1 channels (*Frontiers in Cellular Neuroscience*, 2023, 36 citations), and presented a first-author abstract at the Biophysical Society Annual Meeting.
  - Co-wrote a funded $200K NSF Engineering Research Initiation proposal on protecting against noise-induced hearing loss via Piezo channels.
- **Outputs:** J8, A1

Undergraduate & Early-Career Projects — North South University (2020–2022)

PR18. Smart Motorcycle Helmet with Rear View and Accident Detection

- **Dates:** 2021 – 2022 · **Role:** lead designer (first author)
- **Duties:** designed and built a helmet with a Raspberry Pi rear-view camera, speed indicator, smart turn signals, heart-rate-based crash and drowsiness detection with vibration alerts, and GPS / GSM emergency SMS; integrated and tested the prototype
- **Tech stack:** Raspberry Pi, Pi Camera, Arduino, heart-rate sensor, GPS (NEO-6M), GSM (SIM800 / SIM900), Python, C / C++
- **Highlights:** Designed and prototyped an IoT smart helmet with rear-view camera, crash and drowsiness detection, and automatic GPS / GSM emergency alerts (IEEE CCWC 2022, 14 citations).

PR19. Millimeter-Wave Antennas for Body-Centric Healthcare Communication

- **Dates:** 2021 – 2022 · **Role:** antenna designer (first author on 1, co-author on 1)
- **Duties:** designed and simulated a compact wideband V / W-band antenna (60, 94, 110 GHz) and a transparent 60 GHz glass-substrate patch antenna; evaluated free-space and on-body (skin phantom) performance: return loss, bandwidth, gain, radiation pattern, efficiency
- **Relevance note:** the W-band (75–110 GHz) covers the 77 GHz automotive and traffic radar band; the mm-wave antenna design, simulation, and measurement skills carry over to radar sensors
- **Tech stack:** CST Studio Suite / Ansys HFSS, MATLAB
- **Highlights:** Designed mm-wave antennas for 5G body-centric healthcare networks, including a transparent 60 GHz antenna with 5.95 GHz bandwidth, 7.42 dB gain, and 65.77% efficiency in a 5.85 × 9 mm footprint (IEEE CCWC, ICAEEE 2022).

PR20. Digital Health and Fintech Applications

- **Dates:** 2021 – 2022 · **Role:** full-stack developer (first author on 4 papers)
- **Projects:** smart telemedicine platform (audio / video calls, messaging, payments, consultation notes, e-prescriptions); Info Hospital web and mobile hospital-information system (electronic health record (EHR)-style patient records, admission / discharge / transfer (ADT) and appointment data, clinical notes, role-based access); mental well-being consultation app for COVID-19 (Java, Jitsi video, doctor / patient roles, booking, prescriptions); web-based financial aggregator with custom search / matching algorithms that compared the product portfolio of participating institutions (accounts, credit cards, loans, rates, and fees) to rank options for each user
- **Tech stack:** Java / Android, web stack (HTML / CSS / JavaScript, PHP), MySQL / Firebase, Jitsi API, payment gateway integration, Git
- **Highlights:** Built 4 full-stack web and mobile applications for telemedicine, hospital information, mental-health consultation, and financial product aggregation, published as 4 first-author IEEE / ICCMC papers (2022).

PR21. IoT Wearable Vital-Signs Monitoring System (B.Sc. Thesis)

- See Education → B.Sc. (thesis). Arduino-based wearable sensors streaming vital signs over Bluetooth to a smartphone app; led a 4-person team.

PR22. Fuzzy-Logic Energy-Efficient Induction Motor Control (Directed Research)

- See Research Experience → Undergraduate Researcher. MATLAB / Simulink fuzzy controller that minimizes losses in a three-phase induction motor, benchmarked against V/f and PI control.

Publications & Presentations

Google Scholar (Sep 2026): 144 citations · h-index 6 · i10-index 4 · scholar.google.com/citations?user=BID713QAAAAJ **Totals:** 9 journal articles · 12 conference papers / posters · 1 conference abstract · 1 under review · 1 preprint (24) · 4 public datasets · 1 open-source software release **First-author papers:** 14 · **Corresponding author:** 3 (VQI, Signals, TBIOM children study) **Research funding acknowledged:** CITeR / NSF grants 1650503, 2413228, 2501916

Journal Articles

J1. VQI: A Voice Quality Index for Predicting Speaker Recognition Performance

- **Reference: A. Ahmed**, S. Schuckers, and M. H. Imtiaz, "VQI: A voice quality index for predicting speaker recognition performance," *IEEE Access*, vol. 14, pp. 94103–94121, 2026, doi: 10.1109/ACCESS.2026.3703023.
- **Status:** Published June 2026 · first and corresponding author · code: github.com/ahmedajan/VQI-v1 · funding: CITeR 2413228, NSF 2501916
- **Abstract:** Speaker recognition systems are increasingly deployed in security, forensics, and authentication, yet no standardized metric exists to predict the quality of a voice sample for recognition purposes. Existing audio quality metrics such as the Perceptual Evaluation of Speech Quality (PESQ) and Perceptual Objective Listening Quality Prediction (POLQA) measure perceptual quality rather than recognition utility, defined as a sample's ability to produce accurate recognition outcomes. We propose the Voice Quality Index (VQI), the first utility-based voice quality metric designed to predict speaker verification performance. VQI employs a dual-score architecture producing two complementary integer scores on a 0–100 scale: VQI-S for signal quality and VQI-V for voice distinctiveness. We extract 544 signal-quality and 161 voice-distinctiveness candidate features from each audio sample, selecting a subset of features through a three-stage pipeline. Random Forest classifiers trained on 20,288 balanced samples drawn from 1.2 million utterances across eight datasets produce utility-based quality scores. Evaluation using Error vs. Reject Curves on three test datasets (VoxCeleb1, VCTK, CN-Celeb) with five speaker verification providers demonstrates that rejecting the lowest-quality 20% of comparisons reduces the false non-match rate (FNMR) by up to 38.0% at the 1% FNMR threshold. A Principal Component Analysis (PCA)-based dimensionality reduction pipeline (77% for signal quality, 65% for voice distinctiveness) enables efficient deployment with minimal accuracy loss. VQI generalizes across English datasets and partially across languages, with quality-ordered partitions consistently separating system performance. Per-file processing latency ranges from 1.9 to 36.7 seconds, positioning VQI as an offline and batch quality-assessment tool rather than a real-time enrollment filter. An open-source desktop application is provided for practical deployment.
- **Highlights:**
  - Invented the Voice Quality Index (VQI), the first utility-based voice quality metric that predicts speaker verification performance, aligned with ISO/IEC 29794-1; published as first and corresponding author in *IEEE Access*.
  - Engineered 705 explainable acoustic features (544 signal-quality, 161 voice-distinctiveness) and a 3-stage feature-selection pipeline (variance, Spearman correlation, Random Forest importance).
  - Built a consensus labelling pipeline over 1.2 M utterances from 8 datasets using 3 speaker verification systems (ECAPA-TDNN, ResNet34-SE, ECAPA2) with s-norm scoring, yielding a balanced 20,288-sample training set.
  - Cut false non-match rate by up to 38.0% by rejecting the lowest-quality 20% of comparisons, validated with Error-vs-Reject Curves on 3 test sets and 5 verification systems, including 2 never seen in training.
  - Reduced feature dimensionality by 77% (signal) and 65% (voice) with PCA at minimal accuracy loss, enabling efficient deployment.

- Shipped an open-source desktop app and CLI (github.com/ahmedajan/VQI-v1) with dual 0–100 scores and per-feature explanations; led peer review and revisions as corresponding author.

J2. Quantifying the Relationship Between Speech Quality Metrics and Biometric Speaker Recognition Performance Under Acoustic Degradation

- **Reference: A. Ahmed** and M. H. Imtiaz, "Quantifying the relationship between speech quality metrics and biometric speaker recognition performance under acoustic degradation," *Signals*, vol. 7, no. 1, Art. no. 7, 2026, doi: 10.3390/signals7010007.
- **Status:** Published January 2026 · first and corresponding author · funding: CITeR / NSF 2413228, 2501916
- **Abstract:** Self-supervised learning (SSL) models have achieved remarkable success in speaker verification tasks, yet their robustness to real-world audio degradation remains insufficiently characterized. This study presents a comprehensive analysis of how audio quality degradation affects three prominent SSL-based speaker verification systems (WavLM, Wav2Vec2, and HuBERT) across three diverse datasets: TIMIT, CHiME-6, and Common Voice. We systematically applied 21 degradation conditions spanning noise contamination (SNR levels from 0 to 20 dB), reverberation (RT60 from 0.3 to 1.0 s), and codec compression (various bit rates), then measured both objective audio quality metrics (PESQ, STOI, SNR, SegSNR, fwSNRseg, jitter, shimmer, HNR) and speaker verification performance metrics (EER, AUC-ROC, d-prime, minDCF). At the condition level, multiple regression with all eight quality metrics explained up to 80% of the variance in minDCF for HuBERT and 78% for WavLM, but only 35% for Wav2Vec2; EER predictability was lower (69%, 67%, and 28%, respectively). PESQ was the strongest single predictor for WavLM and HuBERT, while Shimmer showed the highest single-metric correlation for Wav2Vec2; fwSNRseg yielded the top single-metric R2 for WavLM, and PESQ for HuBERT and Wav2Vec2 (with much smaller gains for Wav2Vec2). WavLM and HuBERT exhibited more predictable quality-performance relationships compared to Wav2Vec2. These findings establish quantitative relationships between measurable audio quality and speaker verification accuracy at the condition level, though substantial within-condition variability limits utterance-level prediction accuracy.
- **Highlights:**
  - Led a sole-student-author study (first and corresponding author) quantifying how audio degradation affects self-supervised speaker verification (WavLM, Wav2Vec2, HuBERT); owned conceptualization, methodology, software, analysis, and writing.
  - Designed 21 controlled degradation conditions: additive noise (0–20 dB SNR, MUSAN), reverberation (RT60 0.3–1.0 s, RIR), and telephone / MP3 / AAC codecs (FFmpeg), applied to TIMIT, CHiME-6, and Common Voice.
  - Measured 8 objective quality metrics (PESQ, STOI, SNR, SegSNR, fwSNRseg, jitter, shimmer, HNR) against 4 verification metrics (EER, AUC-ROC, d', minDCF) using Hugging Face models in Python.
  - Showed quality metrics explain up to 80% of minDCF variance (HuBERT) but only 35% for Wav2Vec2, using Spearman correlation and multiple regression, which motivated a utility-based quality metric (VQI).

J3. FPGA-Based Implementations of Biometric Recognition: A Review

- **Reference:** A. Kia, **A. Ahmed**, and M. H. Imtiaz, "FPGA-based implementations of biometric recognition: A review," *Electronics*, vol. 15, no. 10, Art. no. 2145, 2026, doi: 10.3390/electronics15102145.
- **Status:** Published May 2026 · second author
- **Abstract:** Field-programmable gate arrays (FPGAs) are increasingly used to bring biometric recognition from cloud- or GPU-centric deployments to resource-constrained edge devices where latency, power, and privacy are critical. This paper surveys recent (2021–2025) FPGA and FPGA-SoC implementations across five widely deployed modalities: face, fingerprint, iris, speaker (voiceprint), and finger vein. For each modality, we summarize representative implementations and the performance figures commonly reported in the literature (e.g., accuracy or EER, latency/throughput, resource usage, and power), highlighting the algorithm–hardware co-design choices that enable real-time operation. Across modalities, successful designs repeatedly employ streaming/dataflow architectures, aggressive quantization and fixed-point arithmetic, reuse-aware buffering, and heterogeneous CPU–FPGA partitioning, often supported by high-level synthesis and vendor deep learning IP. Beyond throughput, we discuss how FPGAs facilitate privacy-preserving on-device processing and can integrate template protection and presentation attack detection within the same fabric. Finally, we identify open challenges related to scalability to

larger models, memory-bandwidth constraints, and design productivity, and outline research directions enabled by emerging adaptive FPGA architectures and more automated toolflows. Overall, the surveyed evidence indicates that FPGAs are a compelling platform for deterministic, energy-efficient, and secure biometric inference at the sensor edge.

- **Highlights:**
  - Co-led, with the first author, the conceptualization, methodology, drafting, and figures of a 33-page review of 2021–2025 FPGA / FPGA-SoC biometric recognition systems across face, fingerprint, iris, speaker, and finger vein.
  - Synthesized accuracy / EER, latency, throughput, FPGA resource use, and power across implementations, identifying recurring design patterns: streaming dataflow, quantization and fixed-point arithmetic, HLS, and CPU–FPGA partitioning.
  - Wrote the speaker-recognition and evaluation-metric sections, and built comparison tables and figures.

J4. A Comparative Analysis of Deep-Learning-Based Speech Enhancement Models: Assessing Biometric Speaker Verification in Real-World Noisy Environments

- **Reference:** M. J. A. Khondkar, **A. Ahmed**, S. Schuckers, and M. H. Imtiaz, "A comparative analysis of deep-learning-based speech enhancement models: Assessing biometric speaker verification in real-world noisy environments," *Big Data and Cognitive Computing*, vol. 10, no. 3, Art. no. 98, 2026, doi: 10.3390/bdcc10030098. Preprint: arXiv:2506.15000 (2025).
- **Status:** Published March 2026 · second author · funding: CITeR / NSF 1650503, 2413228
- **Abstract:** Speech enhancement through denoising is essential for maintaining signal intelligibility and quality in biometric speaker verification pipelines that operate in acoustically adverse conditions. Despite the proliferation of deep learning (DL) architectures for speech denoising, simultaneously optimizing noise attenuation, perceptual fidelity, and speaker-identity preservation remains an open problem. We address this gap by benchmarking three architecturally distinct DL-based enhancement models—Wave-U-Net, CMGAN, and U-Net—on three independent, domain-diverse corpora (SpEAR, VPQAD, and Clarkson) that the models never encountered during training and by introducing commercial-grade VeriSpeak speaker-verification scores as a biometric evaluation dimension absent from prior comparative studies. Our experiments reveal a clear three-way trade-off: U-Net achieves the highest signal-to-noise ratio (SNR) gains (+61.44% on SpEAR, +67.05% on VPQAD, +235.3% on Clarkson) but sacrifices naturalness; CMGAN yields the best perceptual evaluation of speech quality (PESQ) values (3.33, 1.35, and 2.50, respectively), favoring listening-comfort applications; and Wave-U-Net delivers the strongest biometric fidelity (VeriSpeak improvements of +11.63%, +30.22%, and +29.24%) while offering competitive perceptual quality. These results highlight that model selection must be driven by the target deployment scenario and provide actionable guidance for improving biometric verification robustness under real-world noise.
- **Highlights:**
  - Contributed to evaluating 3 deep-learning speech enhancement models (Wave-U-Net, CMGAN, U-Net) on 3 unseen real-world corpora, adding commercial VeriSpeak speaker-verification scores as a biometric metric.
  - Supplied the VPQAD dataset (my own) as a test corpus and helped analyze the three-way trade-off: U-Net gave the best SNR (up to +235%), CMGAN the best PESQ (up to 3.33), and Wave-U-Net the best speaker-identity preservation (up to +30% VeriSpeak).

J5. Descriptor: Extended-Length Audio Dataset for Synthetic Voice Detection and Speaker Recognition (ELAD-SVDSR)

- **Reference:** R. Vijaykumar, **A. Ahmed**, Y. Sukhdeve, J. Parker, D. Pendyala, A. Collins, S. Schuckers, and M. H. Imtiaz, "Descriptor: Extended-length audio dataset for synthetic voice detection and speaker recognition (ELAD-SVDSR)," *IEEE Data Descriptions*, vol. 3, pp. 93–99, 2026, doi: 10.1109/IEEEDATA.2026.3659062. Preprint: arXiv:2510.00218 (2025).
- **Status:** Published January 2026 · second author · data DOI: 10.21227/ab5w-0c23 · funding: NSF 1650503
- **Abstract:** This article presents the Extended-Length Audio Dataset for Synthetic Voice Detection and Speaker Recognition (ELAD-SVDSR), a comprehensive resource designed to advance both the generation of high-fidelity

synthetic voices and the development of robust detection systems. Unlike existing datasets that typically provide short utterances, single-microphone recordings, or lack synthetic samples, ELAD-SVDSR uniquely integrates three critical features: 45-minute audio recordings from 36 participants, each reading various newspaper articles recorded under controlled conditions and captured via five microphones of differing quality, along with included deepfake voices for direct comparison. By having an extended duration of audio, ELAD-SVDSR captures a richer range of speech attributes such as pitch contours, intonation patterns, and nuanced delivery. This enables models to generate more realistic and coherent synthetic voices. The controlled microphone diversity enables systematic cross-device analysis, while the integrated synthetic samples eliminate the need for independent deepfake generation prior to detection research. Comparative evaluation demonstrates that the extended-duration training data may yield higher-quality deepfakes than those generated from existing datasets. Anonymized metadata accompanies the dataset to provide essential demographic information while preserving participant privacy. This dataset is anticipated to advance research in forensics, synthetic audio detection, biometric security, and voice authentication systems.

- **Highlights:**
  - Curated and led the data collection for a public deepfake / speaker-recognition dataset: 36 participants × 45 minutes of read speech, captured on 5 microphones of differing quality (studio, dynamic, earbuds, handheld recorder, spy pen).
  - Analyzed the dataset and showed that long-form training audio yields higher-quality synthetic voices (Tortoise TTS) than existing short-utterance datasets.
  - Wrote the manuscript and coordinated a 6-person student team, including 2 mentees, through collection, preprocessing, and deepfake generation; published on IEEE DataPort.

J6. Descriptor: Clarkson University Affective Research Dataset (CUADS)

- **Reference:** T. C. Sweeney-Fanelli, **A. Ahmed**, and M. H. Imtiaz, "Descriptor: Clarkson University affective research dataset (CUADS)," *IEEE Data Descriptions*, vol. 2, pp. 41–49, 2025, doi: 10.1109/IEEEDATA.2025.3553824.
- **Status:** Published March 2025 · second author · data DOI: 10.21227/d66d-y256
- **Abstract:** This article introduces the Clarkson University affective dataset (CUADS), a multimodal affective dataset designed to assist in machine learning model development for automated emotion recognition (AER), and the affective research dataset toolkit (ARDT), which contains both a mobile application to facilitate data collection and a Python API providing affective dataset abstractions to simplify data analysis and machine learning implementations. CUADS provides electrocardiogram, photoplethysmogram, and galvanic skin response data from 38 participants, captured under controlled conditions using Shimmer3 ECG and PPG sensors. Each participant used the ARDT mobile application to view and rate 20 movie clips. CUADS also provides Big Five personality traits for each participant. All data have been anonymized to protect participant privacy. CUADS and ARDT have been made public to promote collaborative research and advance research in AER and other applications requiring physiological signal processing.
- **Highlights:**
  - Co-designed the protocol for a multimodal affective-computing dataset (ECG, PPG, GSR via Shimmer3 sensors; 20 emotion-eliciting film clips; Big Five personality survey).
  - Recruited all 38 participants and supervised every data-collection session under an IRB-approved protocol; the dataset is released publicly on IEEE DataPort.

J7. Descriptor: Voice Pre-Processing and Quality Assessment Dataset (VPQAD)

- **Reference: A. Ahmed**, M. J. A. Khondkar, A. Herrick, S. Schuckers, and M. H. Imtiaz, "Descriptor: Voice pre-processing and quality assessment dataset (VPQAD)," *IEEE Data Descriptions*, vol. 1, pp. 146–153, 2024, doi: 10.1109/IEEEDATA.2024.3493798.
- **Status:** Published November 2024 · first author · data DOI: 10.21227/yb1h-hs38 · funding: CITeR / NSF 1650503
- **Abstract:** This article introduces the voice pre-processing and quality assessment dataset (VPQAD), a scalable resource developed to validate various pre-processing techniques and improve voice signal quality in noisy environments. The dataset comprises voice recordings from 50 participants aged 18–40, captured in controlled real-life conditions using Audio Technica AT2020 and SHURE SM58 microphones. These high-quality recordings,

made under diverse noise levels and settings, could be used for testing and developing voice enhancement algorithms. The dataset includes detailed metadata on the environment and participant demographics for analyzing and improving speech clarity and intelligibility, particularly in challenging conditions. To protect privacy, all data have been anonymized. VPQAD has been made public to promote collaborative research and advance research in biometrics, telecommunications, assistive technologies, and other applications requiring clear voice communication.

- **Highlights:**
  - Designed and led a public voice dataset of 50 participants recorded simultaneously on 2 microphones (Audio-Technica AT2020, Shure SM58) in real-world noisy environments, with text-dependent and text-independent speech at 44.1 kHz.
  - Curated, anonymized, and quality-profiled the recordings (SNR, SegSNR); released the data on IEEE DataPort with MATLAB / Python processing scripts on GitHub.
  - Wrote the manuscript as first author; the dataset has since been adopted in follow-up speech-enhancement and deepfake-detection studies.

J8. Inflammation Condition Sensitizes Piezo1 Mechanosensitive Channel in Mouse Cerebellum Astrocyte

- **Reference:** D. Yu, **A. Ahmed**, J. Jayasi, A. Womac, O. Sally, and C. Bae, "Inflammation condition sensitizes Piezo1 mechanosensitive channel in mouse cerebellum astrocyte," *Frontiers in Cellular Neuroscience*, vol. 17, Art. no. 1200946, 2023, doi: 10.3389/fncel.2023.1200946.
- **Status:** Published May 2023 (Brief Research Report) · second author · 36 citations
- **Abstract:** Piezo1 mechanosensitive ion channel (MSC) plays a significant role in human physiology. Despite several research on the function and expression of Piezo1 in the nervous system, its electrophysiological properties in neuroinflammatory astrocytes remain unknown. We tested whether astrocytic neuroinflammatory state regulates Piezo1 using electrical recordings, calcium imaging, and wound healing assays on cultured astrocytes. In this study, we determined whether neuroinflammatory condition regulates astrocytic Piezo1 currents in astrocytes. First, we performed electrophysiological recordings on the mouse cerebellum astrocytes (C8-S) under lipopolysaccharide (LPS)-induced neuroinflammatory condition. We found that LPS treatment significantly increased MSC currents in C8-S. The half-maximal pressure of LPS treated MSC currents was left-shifted but the slope sensitivity was not altered by LPS treatment. LPS-induced increase of MSC currents were further augmented by Piezo1 agonist, Yoda1 but were normalized by Piezo1 inhibitor, GsMTx4. Furthermore, silencing Piezo1 in LPS treated C8-S normalized not only MSC currents but also calcium influx and cell migration velocity. Together, our results show that LPS sensitized Piezo1 channel in C8-S astrocytes. These findings will suggest that astrocytic Piezo1 is a determinant of neuroinflammation pathogenesis and may in turn become the foundation of further research into curing several neuronal illnesses and injury related inflammation of neuronal cells.
- **Highlights:**
  - Contributed to experimental design and data analysis for a neuroscience study showing that neuroinflammation sensitizes Piezo1 mechanosensitive channels in astrocytes (*Frontiers in Cellular Neuroscience*, 36 citations).
  - Analyzed patch-clamp electrophysiology (cell-attached and outside-out, high-speed pressure clamp), confocal calcium imaging, and cell-migration data using pClamp, ImageJ, GraphPad Prism, and Origin.
  - Ran the statistical testing (Mann-Whitney, one-way ANOVA) that showed a left-shifted activation pressure and Piezo1-dependent currents; edited the manuscript.

J9. IoT-Based Real-Time Patients Vital Physiological Parameters Monitoring System Using Smart Wearable Sensors

- **Reference: A. Ahmed**, M. M. Khan, P. Singh, R. S. Batth, and M. Masud, "IoT-based real-time patients vital physiological parameters monitoring system using smart wearable sensors," *Neural Computing and Applications*, vol. 35, no. 7, p. 5595, 2023 (online Apr. 2022), doi: 10.1007/s00521-022-07090-y.
- **Status:** first author · journal version of the B.Sc. thesis (see Education)

- **Summary:** An affordable, easy-to-use IoT system for real-time remote monitoring of patients' vital signs, aimed at improving healthcare access in Bangladesh and similar countries. Wearable sensors connected to an Arduino microcontroller acquire physiological parameters; the microcontroller conditions the analog signals and sends the data over Bluetooth to a smartphone app for patients and physicians.
- **Highlights:**
  - Designed and built a low-cost wearable IoT system for real-time remote monitoring of patients' vital signs, aimed at improving healthcare access in developing countries.
  - Interfaced wearable physiological sensors with an Arduino, implemented analog signal conditioning and firmware, and streamed data over Bluetooth to a smartphone app for patients and physicians.
  - Led a 4-person team through prototyping and validation; wrote the manuscript as first author.

Conference Papers

C1. Bridging the Gap in ECG-Based Emotion Recognition: A Unified Evaluation of Deep Learning Models

- **Reference:** T. C. Sweeney-Fanelli, **A. Ahmed**, and M. H. Imtiaz, "Bridging the gap in ECG-based emotion recognition: A unified evaluation of deep learning models," accepted at *2026 IEEE 17th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*; arXiv:2609.15055, 2026.
- **Status:** Accepted, 2026 · second author
- **Abstract:** Deep learning has led to numerous proposed architectures for Automated Emotion Recognition (AER) from electrocardiogram (ECG) data, but inconsistencies in preprocessing, training, and evaluation make direct comparisons difficult. Most studies train and validate models on individual datasets collected under homogeneous conditions, limiting variability and raising concerns about generalizability. Cross-dataset validation is sometimes used but primarily assesses model adaptability rather than true generalization. This study presents a comparative analysis of prominent deep learning architectures in AER, emphasizing model generalization over dataset adaptability. To enable this benchmark, we introduce two open-source frameworks: Affective Research on Representations and Classifications (ARRC), a standardized benchmarking toolkit, and Affective Research Dataset Toolkit (ARDT), a framework for inter-dataset training and validation. Using ARDT, we consolidate three publicly available AER datasets, CUADS, ASCERTAIN, and DREAMER, into a single dataset, increasing variability in sensor types, recording conditions, and participant demographics. We then use ARRC to evaluate three widely studied deep learning models and two CNN baselines through hyperparameter optimization and 10-fold cross-validation. Our findings provide insights into the trade-offs between classification accuracy and model complexity, establishing a reproducible benchmark for AER research. All source code for ARRC, ARDT, and model evaluation is publicly available to ensure transparency and facilitate further research.
- **Highlights:**
  - Contributed the CUADS dataset (co-designed the protocol, supervised collection) to a unified cross-dataset benchmark of ECG-based emotion recognition (3 datasets, 119 participants, 3,216 trials).
  - Helped design the benchmarking protocol (10-fold cross-validation, hyperparameter optimization) for 3 deep-learning models and 2 CNN baselines, and helped interpret accuracy vs. model-complexity trade-offs; accepted at IEEE UEMCON 2026.

C2. Classical Machine Learning Baselines for Deepfake Audio Detection on the Fake-or-Real Dataset

- **Reference:** F. Ahmad, **A. Ahmed**, and M. H. Imtiaz, "Classical machine learning baselines for deepfake audio detection on the Fake-or-Real dataset," accepted for oral presentation at *35th IEEE Microelectronics Design and Test Symposium (MDTS)*, 2026; arXiv:2604.13400.
- **Status:** Accepted (oral), 2026 · second author (mentee first author)
- **Abstract:** Deep learning has enabled highly realistic synthetic speech, raising concerns about fraud, impersonation, and disinformation. Despite rapid progress in neural detectors, transparent baselines are needed to reveal which acoustic cues reliably separate real from synthetic speech. This paper presents an interpretable classical machine learning baseline for deepfake audio detection using the Fake-or-Real (FoR) dataset. We extract prosodic, voice-quality, and spectral features from two-second clips at 44.1 kHz (high-fidelity) and 16 kHz

(telephone-quality) sampling rates. Statistical analysis (ANOVA, correlation heatmaps) identifies features that differ significantly between real and fake speech. We then train multiple classifiers -- Logistic Regression, LDA, QDA, Gaussian Naive Bayes, SVMs, and GMMs -- and evaluate performance using accuracy, ROC-AUC, EER, and DET curves. Pairwise McNemar's tests confirm statistically significant differences between models. The best model, an RBF SVM, achieves ~93% test accuracy and ~7% EER on both sampling rates, while linear models reach ~75% accuracy. Feature analysis reveals that pitch variability and spectral richness (spectral centroid, bandwidth) are key discriminative cues. These results provide a strong, interpretable baseline for future deepfake audio detectors.

- **Highlights:**
  - Mentored a graduate first author to an accepted oral paper at IEEE MDTS 2026 on interpretable deepfake-audio detection.
  - Guided feature engineering (pitch, jitter, shimmer, spectral centroid / bandwidth), classifier benchmarking (LR, LDA, QDA, Naive Bayes, SVM, GMM), and statistical validation (ANOVA, McNemar's test); the best RBF-SVM reached ~93% accuracy and ~7% EER.
  - Reviewed code and results and edited the manuscript.

C3. Automatic Detection of Anger Based on Gait and Balance From Wearable Mobility Sensors

- **Reference: A. Ahmed**, S. Crowell, A. Boolani, and M. H. Imtiaz, "Automatic detection of anger based on gait and balance from wearable mobility sensors," in *Proc. 2024 IEEE Healthcare Innovations and Point of Care Technologies (HI-POCT)*, 2024, pp. 69–72, doi: 10.1109/HI-POCT64255.2024.10876435.
- **Status:** Published 2024 · first author
- **Abstract:** Anger affects muscle tension, cognitive performance, and overall well-being. Gait and balance are possible indicators of many emotional states. Computer models can be used to study their relationship with anger. This project intends to construct Machine Learning (ML) algorithms that can identify anger from gait and balance data, improving our understanding of anger's physical manifestations. One hundred seven participants completed the Profile of Mood Survey-Short Form (POMS-SF) to assess anger. Afterward, participants performed a modified Clinical Test for Sensory Interaction on Balance (mCTSIB) and a two-minute walk on a 6 m track wearing APDM mobility sensors. The collected data were utilized to optimize and train various ML models using decreased feature sets. To validate the models, an ANCOVA was performed to account for age, height, and weight. A support vector classifier using the top five features effectively classified people by anger with 85.3 % accuracy, demonstrating the potential of gait and balance data to determine emotional states. This study shows that gait and balance metrics can identify anger, opening new paths for physical and mental health interventions.
- **Highlights:**
  - Built machine-learning models that detect anger from wearable gait and balance data (APDM IMU sensors) collected from 107 participants, validated against POMS-SF mood scores.
  - Engineered and selected features from balance (mCTSIB) and 2-minute-walk tests, controlled for age, height, and weight with ANCOVA, and reached 85.3% accuracy with a support vector classifier using only 5 features.
  - Wrote the paper as first author and presented it at IEEE HI-POCT 2024.

C4. Smart Helmet with Rear View and Accident Detection System for Increased Safety

- **Reference: A. Ahmed**, M. M. Khan, R. Dey, and I. Nanda, "Smart helmet with rear view and accident detection system for increased safety," in *Proc. 2022 IEEE 12th Annual Computing and Communication Workshop and Conference (CCWC)*, Las Vegas, NV, USA, 2022, pp. 673–678, doi: 10.1109/CCWC54503.2022.9720833.
- **Status:** Published 2022 · first author · 14 citations
- **Abstract:** This paper presents a smart helmet to show the rider his rear view using a raspberry pi camera. A speed limit indicator will help the rider stay within safe speed limits. Accident detection system is implemented by means of a heart rate sensor. In case of an accident an automated SMS will be sent to the relevant authorities using a GPS and GSM module. The heart rate sensor is also used to detect drowsiness and a vibrator alerts the rider to stay awake. A smart turn indicator warns the vehicles behind motorcycle of the rider's intention to turn. This proposed smart helmet aims to make motorcycle riding safer.
- **Highlights:**

- Designed and prototyped a smart motorcycle helmet with a Raspberry Pi rear-view camera, speed-limit indicator, smart turn signals, and heart-rate-based accident and drowsiness detection (14 citations).
- Implemented automatic GPS / GSM emergency SMS alerts after crashes and vibration alerts for drowsy riders; integrated, tested, and presented the system at IEEE CCWC 2022 as first author.

C5. Design and Analysis of a Compact Wideband V-Band and W-Band Antenna for Healthcare Applications

- **Reference: A. Ahmed**, H. M. A. Rahman, and M. M. Khan, "Design and analysis of a compact wideband V-band and W-band antenna for healthcare applications," in *Proc. 2022 IEEE 12th Annual Computing and Communication Workshop and Conference (CCWC)*, Las Vegas, NV, USA, 2022, pp. 1043–1048, doi: 10.1109/CCWC54503.2022.9720767.
- **Status:** Published 2022 · first author
- **Abstract:** An antenna design has been proposed for operating in the millimeter wave (Extremely High Frequency) spectrum. Free space simulations have been done to determine and analyze the performance of the antenna in three different frequencies (60 GHz, 94 GHz, & 110 GHz) within the W-band of millimeter wave (mmWave) region. 60 GHz operations are a part of 5G data communication that can achieve a data transfer rate of several gigabits per second, which can be utilized in Body Centric Networking (BCN) in healthcare services for real-time patient monitoring. However, 94 GHz and 110 GHz are already utilized in radar, image sensing, and Radio over Fiber (RoF) applications. Simulations produced promising results in the observed frequencies, which indicates the antenna could be a good candidate for such applications.
- **Highlights:**
  - Designed a compact wideband mm-wave antenna and simulated its free-space performance at 60, 94, and 110 GHz (return loss, bandwidth, gain, radiation pattern, efficiency) for 5G body-centric healthcare networks, radar, and imaging.
  - Wrote and presented the paper at IEEE CCWC 2022 as first author.

C6. Development of Smart Telemedicine System

- **Reference: A. Ahmed** and M. M. Khan, "Development of smart telemedicine system," in *Proc. 2022 IEEE 12th Annual Computing and Communication Workshop and Conference (CCWC)*, Las Vegas, NV, USA, 2022, pp. 561–567, doi: 10.1109/CCWC54503.2022.9720832.
- **Status:** Published 2022 · first author
- **Abstract:** This paper presents the implementation of Telemedicine Service in Bangladesh. Nowadays, people are doing a lot of work sitting at home. So, we thought it would be better if the medical service could be taken at home. Although there are several telemedicine services in this country but they are not so popular. We wanted to create better services from others. We took up this project with that goal in mind. Moreover, we are doing this course at the time of epidemic. Where going out of the house is risky. At this point, it is best if we can take a service like medical service without going out. It seemed logical to us to work with this service at such a time. We've added a lot of features to this service like, audio/ video calling, messaging, payment option etc. We've added a lot to this service. We wish to further improve this system in the future. Telemedicine services are gaining popularity in foreign countries. Although it is not yet common in our country, we hope that our service will be popular with the people.
- **Highlights:**
  - Developed a telemedicine platform for Bangladesh during COVID-19 with audio / video calling, messaging, and online payment, enabling remote doctor consultations.
  - Owned requirements, design, implementation, and testing; presented the work at IEEE CCWC 2022 as first author.

C7. Info Hospital: Web/Mobile Application Based Health Care System

- **Reference: A. Ahmed**, M. T. Mahmud, and M. M. Khan, "Info Hospital: Web/mobile application based health care system," in *Proc. 2022 6th International Conference on Computing Methodologies and Communication (ICCMC)*,

Erode, India, 2022, pp. 1546–1552, doi: 10.1109/ICCMC53470.2022.9753895.

- **Status:** Published 2022 · first author
- **Abstract:** This paper presents the detailed design and implementation of an online platform which is available to provide all medical information via its website and mobile application. This platform can be categorized as a web-based information providing and storing system that captures all the hospital's and its service-based information. It will be a very effective way to spread information among medical help-seeking people. This digitalized system will be helpful for all patients who can get medical services from this web-based platform. This system is not expensive but rather effective from many viewpoints. A mobile application and a web application have been developed in this research. The goal is to build a straightforward, reasonable, and productive service system within the healthcare system. This system will help to get the hospital information available to people very easily.
- **Highlights:**
  - Built a web application and a mobile application that centralize hospital and medical-service information, making it easy for patients to find care.
  - Designed the system architecture and database, and implemented and tested the platform; published at ICCMC 2022 as first author.

C8. Development of an Online Mental Well-being Mobile Application for Covid-19 Pandemic

- **Reference: A. Ahmed**, P. Dixit, and M. M. Khan, "Development of an online mental well-being mobile application for Covid-19 pandemic," in *Proc. 2022 6th International Conference on Computing Methodologies and Communication (ICCMC)*, Erode, India, 2022, pp. 1553–1558, doi: 10.1109/ICCMC53470.2022.9754112.
- **Status:** Published 2022 · first author
- **Abstract:** Mental illness is a combination of a number of emotional, physical, or behavioral problems or abnormalities that cause a person pain or disrupt his social and daily activities. Just like any other illness, it can affect people of any age and occupation. There is a treatment for mental illness like many diseases. This disease can cure if proper treatment and advice are taken in time. In Bangladesh, 18.5% of adults and about 13% of children and adolescents have some kind of mental illness. But 92% of them do not take any kind of medical service or advice. Because having a mental illness in our society means that he is insane. However, using such app technology, experienced psychologists can be consulted sitting at home. But the majority of the existing systems are a wearable device to measure mental condition or app technologies does not provide such service as a direct connection with doctors. This paper presents the building of an app that anyone who uses can receive psychiatric services from experienced psychologists sitting at home. The overall system has separate login arrangements for doctors and patients. After log in the doctors will see the patient's appointments with him and the patients will see which doctor is free when. A patient can choose the doctor of his choice and have a video consultation or chat with him but he must pay in advance and at the end of the consultation he will be given a prescription via message on his mobile phone. The whole system has been developed using 'Java' and for handling video consultation, 'Jitsi' API has been used. The system has been tested in a trial stage for some time after its development and it works as expected. If people welcome this system, they will not have to go to the hospital and stand in long lines for doctors, they will be able to receive modern and quality services sitting at home.
- **Highlights:**
  - Developed a Java mobile app connecting patients with psychologists during COVID-19: separate doctor / patient logins, availability-based booking, chat, Jitsi video consultations, prepayment, and prescriptions sent by message.
  - Ran trial-stage testing to confirm the end-to-end consultation flow; published at ICCMC 2022 as first author.

C9. Development of a Web Based Online Financial Aggregator Service

- **Reference: A. Ahmed** and M. M. Khan, "Development of a web based online financial aggregator service," in *Proc. 2022 6th International Conference on Computing Methodologies and Communication (ICCMC)*, Erode, India, 2022, doi: 10.1109/ICCMC53470.2022.9753701.
- **Status:** Published 2022 · first author

- **Abstract:** Based on the joint realization of financial technology and search algorithms, an online banking aggregation service was developed. A large database containing all data from participating financial institutions and precise search algorithms is the primary key behind the specific aggregation results. Users, regardless of occupation, position, or age, need to handle bank accounts, credit cards, and loans for different purposes. Using this algorithm, users can find the most suitable option for them.
- **Highlights:**
  - Developed a fintech web platform that brings together bank accounts, credit cards, and loans from multiple institutions into a single searchable database.
  - Designed search and matching algorithms that recommend the best financial product for each user; published at ICCMC 2022 as first author.

C10. Design Evaluation of a Millimeter-Wave 60 GHz Transparent Antenna for Body-Centric Communication in Healthcare Application

- **Reference:** K. Islam, **A. Ahmed**, and M. M. Khan, "Design evaluation of a millimeter-wave 60 GHz transparent antenna for body-centric communication in healthcare application," in *Proc. 2022 International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, Gazipur, Bangladesh, 2022, doi: 10.1109/ICAEEE54957.2022.9836357.
- **Status:** Published 2022 · second author
- **Abstract:** A 60 GHz millimeter-wave patch antenna design is evaluated in both free space and on a skin-equivalent phantom. For achieving transparency, a glass substrate of a thickness of 2 mm is used. At 60 GHz, the relative permittivity of the substrate is 4.7. The antenna achieved a resonant frequency of 60.01 GHz in free space with an impedance bandwidth of 5.951 GHz. The impedance bandwidth increased by 0.178 GHz when the antenna was placed 4 mm away from a skin-equivalent phantom. The peak gain at 60 GHz is 7.42 dB and the radiation efficiency is 65.77%. The overall size of the patch antenna is 5.85 mm × 9 mm.
- **Highlights:**
  - Co-designed and simulated a transparent 60 GHz patch antenna on glass for wearable healthcare communication, reaching 60.01 GHz resonance, 5.95 GHz bandwidth, 7.42 dB gain, and 65.77% efficiency in a 5.85 × 9 mm footprint.
  - Evaluated on-body performance against a skin-equivalent phantom; co-authored the paper (ICAEEE 2022).

C11. AIDrawPen: An AI-Powered Gesture-Based Drawing Tool for Pediatric Rehabilitation

- **Reference:** L. Akanji, **A. Ahmed**, and M. H. Imtiaz, "AIDrawPen: An AI-powered gesture-based drawing tool for pediatric rehabilitation," poster presented at *34th IEEE Microelectronics Design and Test Symposium (MDTS)*, Albany, NY, USA, May 2025; preprint on *TechRxiv*, doi: 10.36227/techrxiv.174785218.86760927/v1.
- **Status:** Poster, IEEE MDTS 2025 · second author
- **Abstract:** Children with developmental disorders, such as ADHD, Autism Spectrum Disorder (ASD), and Developmental Coordination Disorder (DCD), often face challenges with fine motor skills, affecting their daily activities and overall quality of life. Existing rehabilitation tools are often expensive, complex, and lack engaging features. This study presents AIDrawPen, an AI-powered, low-cost, wireless gesture-based drawing tool designed to support pediatric rehabilitation. Powered by a coin-cell battery, a custom PCB, a BLE Microcontroller (EFR32MG32), and an Inertial Measurement Unit (IMU), the tool enhances motor and cognitive skills through tailored exercises and realtime feedback. A custom deep learning model interprets drawing gestures and provides immediate guidance. Embedded sensors process data directly on the device, while a companion mobile app offers features like progress tracking, personalized exercises, and real-time feedback to make therapy interactive and effective. With a gesture recognition accuracy exceeding 95%, AIDrawPen demonstrates its potential as a transformative therapeutic tool, combining accessibility, personalization, and open-source collaboration to revolutionize pediatric rehabilitation.
- **Highlights:**
  - Mentored the first author (Leonel Akanji) on AIDrawPen, a low-cost wireless IMU drawing pen (EFR32MG BLE microcontroller, custom PCB) for pediatric fine-motor rehabilitation, with over 95% gesture-recognition

accuracy.
    - Guided the deep-learning gesture model, IMU data pipeline, and evaluation; reviewed the manuscript; presented as a poster at IEEE MDTS 2025.

C12. Synthetic Fingerprints for Children Under Four: Generation and Biometric Evaluation

- **Reference:** F. Ahmad, **A. Ahmed**, M. R. Sumi, S. Schuckers, and M. H. Imtiaz, "Synthetic fingerprints for children under four: Generation and biometric evaluation," accepted at *2026 IEEE 17th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*; arXiv:2609.20621, 2026.
- **Status:** Accepted, 2026 · second author (mentee first author)
- **Abstract:** Fingerprint recognition in children under four is of interest for longitudinal identity applications, but research in this age range is constrained by the limited availability and sensitivity of real fingerprint data. Synthetic data may provide a useful complementary resource if generated samples are carefully evaluated for biometric quality, similarity to the real training data, and identity diversity. This paper presents an evaluation and selection protocol for synthetic fingerprints generated from a small pediatric dataset. The protocol was applied to 32,000 candidates produced by an age-conditioned generator fine-tuned with 205 fingerprints from nine children. Candidates were evaluated using NFIQ 2, NBIS minutiae extraction and matching, fingerprint-pattern classification, similarity to the complete real reference set, and pairwise similarity among retained synthetic samples. After candidate filtering and a final symmetric pairwise verification, 1,985 synthetic fingerprints remained. The youngest age group continued to produce retained samples within the sampling budget, whereas the oldest original age group produced 24 retained prints. A data-derived two-group age representation improved pattern-distribution agreement for the younger group. Repeated renderings of fixed synthetic identities produced minutiae-based mated scores comparable to the real mated scores, although a DINOv2 texture representation showed substantially greater within-generator similarity. The results indicate that synthetic pediatric fingerprints can support controlled research use, but their evaluation should include both real-to-synthetic similarity and synthetic-to-synthetic diversity, and conclusions should remain specific to the matchers and measurements used.
- **Highlights:**
    - Co-developed an evaluation and selection protocol for synthetic pediatric fingerprints, screening 32,000 candidates from an age-conditioned generator (fine-tuned on 205 real prints from 9 children) down to 1,985 high-quality, identity-diverse samples.
    - Guided the biometric evaluation using NFIQ 2 quality, NBIS minutiae extraction and matching, pattern classification, and DINOv2 texture similarity for real-to-synthetic and synthetic-to-synthetic analysis.
    - Mentored the graduate first author to acceptance at IEEE UEMCON 2026.

Conference Abstracts

A1. A Novel Activator of Piezo1: Wound Healing Assay

- **Reference: A. Ahmed**, O. Sally, and C. Bae, "A novel activator of Piezo1: Wound healing assay," *Biophysical Journal*, vol. 122, no. 3, suppl. 1, p. 523a, Feb. 2023, doi: 10.1016/j.bpj.2022.11.2779. (67th Biophysical Society Annual Meeting, 2023)
- **Status:** Published abstract, 2023 · first author
- **Summary:** Meeting abstract from Dr. Chilman Bae's lab (SIUC) reporting wound-healing (cell-migration) assays that test a novel activator of the Piezo1 mechanosensitive ion channel. (Full abstract text not retrievable online; add from your copy if available.)
- **Highlights:**
    - Led wound-healing (scratch) cell-migration assays that test a novel Piezo1 channel activator in astrocytes, quantifying migration with time-lapse microscopy and ImageJ.
    - Presented first-author work at the Biophysical Society Annual Meeting (published in *Biophysical Journal*).

Under Review

R1. Speaker Verification Performance in Children: A Longitudinal Study

- **Reference: A. Ahmed**, F. Ahmad, Y. Sukhdeve, S. Schuckers, and M. H. Imtiaz, "Speaker verification performance in children: A longitudinal study," *IEEE Transactions on Biometrics, Behavior, and Identity Science (TBIOM)* (under review; revision submitted).
- **Status:** Under review at IEEE TBIOM: first-round reviewer comments addressed and revision submitted · first and corresponding author · funding: CITeR, NSF 2413228 and 2501916
- **Abstract:** Speaker verification is increasingly considered for applications in which users may be enrolled as children and verified years later. The main challenge is that children's voices change substantially during growth, which can weaken a fixed biometric reference. Current speaker verification systems usually rely on adult-trained backends and fixed references, but their behavior on developing voices has not been characterized in a controlled longitudinal setting, and most systems provide limited insight into the acoustic sources of failure. This paper presents a 9-year longitudinal study of 292 children aged 4 to 18, recorded across 16 semiannual collections. A commercial system, three deep-learning backends, and ten interpretable i-vector pipelines are evaluated, and the effects of time gap, enrollment age, and acoustic covariates are modeled using LMER and GLMM frameworks. A controlled age-normalization intervention is then applied to test whether feature-level compensation can recover lost performance. At FMR = 0.1%, mean FNMR reaches roughly 33% at five years and approaches 60% beyond seven years, with some enrollment-age groups reaching 70%. Age-normalizing the top-ranked features recovers up to 35% of the lost performance on the interpretable i-vector pipeline and 9–12% on ECAPA-TDNN and ResNet. These results show that vocal development is a major source of long-term verification degradation in children and that targeted age compensation can reduce, but not eliminate, the need for periodic reference update or re-enrollment.
- **Highlights:**
  - Led a 9-year longitudinal study of speaker verification in 292 children (ages 4–18, 16 semiannual collections) from an IRB-approved Clarkson biometrics study, to the authors' knowledge the largest and longest such dataset.
  - Evaluated a commercial system (VeriSpeak), 3 deep-learning backends (ECAPA-TDNN, ResNet, ECAPA2), and 10 interpretable i-vector / PLDA pipelines, reporting FNMR at FMR = 0.1% with bootstrapped confidence intervals.
  - Modeled the effects of time gap, enrollment age, pitch, and formant drift with nested LMER and GLMM mixed-effects models; showed FNMR climbs to ~33% at 5 years and ~60% beyond 7 years.
  - Designed an intervention-based ablation and age-normalization method that recovers up to 35% of lost performance on i-vector pipelines and 9–12% on ECAPA-TDNN and ResNet.
  - Addressed first-round reviewer comments and submitted the revision as corresponding author; mentored 2 student co-authors.

Preprints

P1. Beyond ROC-AUC: Operating-Point Performance Reporting for Biometric Verification

- **Reference: A. Ahmed** and M. H. Imtiaz, "Beyond ROC-AUC: Operating-point performance reporting for biometric verification," arXiv:2606.20680, 2026.
- **Status:** Preprint, June 2026 · first author
- **Abstract:** A biometric verifier is often deployed with a strict false match budget, so only a narrow, low false match rate (FMR) slice of the score range is used. A reporting standard for this setting already exists. ISO/IEC 19795-1 asks for error rates at stated operating points, for the detection error tradeoff (DET) curve as the view of the trade-off between FMR and the false non-match rate (FNMR), and for an interval of uncertainty on every value. In practice, a single area under the receiver operating characteristic curve (ROC-AUC), the equal error rate (EER), or a verification accuracy is still reported as the resolution, which is a threshold-independent summary that the standard does not endorse. The full ROC-AUC averages the true match rate (TMR) with equal weight over the

whole FMR range from 0 to 1, so almost all of its weight is placed where the system is never operated; low-FMR behavior can then be hidden, and the order of two systems can even be reversed. The guideline is revisited in this paper and tested against seven pretrained matchers across four modalities, face, voice, iris, and fingerprint, each reported with bootstrap confidence intervals and paired bootstrap tests. A system that looks stronger on full ROC-AUC is shown to be significantly worse at FMR = 10^-3. For face, a higher full AUC was obtained by FaceNet, whereas a higher TMR at FMR = 10^-3 was obtained by ArcFace, and both gaps were significant with non-overlapping intervals. Hence, the DET curve and the FNMR at a fixed FMR are re-iterated in this paper as the primary report, with ROC-AUC and EER retained as supplementary context.

- **Highlights:**
  - Benchmarked 7 pretrained biometric matchers across 4 modalities (face: ArcFace, FaceNet; voice: ECAPA-TDNN, x-vector; iris: open-iris, log-Gabor; fingerprint: SourceAFIS) on LFW, VoxCeleb1, CASIA-Iris, and FVC2002.
  - Implemented ISO/IEC 19795-1 operating-point reporting (TMR / FNMR at FMR 10^{-1}–10^{-4}, DET, EER, minDCF) with bootstrap confidence intervals and paired significance tests.
  - Showed that ROC-AUC can reverse system rankings (FaceNet ahead on AUC, ArcFace significantly better at FMR = 10^{-3}), and made the case for DET / FNMR@FMR as the primary metric; first author.

Invited Talks & Seminars

T1. Formatting Scientific Articles, Capstone Reports, Resumes, and Presentations with LaTeX and Overleaf

- **Type:** Invited seminar
- **Host:** Department of Electrical and Computer Engineering, Clarkson University (online, via Zoom; announced to the EE 502 graduate seminar course by Dr. Masudul Imtiaz)
- **Date:** September 3, 2026, 12:15–1:15 pm ET
- **Audience:** ECE graduate students and faculty, ~50 attendees
- **Abstract:** LaTeX is a powerful typesetting system widely used in academia, research, and technical fields to create well-structured and professional documents. This seminar introduces the practical use of LaTeX through Overleaf, a collaborative online platform that makes document preparation easier and more accessible. Participants will learn how LaTeX can help improve consistency, organization, readability, and overall presentation while reducing the effort required for manual formatting. The seminar will also highlight useful features such as document templates, citation and reference management, figures, tables, equations, and collaborative editing. Through practical demonstrations, participants will gain a better understanding of how LaTeX and Overleaf can support efficient and professional academic and technical communication.
- **Supporting material:** public Overleaf / LaTeX example repositories on GitHub (ajans-overleaf-examples, Ajans_overleaf_examples)
- **Highlights:**
  - Delivered an invited ECE department seminar (a follow-up to my 2025 Overleaf seminar) on professional technical writing with LaTeX and Overleaf for scientific articles, capstone reports, resumes, and presentations.
  - Ran live demonstrations of templates, BibTeX citation management, figures, tables, equations, and collaborative editing, and shared reusable example templates on GitHub.

T2. VQI: A Voice Quality Index for Predicting Speaker Recognition Performance

- **Type:** Invited seminar (research talk, 30 minutes)
- **Host:** Department of Electrical and Computer Engineering, Clarkson University: EE 610 ECE Graduate Seminar (online, via Zoom)
- **Date:** Thursday, March 5, 2026, 12:30–1:00 pm ET
- **Audience:** ECE graduate students and faculty, ~50 attendees
- **Abstract:** Speaker recognition is increasingly used in security, forensics, and authentication, yet no standardized metric exists to predict whether a voice recording is suitable for reliable recognition. We propose the Voice Quality

Index (VQI), a utility-based voice quality metric designed to predict speaker verification performance directly, defined as a sample's ability to produce accurate recognition outcomes. VQI outputs two complementary integer scores on a 0 to 100 scale. VQI-S reflects signal quality factors such as background noise, channel distortion, and clipping, while VQI-V reflects voice distinctiveness, capturing how strongly speaker-specific traits are expressed in a recording. For each audio sample, we compute a large set of candidate properties describing both signal condition and speaker individuality, then apply a three-stage feature selection pipeline to retain the most informative subset for modeling. Random Forest classifiers trained on a balanced set of 20,288 samples drawn from 1.2 million utterances across eight datasets map these properties to the two VQI scores. Evaluation on three test datasets using five speaker verification providers shows that rejecting the lowest-quality 20 percent of comparisons reduces verification errors, with false non-match rate improvements of up to 38.0 percent at a 1 percent operating point. A PCA-based reduction enables efficient deployment with minimal accuracy loss, and an open-source desktop application supports practical use.

- **Highlights:**
  - Gave an invited seminar on my dissertation research, the Voice Quality Index (VQI), at the ECE Graduate Seminar, explaining its dual signal-quality / voice-distinctiveness scores, feature-selection pipeline, and Random Forest models trained on 1.2 M utterances.
  - Communicated key results to a mixed technical audience: up to 38.0% fewer false non-matches when the lowest-quality 20% of samples are rejected, and an open-source desktop app for practical use.

T3. Evaluating and Progressing the State-of-the-Art in Deep Fake Audio Generation and Detection (CITeR Project 23F-03C)

- **Type:** Invited talk (webinar)
- **Host:** CITeR (Center for Identification Technology Research, NSF IUCRC): Fall 2025 Webinar
- **Date:** Fall 2025
- **Speaker:** A. Ahmed (sole presenter); project team: Y. Sukhdeve, A. Ahmed, R. Vijaykumar, M. Imtiaz (Clarkson University)
- **Audience:** CITeR industry and government member organizations
- **Summary:** Progress report on a CITeR-funded project to build a deepfake-ready voice dataset recorded in real noisy environments and to evaluate deepfake generation and detection. Covered deepfake generation with open-source and commercial TTS (Tortoise TTS, ElevenLabs, YourTTS, XTTS, MeloTTS, MetaVoice); multi-microphone data collection (pilot of 8 subjects × 3 × 15 min on 5 microphones, scaled to ELAD with 36 subjects × 45 min); annotation and cleanup in Audacity; denoising comparisons (spectral gating, VAD-based adaptive, spectral-statistics and multi-band noise profiling, Meta / Facebook's deep-learning Denoiser); preprocessing with VAD and speaker diarization; and evaluating deepfakes against VeriSpeak (thresholds tied to FAR) and WeSpeaker (cosine similarity 0.74–0.81 mean). Lessons learned: longer audio and trimmed pauses give better deepfakes, and real environmental noise degrades TTS models (Tacotron 2, WaveNet) more than artificial noise.
- **Highlights:**
  - Gave an invited CITeR webinar talk to industry and government member organizations on a CITeR-funded deepfake-audio project, reporting progress on dataset creation, deepfake generation, and detection evaluation.
  - Generated voice deepfakes with 6+ TTS / voice-cloning systems (Tortoise TTS, ElevenLabs, YourTTS, XTTS, MeloTTS, MetaVoice) and showed they can fool a commercial speaker-recognition engine (VeriSpeak match scores up to 91; all 6 deepfake comparisons matched for 2 of 3 test subjects).
  - Scaled a pilot 5-microphone collection (8 subjects) to the 36-subject, 45-minute-per-subject ELAD dataset, and compared 4 signal-processing denoisers against a deep-learning denoiser for deepfake-ready preprocessing.

T4. Creating Professional Reports, Articles, Slides, and Posters with Overleaf

- **Type:** Invited seminar (hands-on workshop format)
- **Host:** Electrical & Computer Engineering Seminar series, Clarkson University (online, via Zoom)

- **Date:** Thursday, September 4, 2025, 12:15–1:15 pm ET
- **Audience:** ECE students and faculty, announced department-wide, ~50 attendees
- **Abstract:** A hands-on seminar guiding students through a start-to-finish Overleaf workflow for producing publication- and presentation-ready materials: technical reports, journal and conference papers, presentation slides, and research posters. The session demystifies LaTeX essentials; choosing and customizing templates to meet IEEE/ACM/thesis guidelines; managing citations with BibTeX/BibLaTeX; and consistent style with table and figure packages. The seminar also covers collaboration tools (track changes, comments, and sharing), best practices for graphics, and approaches to math and typography drawn from the presenters' own experience that enhance readability and reviewer perception. Attendees will leave with a reusable Overleaf project skeleton for reports, papers, slides, and posters. No prior LaTeX experience is required; please create an Overleaf account in advance if you'd like to follow along.
- **Highlights:**
  - Delivered an invited hands-on ECE department seminar that took students through a complete Overleaf workflow for technical reports, IEEE/ACM papers, slides, and research posters.
  - Taught template customization for IEEE / ACM / thesis guidelines, BibTeX / BibLaTeX citation management, table and figure packages, math typography, and collaboration tools (track changes, comments, sharing).
  - Built and distributed a reusable Overleaf project skeleton for reports, papers, slides, and posters; the department invited the seminar back in 2026.

T5. Modified EKF-ResNet Architecture for Automatic Speaker Recognition from Noisy Environment (CITeR Project 23S-07C)

- **Type:** Invited talk
- **Host:** International Martigny Biometrics Workshop 2025, organized by the European Association for Biometrics (EAB), CITeR, and the Idiap Research Institute; Idiap Research Institute, Martigny, Switzerland
- **Date:** May 19–20, 2025 (co-located with the CITeR Spring 2025 Program Review, May 21–22)
- **Team: A. Ahmed**, S. Purnapatra, M. Imtiaz (Clarkson University). Lead student researcher.
- **Summary:** Progress report on a CITeR-funded project to improve speaker recognition in noisy real-world environments with a deep speaker-recognition system (text-dependent and text-independent) and a robust data-processing toolchain. Expanded the CITeR child voice dataset with recordings from 100+ adults in genuinely noisy environments, instead of artificially added noise, and compared performance on child vs. adult voices. Denoised with Kalman filtering (extended Kalman filter, EKF) and Wave-U-Net using frame sizes of 1.5–46.5 ms with 50% overlap, computed frame-wise SNR from FFT power spectra, and validated with VeriSpeak match scores, which improved after filtering. Future direction: a Meta-Denoiser using model-agnostic meta-learning (MAML) to adapt quickly to new noise conditions.
- **Highlights:**
  - Gave an invited talk at the International Martigny Biometrics Workshop 2025 (European Association for Biometrics, CITeR, Idiap Research Institute) in Switzerland, presenting a CITeR-funded project on speaker recognition in noisy environments that I led as student researcher.
  - Built a denoising toolchain (extended Kalman filter, Wave-U-Net) with frame-wise FFT-based SNR analysis, and improved commercial VeriSpeak match scores on child and adult voice data.
  - Planned expansion of the CITeR child voice corpus with 100+ adult subjects recorded in real noisy environments, and proposed a meta-learning (MAML) denoiser for fast adaptation to new noise.

T6. Improving Speaker Recognition in Real-World Scenarios: Datasets, Challenges, and Quality Assessment

- **Type:** Invited seminar (research talk)
- **Host:** ECE Graduate Seminar (EE 610), Department of Electrical and Computer Engineering, Clarkson University
- **Date:** Thursday, September 26, 2024
- **Audience:** ECE graduate students and faculty, ~50 attendees
- **Summary:** Introduced biometrics and speaker recognition (verification vs. identification), why voice is a useful modality, and the main real-world challenges (noise, speaker variability, language, emotion, compression,

spoofing). Presented early-PhD work on an open-source voice preprocessing pipeline (high-pass Butterworth filtering, Kalman filtering, VAD-based segmentation, MFCC extraction) and on deep-learning denoising with Wave-U-Net. Evaluated with SNR and Neurotechnology VeriSpeak 1:1 match scores on the in-house adult noisy-environment dataset (51 subjects), the Clarkson longitudinal child voice dataset (14+ collections over 7+ years, ages 4–18), and the public SpEAR corpus. Closed with an early analysis of speech-quality metrics (MOS, PESQ, segmental / weighted / frequency-weighted SNR, intelligibility / STI) against VeriSpeak match scores on the CABank CallHome and ESD corpora, the groundwork for the Voice Quality Index, and invited the audience to take part in data collection.

- **Key results presented:**
  - Wave-U-Net denoising raised SNR on child voice samples by 26.99–53.88 dB and improved VeriSpeak match scores across collections
  - On a SpEAR speaker-recognition test with a neural network model, accuracy rose from 30% (unfiltered) to 60% (Kalman filter) to 70% (Wave-U-Net)
  - Early finding: segmental SNR and intelligibility tracked recognition performance better than MOS or PESQ
- **Highlights:**
  - Gave an invited ECE Graduate Seminar talk on improving real-world speaker recognition, covering data-collection challenges, a voice preprocessing pipeline (Kalman filtering, VAD segmentation, MFCC extraction), and Wave-U-Net deep-learning denoising.
  - Reported that Wave-U-Net denoising raised a neural speaker-recognition model's accuracy from 30% to 70% on the SpEAR corpus and increased SNR on child voice samples by up to 53.9 dB, validated with Neurotechnology VeriSpeak.
  - Presented early evidence linking speech-quality metrics (segmental SNR, intelligibility) to recognition performance, which laid the groundwork for the Voice Quality Index; recruited participants for ongoing data collection.

T7. Open-Source Facial Image Quality Toolbox Based on ISO/IEC 29794-5 (CITeR Project 25F-04W), Final Report

- **Type:** Program review presentation (final report)
- **Host:** CITeR (Center for Identification Technology Research, NSF IUCRC): Fall 2026 Program Review
- **Date:** Fall 2026
- **Speaker:** A. Ahmed; project team: A. Ahmed, A. Storey (Clarkson), P. Gyawali (WVU), M. Imtiaz (Clarkson)
- **Audience:** CITeR industry and government member organizations
- **Summary:** Presented FUQI, a face utility quality index built on the ISO/IEC 29794-5 (OFIQ) components and learned from real face-recognition results; the data, consensus utility labels, models, and ISO/IEC 29794-1 evaluation against OFIQ UQS and CR-FIQA; relevance for passport, ID, and border enrolment, forensic triage, and vendor benchmarking; and the open-source release plan.
- **Highlights:**
  - Presented the final report of a CITeR face image quality project to industry and government members at the Fall 2026 CITeR Program Review.

Public Datasets & Software

- **VPQAD:** Voice Pre-Processing and Quality Assessment Dataset, IEEE DataPort, doi: 10.21227/yb1h-hs38 (lead creator; 50 participants, dual microphone)
- **ELAD-SVDSR:** Extended-Length Audio Dataset for Synthetic Voice Detection and Speaker Recognition, IEEE DataPort, doi: 10.21227/ab5w-0c23 (curated the collection; 36 participants × 45 min, 5 microphones, deepfakes included)
- **CUADS:** Clarkson University Affective Research Dataset, IEEE DataPort, doi: 10.21227/d66d-y256 (co-designed the protocol, recruited participants, supervised collection; 38 participants, ECG / PPG / GSR)

- **SPEAD:** Y. Sukhdeve, **A. Ahmed**, D. Pendyala, and M. Imtiaz, "Speech Phoneme-Enriched Audio Dataset (SPEAD)," IEEE DataPort, 2025, doi: 10.21227/f48c-2934 (co-creator; 100 speakers, phoneme-rich narrations, dialogues, and tongue twisters; mentee-led)
- **VQI software:** open-source Voice Quality Index desktop app and command-line tool, github.com/ahmedajan/VQI-v1

Teaching Experience

Summary: 5 university courses at 2 universities (Clarkson University; Southern Illinois University Carbondale), plus programming instruction at ImransLab and independent O / A Level tutoring; 9+ semester-offerings at Clarkson (Spring 2024 – Fall 2026), from first-year engineering science to graduate biometrics; lab instructor for BME 338L at SIUC (Spring 2023); ~665 students taught in total (not counting Fall 2026); class sizes from 12 to 120; promoted from TA to co-instructor for EE 402/502. **Student evaluations:** average 4.95 out of 5 across all teaching assistant roles (user-reported, Sep 2026).

Co-Instructor — EE 402/502 Machine Learning on Biomedical Signals, Clarkson University

- **Dates / Semesters:** Fall 2025, Fall 2026 (current)
- **Level & enrollment:** combined senior undergraduate / graduate; 42 students
- **Responsibilities:** delivered 12 lectures (the course's voice and speech module); designed assignments and course projects; grading; office hours
- **Lecture topics (voice & speech processing + machine learning):**
  - Speech production and the source–filter model; vocal-fold physiology; acoustic phonetics (vowels, consonants, voicing)
  - Digital audio fundamentals: sampling, quantization, bit depth, file formats, microphones and recording conditions
  - Time-domain analysis: energy, zero-crossing rate, autocorrelation, voice activity detection
  - Frequency and time-frequency analysis: Fourier transform, STFT, windowing, spectrograms, mel scale
  - Pitch (F0) estimation, jitter, shimmer, harmonics-to-noise ratio; formant analysis
  - Linear prediction (LPC / LPCC), cepstral analysis, MFCCs, filter banks, delta features
  - Speech preprocessing: noise, reverberation, denoising and speech enhancement, normalization
  - Speech and voice quality assessment (SNR, PESQ, STOI, perceptual measures)
  - Voice as a biomedical signal: pathological voice analysis, voice disorders, neurological disease markers (e.g. Parkinson's), respiratory sounds and cough analysis, emotion, stress and mental-health cues in speech
  - Classical ML for speech: feature pipelines, k-NN, SVM, Random Forest, Gaussian Mixture Models, Hidden Markov Models
  - Deep learning for speech: CNNs on spectrograms, RNNs / LSTMs, attention and transformers, self-supervised models (wav2vec 2.0, HuBERT, WavLM), transfer learning
  - Applications: speaker recognition and verification, speech emotion recognition, automatic speech recognition basics (e.g. Whisper), voice biometrics, deepfake / spoofed speech detection
  - Evaluation: train/test splits by speaker, cross-validation, accuracy, precision/recall/F1, ROC/AUC, EER, confusion matrices; ethics, privacy, and bias in voice data
  - Clinical data examples: PhysioNet / MIMIC waveforms and de-identified clinical notes for classification and text-mining exercises
  - Wearable health course projects (designed and mentored): sleep staging from heart rate, heart rate variability (HRV), and accelerometer signals; stress detection from EDA / GSR and HRV (WESAD); and glucose trend forecasting from continuous glucose monitor (CGM) data (OhioT1DM) with time-series models, including per-person baselines (personalization) and subject-wise evaluation

- **Curriculum / course development:** built the voice and speech module's lecture materials, assignments, and projects
- **Learning outcomes and course design:** wrote course learning outcomes for the voice and speech module and aligned each lecture, lab, assignment, and project rubric with them (constructive alignment); revised materials each offering from student feedback
- **Flipped and active learning:** assigned booklet chapters and Jupyter notebooks before class, then used class time for worked problems, live coding, and team activities (flipped classroom); applied and experiential learning with real voice and biomedical data
- **Educational technology:** managed course pages, quizzes, gradebooks, and announcements on Moodle (learning management system); used Microsoft Teams, SharePoint, and OneDrive for course coordination with the instructor and TAs, and Word, Excel, and PowerPoint for course materials, grade analysis, and lectures
- **Responsible and ethical AI:** taught ethics, privacy, consent, and bias in voice and health data, and responsible use of generative AI tools for coding and writing
- **Teaching methods used:** lectures with worked examples; live coding demonstrations; Jupyter notebook labs; hands-on coding assignments; in-class activities and think-pair-share discussions; project-based learning with open-ended team projects; real-data case studies (voice, ECG, EEG); formative feedback through office hours and project check-ins
- **Highlights:**
  - Co-taught a 42-student cross-listed senior undergraduate / graduate course, designing and delivering 12 lectures on voice and speech processing and machine learning for health.
  - Built the module's curriculum from scratch: speech production, spectral and cepstral features (MFCC, LPC), pitch and formants, voice pathology, classical ML (GMM, HMM, SVM), and deep learning (CNNs, transformers, wav2vec 2.0).
  - Designed hands-on Python assignments and open-ended course projects, graded them against rubrics, and mentored project teams in office hours.
  - Wrote the course's ~120-page booklet, *Machine Learning on Audio Signals* (see Course Materials Developed), and published teaching notebooks on GitHub (e.g. audio classification with MFCCs and decision trees, keyword detection with Random Forest, speech emotion recognition, SVM speaker classification, HMM / SVM text-independent speaker verification, voice-based gender identification).
  - Promoted from teaching assistant (Fall 2024) to co-instructor (Fall 2025, Fall 2026) based on performance.

Lab Instructor — BME 338L Biomedical Measurements Lab, Southern Illinois University Carbondale

- **Dates / Semesters:** Spring 2023
- **Level & enrollment:** undergraduate biomedical engineering; ~35 students
- **Note:** also completed BME 338 / 338L for credit the same term (grade A)
- **Responsibilities:**
  - Ran weekly lab sessions as the lead instructor in the room: pre-lab briefings, demonstrations, supervising experiments
  - Taught sensors and transducers (strain gauges, thermistors / thermocouples, pressure and optical sensors), Wheatstone bridges, signal conditioning, amplifiers, filters, and data acquisition
  - Guided students through biosignal measurement (ECG, EMG, pulse / PPG, blood pressure, respiration) using BIOPAC, NI LabVIEW / DAQ, and bench instruments
  - Taught calibration, measurement uncertainty, error analysis, and data analysis in MATLAB / Excel
  - Graded lab reports and pre-lab / post-lab assignments with written feedback on technical writing
  - Held office hours; troubleshot equipment, circuits, and software during labs
  - Prepared and maintained lab equipment, kits, and handouts; enforced lab and electrical safety with human subjects
- **Highlights:**

- Taught a hands-on biomedical measurements lab to ~35 undergraduates as the instructor in charge: sensors and transducers, bridges, signal conditioning, data acquisition, and recording biosignals (ECG, EMG, PPG, blood pressure) with BIOPAC and LabVIEW.
- Coached students on calibration, uncertainty, and error analysis in MATLAB / Excel; graded lab reports with detailed feedback on technical writing.
- Maintained lab equipment and enforced electrical and human-subject safety during biosignal recordings.

Graduate Teaching Assistant — EE 402/502 Machine Learning on Biomedical Signals, Clarkson University

- **Dates / Semesters:** Fall 2024
- **Level & enrollment:** combined senior undergraduate / graduate; ~40 students
- **Responsibilities:**
  - Graded homework, coding assignments, quizzes, exams, and course projects using consistent rubrics, with written feedback
  - Held weekly office hours; helped students debug Python / MATLAB code and understand signal processing and ML concepts
  - Led tutorial / recitation and hands-on lab sessions on biomedical signal processing (ECG, EEG, EMG, PPG, voice) and ML pipelines
  - Built and ran a multichannel EMG and IMU lab: surface EMG gesture and muscle-activation detection (hand and jaw clench), head-motion detection from IMU, EMG envelope / RMS and onset detection, sliding-window segmentation, and 1D CNN / TCN classifiers with subject-wise cross-validation, including the effects of electrode placement, skin contact, and fatigue drift across sessions
  - Prepared course materials: Jupyter notebook tutorials, starter code, datasets, assignment solutions, and answer keys
  - Helped design and test assignments and projects; checked solutions and difficulty
  - Mentored student project teams on data selection, feature extraction, model choice, evaluation, and report writing
  - Answered student questions on the course forum and by email; managed the course page and gradebook on Moodle
  - Proctored exams and managed submissions, extensions, and academic-integrity checks
  - Gave guest / substitute lectures and in-class demonstrations
  - Coordinated with the instructor on course logistics, grading standards, and student progress
- **Highlights:**
  - Supported ~40 senior undergraduate and graduate students in machine learning on biomedical signals (ECG, EEG, EMG, PPG, voice): grading, office hours, tutorials, and project mentoring.
  - Developed Jupyter-notebook tutorials, starter code, datasets, and solution keys for signal processing and ML assignments.
  - Mentored student teams through end-to-end ML projects on physiological and voice data, from feature extraction to evaluation.

Graduate Teaching Assistant — EE 622 Advanced Biometrics, Clarkson University

- **Dates / Semesters:** Summer 2025
- **Level & enrollment:** graduate; ~12 students
- **Role:** in-lecture TA (also completed the course for credit the same term, grade A)
- **Responsibilities:**
  - Supported the instructor during live lectures: answered student questions and clarified concepts in real time
  - Assisted with in-class demonstrations of biometric systems (fingerprint, face, iris, voice), and helped students during in-class exercises and discussions

- Managed attendance, in-class activities, and handouts; handled logistics for summer-session lectures
- Helped with grading, student questions outside class, and course materials (slides, datasets, demo code)

- **Highlights:**
  - Served as in-lecture TA for a 12-student graduate Advanced Biometrics course, supporting live instruction, biometric system demonstrations (fingerprint, face, iris, voice), and real-time student questions.
  - Assisted with grading, course materials, and demo code during the summer session.

Graduate Teaching Assistant — EE 260 Embedded Systems, Clarkson University

- **Dates / Semesters:** Spring 2025
- **Level & enrollment:** undergraduate; ~60 students
- **Responsibilities:**
  - Ran hands-on lab sessions on microcontroller programming: GPIO, interrupts, timers, PWM, ADC, UART / SPI / I2C, sensor and actuator interfacing
  - Helped students debug C / Assembly code and hardware wiring using debuggers, multimeters, oscilloscopes, and logic analyzers
  - Taught efficient, low-level C / C++ programming for real-time embedded systems: interrupts, timers, DMA, fixed-point math, and memory-constrained code
  - Graded lab reports, programming assignments, quizzes, exams, and final projects with written feedback
  - Held office hours and open-lab help sessions
  - Prepared and tested lab handouts, starter code, and solution code; set up and maintained lab kits and development boards
  - Mentored student final projects (embedded system design, sensors, IoT prototypes)
  - Enforced lab safety and proper equipment handling
  - Managed submissions and grades on Moodle; coordinated with the instructor on lab schedules and grading
- **Highlights:**
  - Led hands-on embedded systems labs for ~60 undergraduates on microcontroller programming in C / Assembly: GPIO, interrupts, timers, PWM, ADC, and UART / SPI / I2C.
  - Taught systematic hardware and firmware debugging with oscilloscopes, multimeters, logic analyzers, and IDE debuggers.
  - Graded labs, assignments, and exams; maintained development kits; mentored student embedded / IoT final projects.

Graduate Teaching Assistant — ES 250 Electrical Science, Clarkson University

- **Dates / Semesters:** Spring 2024, Fall 2024, Fall 2025, Spring 2026 (4 semesters)
- **Level & enrollment:** undergraduate, core engineering course taken by students from many engineering majors; ~120 students per semester (~480 total over 4 semesters)
- **Responsibilities:**
  - Led recitations / problem-solving sessions on circuit analysis (KVL / KCL, nodal and mesh analysis, Thevenin / Norton, transients, AC phasors, op-amps)
  - Ran hands-on labs: circuit building on breadboards, using multimeters, oscilloscopes, function generators, and power supplies, and comparing measurements with theory and SPICE simulation
  - Graded homework, lab reports, quizzes, and exams with consistent rubrics and written feedback
  - Held office hours and exam review sessions
  - Prepared worked solutions, lab handouts, and practice problems
  - Helped students from non-electrical majors (mechanical, civil, chemical, and others) build confidence with electrical concepts
  - Proctored exams, managed submissions and gradebooks on Moodle, and enforced lab safety

- Coordinated with the instructor and other TAs on grading consistency across sections

- **Highlights:**
  - Taught recitations and circuit labs for ~120 students per semester over 4 semesters (~480 students) in a core electrical science course taken by all engineering majors.
  - Taught circuit analysis (KVL / KCL, nodal / mesh, Thevenin, transients, AC phasors, op-amps) and lab skills with breadboards, oscilloscopes, function generators, and SPICE simulation.
  - Graded homework, labs, and exams consistently across sections; ran office hours and exam reviews; helped non-electrical majors build confidence with circuits.

Instructor — ImransLab (programming education platform), Dhaka, Bangladesh

- **Dates:** January 2018 – November 2023 (about 6 years; in person in Dhaka, then online alongside the M.Eng.)
- **Context:** ImransLab is an ed-tech platform offering programming and computer-science courses (e.g. Python, Java, data structures and algorithms) with instructors from Bangladesh and Canada.
- **Highlights:**
  - Taught programming and computer-science courses to learners on an online and in-person ed-tech platform, covering fundamentals through applied problem solving.
  - Developed lesson plans, code examples, exercises, and assessments, and gave individual feedback on students' code.
  - Held live sessions and doubt-clearing hours, adapting explanations for beginners and career-switchers.

Independent Tutor — O Level & A Level Students, Dhaka, Bangladesh

- **Highlights:**
  - Tutored O Level and A Level students one-on-one and in small groups in science and mathematics, preparing them for Cambridge / Edexcel international examinations.
  - Diagnosed each student's gaps and built personalized study plans, worked solutions, and past-paper practice schedules.
  - Tracked progress with regular mock tests and communicated with parents on performance and goals.

Course Materials Developed

B1. Machine Learning on Audio Signals: Course Booklet

- **Authors:** M. H. Imtiaz, **A. Ahmed**, and Y. Sukhdeve
- **For:** EE 502 Machine Learning on Biomedical Signals, Wallace H. Coulter School of Engineering and Applied Sciences, Clarkson University, Fall 2025 / 2026
- **Format:** ~120-page self-contained course booklet (9 chapters), grown from EE 502 lecture notes; every major concept comes with a Python script (librosa, NumPy, scikit-learn), exercises, and Jupyter notebooks. Modular, so it can be used for a 10-week special topic, a semester survey, or a short professional workshop. Institutional course material (not commercially published).
- **Chapters:**
  1. Introduction to Speech Signals: analog vs. digital, amplitude / frequency / phase, time vs. frequency domain, Nyquist–Shannon sampling, quantization and bit depth, Fourier transform, STFT, mel spectrograms, practical audio systems
  2. Speech Preprocessing and Feature Extraction: resampling, noise reduction, silence trimming, normalization, framing and windowing, spectrograms, ZCR, energy and spectral features, MFCCs, log-mel and pitch, formants, feature scaling, feature comparison and fusion
  3. Machine Learning on Audio Signals, a Summary: supervised / unsupervised learning, evaluation metrics, batch vs. online, instance- vs. model-based learning, challenges of speech ML, validation and hyperparameter tuning

4. Audio Classification Using MFCCs and Decision Tree Models (acoustic event and environmental sound classification): decision trees and Random Forest, embedded-system constraints, denoising, feature selection, effect of MFCC dimensionality, confusion matrices
5. Keyword Detection Using Random Forest
6. SVM-Based Speaker Classification: ANOVA, k-fold cross-validation, chi-squared tests
7. Speech Emotion Recognition Using MFCC and Machine Learning
8. Voice-Based Gender Identification: data harmonization, EDA, dimensionality analysis, hyperparameter search, model compression and deployment
9. Deepfake Audio Detection via MFCC Features Using Machine Learning: Fake-or-Real (FoR) variants, noise augmentation, model comparison

- **My contribution:** wrote the entire booklet (all 9 chapters, code, and exercises), using LLMs (ChatGPT, Claude) to draft and refine explanations, exercises, and example code before verifying them; co-authors Dr. Imtiaz and Y. Sukhdeve reviewed and cross-checked it
- **Highlights:**
  - Wrote a ~120-page, 9-chapter course booklet, *Machine Learning on Audio Signals*, for Clarkson's EE 502 as lead author (all chapters, code, and exercises; reviewed by co-authors), covering speech signal fundamentals, feature extraction, and applied ML (keyword detection, speaker classification, emotion recognition, gender identification, deepfake detection).
  - Paired every chapter with runnable Python code (librosa, NumPy, scikit-learn), exercises, and Jupyter notebooks, turning lecture notes into a modular resource usable for a semester course, a special topic, or a professional workshop.

B2. Embedded Machine Learning: Using the Silicon Labs EFR32XG24 BLE Microcontroller (Course Booklet)

- **Authors:** M. H. Imtiaz, A. Storey, **A. Ahmed**, and A. Kia
- **For:** EE 513 Intelligent System Design, Wallace H. Coulter School of Engineering and Applied Sciences, Clarkson University, Spring 2026
- **Format:** ~150-page self-contained course booklet (11 chapters), grown from EE 513 lecture notes; every concept comes with code tested on the EFR32XG24 development kit, plus exercises and lab projects. Modular, for a 10-week special topic, a semester survey, or a professional workshop. Institutional course material (not commercially published).
- **Chapters:**
  1. Introduction to Embedded Machine Learning: applications, the EFR32MG24 microcontroller
  2. The Art and Science of Machine Learning: learning paradigms, uncertainty, regularization, building learning systems, ethics and governance (fairness, transparency, safety)
  3. Model Preparation and Conversion for Microcontroller Deployment: data collection, model design, TensorFlow Lite conversion, model optimization and quantization
  4. The "Hello World" of TinyML: a sine-wave model in Google Colab, quantization, conversion to C, deployment with Simplicity Studio and the Gecko SDK, LEDs / LCD / buttons, power optimization, profiling
  5. Building a TinyML Application: Gecko SDK workflow, PWM output, debugging, memory / power / timing optimization, custom TinyML components
  6. Handwriting Digit Recognition: on-device CNN, quantization effects, resource use, comparison with cloud approaches
  7. IMU-Based Gesture Recognition: CNN, post-training quantization, sensor fusion, motion detection
  8. Real-Time Posture Detection Using Neural Networks: data acquisition, firmware architecture, wireless communication
  9. Real-Time Gesture and Anomaly Detection: impulse design, spectral features, anomaly detection (Edge Impulse workflow), deployment on EFR32MG24
  10. Real-Time Flame Detection Using a Thermal Infrared Sensor: data-streaming firmware, Python data collection, training, on-device inference

11. Keyword Spotting: mel-spectrogram dataset creation, CNN, optimization and conversion, deployment on EFR32xG24

- **Hardware acceleration covered:** the EFR32MG24's on-chip Matrix Vector Processor (MVP) AI / ML accelerator and Arm Cortex-M33 DSP instructions (CMSIS-DSP, CMSIS-NN) for fast, low-power on-device inference; on-device vs. cloud trade-offs (latency, privacy, power)
- **My contribution:** wrote the entire booklet text (all 11 chapters), using LLMs (ChatGPT, Claude) to draft and refine explanations; A. Storey developed the code examples; Dr. Imtiaz and A. Kia reviewed and cross-checked it
- **Highlights:**
  - Wrote a ~150-page, 11-chapter course booklet, *Embedded Machine Learning*, for Clarkson's EE 513 Intelligent System Design as lead writer (all chapter text; code by a co-author), teaching TinyML deployment on the Silicon Labs EFR32XG24 BLE microcontroller.
  - Structured hands-on, hardware-tested labs covering model quantization and TensorFlow Lite conversion, and on-device applications for keyword spotting, gesture and posture recognition, anomaly detection, handwriting recognition, and thermal flame detection.

Research Mentor — AVHBAC Lab, Clarkson University, Potsdam, NY, USA

- **Dates:** 2024 – present
- **Level & number:** 5 mentees: 4 graduate students (including 1 master's student) and 1 undergraduate researcher
- **Context:** informal research supervision alongside PhD work, covering voice biometrics, speech processing, deepfake detection, pediatric biometrics, and research communication
- **Mentees and outcomes:**
  - **Faheem Ahmad** (graduate): voice deepfake detection, pediatric fingerprint synthesis, children's speaker verification. Outcomes: 2 first-author IEEE conference papers (MDTS 2026 oral; UEMCON 2026) and co-author on the IEEE TBIOM children study (under review).
  - **Yash Sukhdeve** (graduate): synthetic voice detection and emotion-aware deepfake detection. Outcomes: first author of the SPEAD dataset (IEEE DataPort, 2025); co-author on ELAD-SVDSR (IEEE Data Descriptions, 2026) and the IEEE TBIOM children study; collaborator on CUWavestarGAN.
  - **John Parker** (undergraduate): extended-length audio data collection and preprocessing. Outcome: co-author on ELAD-SVDSR (IEEE Data Descriptions, 2026).
  - **Leonel Akanji** (master's student): AIDrawPen, an AI-powered wireless gesture-based drawing pen for pediatric rehabilitation (BLE microcontroller, IMU, deep-learning gesture recognition above 95% accuracy). Outcome: first-author poster at IEEE MDTS 2025 and a TechRxiv preprint.
  - **Batuhan Uzunoglu** (graduate): professional and academic development, including presentation skills, research proposal writing, and proposal-defense preparation (mock defenses, anticipating committee questions, slide design, storytelling with data).
- **Mentoring activities:** defining research problems, literature review, dataset preparation, audio preprocessing and feature extraction, experiment design, model training and evaluation, code review and debugging, figures, paper writing and revision, presentation and defense practice, and responsible use of LLM tools for coding and writing
- **Highlights:**
  - Mentored 5 junior researchers in speech processing, voice biometrics, deepfake detection, pediatric biometrics, and assistive technology, producing 2 accepted IEEE conference papers (1 oral), 1 IEEE conference poster, 2 published datasets, and mentee co-authorships on 2 journal papers.
  - Guided mentees end to end: problem definition, literature review, experiment design, code review, statistical evaluation, figures, and manuscript writing.
  - Brought an undergraduate researcher to co-authorship on a peer-reviewed IEEE dataset paper.
  - Coached graduate students on presentations and research proposal defenses through mock defenses and structured feedback on slides and storytelling.

Research Mentor — Dr. M. M. Khan's Group, North South University, Dhaka, Bangladesh

- **Dates:** September 2021 – July 2022
- **Level & number:** undergraduate co-authors on 7 research projects
- **Context:** guided undergraduate students working on RA-era projects (smart helmet, mm-wave antennas, telemedicine, hospital information, mental-health and fintech apps)
- **Highlights:**
  - Guided and coordinated undergraduate co-authors across 7 IEEE conference projects, covering task planning, hands-on help with hardware, antenna simulation, and app development, and paper writing.
  - Helped undergraduates publish at international IEEE conferences (CCWC, ICCMC, ICAEEE) within a single year.

Pedagogy Training

- (none yet; consider a short certificate before teaching-track applications)

Teaching Interests

- Courses you could teach: (based on coursework + research + teaching; confirm) Biomedical Measurements / Instrumentation (with lab), Neuroengineering, Biomedical Imaging, Electrical Science / Circuit Analysis, Embedded Systems / Microcontrollers, Machine Learning on Biomedical Signals, Biomedical Signal Processing, Introduction to Biometrics, Advanced Biometrics, Machine Learning, Deep Learning, Speech and Audio Signal Processing, Pattern Recognition, Intelligent Systems, Introduction to Programming (Python), Digital Signal Processing, Computer Vision / Image Processing, Instrumentation and IoT, Computer Networks, Computer Architecture, Digital Control, Engineering Statistics, Circuits and Electronics
- Applied computing and systems courses (evidence in brackets): Algorithm Design and Software Engineering (data structures and algorithms taught at ImransLab; VQI software engineering, CI/CD, testing); Computer Hardware and Operating Systems (EE 260 Embedded Systems TA; EE 566 Computer Architecture; ECE 536 Embedded Systems, RTOS); Database Design and Optimization (SQL / MySQL / PostgreSQL / SQLite in apps, ETL and research data stores); Computer Network Design, Operation, and Optimization (EE 507 Computer Networks; McGill lab network design and administration); Cloud Computing Fundamentals (AWS, Azure, GCP, Docker / Kubernetes; ECE 517 Edge Computing); Software Testing Automation and Optimization (pytest, GitHub Actions CI/CD, automated tests and releases); Applied AI (machine learning, deep learning, NLP / LLMs, computer vision, generative AI)
- Graduate AI courses (MSc level; evidence in brackets): Machine Learning and Deep Learning (EE 402/502 co-instructor; CS 570); Natural Language Processing and Large Language Models (lab RAG assistant, agents, post-training); Generative AI (TTS / voice cloning, GANs, LLM data synthesis); Data Engineering for AI (Spark / Parquet / SQL pipelines, Airflow, Databricks, MLflow); AI Security and Cybersecurity (deepfakes, presentation attacks, LLM red-teaming, McGill network security); Human-Computer Interaction for AI (VQI desktop app usability and feedback text, doctor-facing GUI, assistive AIDrawPen mentoring); Trustworthy and Responsible AI (fairness, explainability, model cards); AI Applications in Industry (ImransLab product, CITeR industry projects); Edge AI (B2 booklet)
- Courses you'd like to develop:

Technical Skills

Summary (core stack; rebuilt from the sections below)

- **Languages:** Python, MATLAB, R, C/C++, SQL, Bash, LaTeX, JavaScript / TypeScript
- **ML / Data:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, LightFM, DoubleML / EconML, Prophet, RapidFuzz, NumPy, pandas, SciPy, statsmodels, Matplotlib, seaborn
- **Speech / Audio:** SpeechBrain, Hugging Face Transformers, torchaudio, librosa, Praat/Parselmouth, openSMILE, Kaldi, Neurotechnology VeriSpeak SDK

- **Computer Vision / Embedded:** face recognition (AdaFace, ArcFace, LVFace, EdgeFace, FaceNet), face image quality (OFIQ / ISO/IEC 29794-5, CR-FIQA), OpenCV, scikit-image, Arduino, TI MSP430 / Tiva C, STM32, Raspberry Pi, ESP32, LabVIEW, Verilog/VHDL, MATLAB/Simulink
- **Biomedical Signals:** NeuroKit2, MNE-Python, WFDB, BioSPPy, EEGLAB, PhysioNet, EMG / IMU gesture detection, TCN, Lab Streaming Layer (pylsl)
- **Speech Applications:** OpenAI Whisper, ESPnet, NVIDIA NeMo, hmmlearn, Praat, openSMILE (eGeMAPS)
- **LLMs / Generative AI:** OpenAI GPT-4o / ChatGPT, Anthropic Claude (Claude Code), Hugging Face TRL (SFT, DPO, PPO, GRPO), Qwen2-VL (VLM fine-tuning), DSPy (agent self-improvement), Llama Guard / Detoxify (content safety), Label Studio (human-in-the-loop), GitHub Copilot, prompt engineering, OpenAI / Anthropic APIs, Hugging Face Transformers, PEFT (LoRA), LangChain, small language models on device (llama.cpp / GGUF, Ollama, quantization), RAG, knowledge graphs (Neo4j), agents (LangGraph, Google ADK), LLM evaluation, responsible AI (Fairlearn, SHAP; ISO/IEC 42001)
- **Circuits / Lab:** LTspice, Multisim, PSpice, oscilloscopes, multimeters, function generators, NI LabVIEW / DAQ, BIOPAC
- **Biomedical Imaging:** 3D Slicer, ImageJ / Fiji, SimpleITK, pydicom, NiBabel, MONAI, nnU-Net, ITK-SNAP, CVAT, YOLO / Detectron2
- **Neuroscience Wet Lab:** patch-clamp electrophysiology (pClamp, MultiClamp 700B, Digidata, HSPC-2), cell culture, siRNA, GraphPad Prism, Origin
- **Microscopy:** confocal (Olympus FV3000), spectral (lambda-scan) imaging and spectral unmixing, epifluorescence, phase contrast, DIC, live-cell / time-lapse, calcium imaging (incl. ratiometric Fura-2), immunofluorescence; Fiji / ImageJ, Imaris, CellProfiler, NIS-Elements, ZEN, FluoView
- **RF / Antennas:** CST Studio Suite, Ansys HFSS, Keysight ADS, FEKO, VNA, anechoic / OTA chambers (TRP / TIS), SAR testing
- **Web / Mobile:** Android (Java / Kotlin), Flutter, React, Node.js, PHP / Laravel, Django / Flask, MySQL, Firebase
- **Edge AI:** Silicon Labs EFR32 (on-device audio and IMU models), TensorFlow Lite, ONNX Runtime, TensorRT, NVIDIA Triton Inference Server, OpenVINO, NVIDIA Jetson, Hailo-8L (Hailo Dataflow Compiler), Google Coral, Edge Impulse, FreeRTOS, ExecuTorch, NPU / DSP delegates, local–cloud arbitration
- **VLSI / MEMS / Power:** Cadence, Synopsys, Mentor Tessent, ModelSim, COMSOL, ETAP, PowerWorld, OpenDSS
- **Teaching Tech:** Moodle, Gradescope, Canvas, Zoom / Teams, GitHub Classroom
- **IT / Systems:** Windows Server, Active Directory, Linux administration, TCP/IP, DHCP / DNS, VLANs, VPN, firewalls, NAS / RAID, multi-site backup, PowerShell / Bash
- **Tools:** Git, GitHub, Linux, HPC/SLURM, CUDA, Docker, Jupyter, VS Code, Overleaf, IEEE DataPort, Databricks, Spark, MLflow, Azure AI Search

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Programming Languages

- Rust — PhD (VQI feature-extractor prototype via PyO3)
- JavaScript / TypeScript — RA, NSU (web apps); PhD (web dashboards, the lab annotation and review front end in TypeScript / React)
- Python — PhD (analysis, ML, feature extraction, app)
- MATLAB — PhD (signal processing, speaker recognition toolkits)
- R — PhD (mixed-effects models, statistics)
- C / C++ — Kaldi, SDK integration, embedded, FPGA review
- Java / C# — VeriSpeak SDK bindings
- Julia — numerical computing
- Bash / shell scripting / PowerShell — batch processing, cluster jobs
- SQL — data management
- Verilog / VHDL — FPGA review
- LaTeX — papers, dissertation, slides

- Markdown — documentation, READMEs

Machine Learning & Data Science

- **Supervised learning, classification and regression; predictive analytics (predicting verification outcomes, sepsis onset, recruitment and demand)** — PhD (VQI, lab operations); ImransLab (sepsis early warning)
  - Languages: Python, MATLAB, R
  - Libraries: scikit-learn, NumPy, pandas, SciPy, statsmodels; MATLAB Statistics and Machine Learning Toolbox; R caret / tidymodels
  - Software: Jupyter Notebook / JupyterLab, Google Colab, MATLAB Classification Learner / Regression Learner apps, Weka
- **Random Forest, Ridge Regression, Logistic Regression** — PhD (VQI)
  - Languages: Python, MATLAB, R
  - Libraries: scikit-learn; MATLAB TreeBagger / fitcensemble / ridge / fitglm; R randomForest / ranger / glmnet
- **Support Vector Machines (SVC, SVR, RBF kernels, Platt scaling)** — PhD (VQI)
  - Languages: Python, MATLAB, R
  - Libraries: scikit-learn, LIBSVM, ThunderSVM / cuML (GPU); MATLAB fitcsvm / fitrsvm; R e1071 / kernlab
- **Gradient boosting: XGBoost, LightGBM** — PhD (VQI)
  - Languages: Python, R
  - Libraries: xgboost, lightgbm, CatBoost; R xgboost
- **Neural networks: MLP (batch normalization, dropout), TabNet** — PhD (VQI)
  - Languages: Python, MATLAB
  - Libraries: PyTorch, PyTorch Lightning, pytorch-tabnet, TensorFlow, Keras, scikit-learn MLPClassifier; MATLAB Deep Learning Toolbox
  - Software: TensorBoard, Weights & Biases
- **Generative adversarial networks (GANs), deepfake detection** — PhD (CUWavestarGAN)
  - Languages: Python
  - Libraries: PyTorch, TensorFlow / Keras, torchaudio, CUDA / cuDNN
  - Software: TensorBoard, Weights & Biases
- **Model selection across 8 model families; hyperparameter tuning (random and grid search); k-fold cross-validation** — PhD (VQI)
  - Languages: Python, MATLAB, R
  - Libraries: scikit-learn GridSearchCV / RandomizedSearchCV, Optuna, Hyperopt, Ray Tune; MATLAB bayesopt / Experiment Manager; R caret
- **Feature engineering and feature selection (variance, correlation, importance-based pruning)** — PhD (VQI)
  - Languages: Python, MATLAB, R
  - Libraries: scikit-learn feature_selection, pandas, SciPy; MATLAB fscmrmr / relieff; R Boruta
- **Dimensionality reduction: PCA, Factor Analysis (BIC), ICA** — PhD (VQI)
  - Languages: Python, MATLAB, R
  - Libraries: scikit-learn decomposition (PCA, FactorAnalysis, FastICA), factor_analyzer; MATLAB pca / factoran / rica; R psych / fastICA
- **Class balancing, label design, train/validation/held-out splits** — PhD (VQI)
  - Languages: Python

- Libraries: pandas, scikit-learn, imbalanced-learn

- **Explainable ML (interpretable features, per-feature score explanations)** — PhD (VQI)
  - Languages: Python
  - Libraries: SHAP, LIME, scikit-learn permutation / Gini importance
- **Extending and optimizing ML libraries: custom scikit-learn transformers and estimators (fit / transform API, Pipeline and GridSearchCV compatible), custom XGBoost / LightGBM objectives (gradient and Hessian), custom PyTorch / SpeechBrain modules (s-norm scoring, pair samplers), C++ (pybind11) and Numba extensions for hot loops** — PhD (VQI)
  - Languages: Python, C++
  - Libraries: scikit-learn (BaseEstimator, TransformerMixin, Pipeline), xgboost, lightgbm, PyTorch, SpeechBrain, NumPy, Numba, Cython, pybind11, CuPy
  - Software: Jupyter, VS Code, Git / GitHub, pytest, PyTorch Profiler, py-spy
- **Model persistence, deployment, and inference latency benchmarking** — PhD (VQI)
  - Languages: Python
  - Libraries: joblib, pickle, ONNX / ONNX Runtime, timeit
  - Software: Docker
- **Large-scale data processing (~2.6 M utterances, 11 datasets, ~29 M verification pairs)** — PhD (VQI)
  - Languages: Python, Bash, SQL
  - Libraries: pandas, NumPy, Polars, Dask, multiprocessing / joblib, h5py (HDF5), PyArrow (Parquet), tqdm
  - Software: HPC cluster, SLURM, NVIDIA GPUs, CUDA, SQLite / PostgreSQL
- **Data augmentation (additive noise at random SNR, room impulse response convolution)** — PhD (VQI)
  - Languages: Python, MATLAB
  - Libraries: torchaudio, audiomentations, torch-audiomentations, pyroomacoustics, SciPy signal (fftconvolve); MATLAB Audio Toolbox audioDataAugmenter
  - Data: MUSAN corpus, RIR corpus
- **Data visualization (ROC, DET, ERC, heatmaps, box plots, scatter plots)** — PhD
  - Languages: Python, MATLAB, R
  - Libraries: Matplotlib, seaborn, Plotly; MATLAB plotting; R ggplot2
  - Software: Excel, Tableau / Power BI

Speech, Audio & Signal Processing

- **Acoustic AI: acoustic event and environmental sound classification, acoustic scene analysis, room acoustics (reverberation, RT60, C50), microphone and channel effects, cough and respiratory sound analysis** — B1 booklet; EE 402/502 lectures; PhD (degradation study, VQI features, multi-microphone datasets)
  - Libraries: librosa, torchaudio, pyroomacoustics, PANNs, YAMNet / VGGish, scikit-learn, PyTorch
- **Speaker recognition / verification / identification** — PhD
  - Also on device: text-dependent speaker recognition on the Silicon Labs EFR32 board (MFCC with CMSIS-DSP, small CNN, INT8, TFLite Micro, CMSIS-NN) — EE 513 booklet / PhD
  - Languages: Python, MATLAB, C++
  - Libraries: SpeechBrain, WeSpeaker, NVIDIA NeMo, pyannote.audio, Resemblyzer, 3D-Speaker; MATLAB Audio Toolbox speakerRecognition / ivectorSystem
  - Software: Neurotechnology VeriSpeak SDK, Kaldi, ALIZÉ
- **Deep speaker embedding systems: ECAPA-TDNN, ECAPA2, ResNet34-SE, x-vector, WavLM-SV** — PhD

- Languages: Python
- Libraries: SpeechBrain (ECAPA-TDNN, x-vector), Hugging Face Hub / Transformers (ECAPA2, WavLM-SV), WeSpeaker (ResNet34), PyTorch, torchaudio
- Software: Kaldi (x-vector recipes)

- **Classical speaker recognition: i-vector / PLDA** — PhD (children study)
  - Languages: Python, MATLAB, C++
  - Libraries: MSR Identity Toolbox (MATLAB), MATLAB ivectorSystem, SIDEKIT, scikit-learn GMM
  - Software: Kaldi, ALIZÉ
- **Self-supervised speech models: WavLM, Wav2Vec2, HuBERT** — PhD (metrics study)
  - Languages: Python
  - Libraries: Hugging Face Transformers, PyTorch, torchaudio pipelines, fairseq, s3prl
- **Audio preprocessing: resampling, voice activity detection, loudness normalization, format conversion** — PhD
  - Languages: Python, MATLAB, Bash
  - Libraries: librosa, torchaudio, SoundFile, pydub, resampy, webrtcvad, Silero VAD, pyannote VAD, pyloudnorm; MATLAB Audio Toolbox detectSpeech / resample
  - Software: FFmpeg, SoX, Audacity
- **Acoustic feature extraction: MFCC, LPCC, formants (F1–F4), F0/pitch, HNR, SNR, clipping, reverberation (C50), bandwidth, spectral features** — PhD
  - Languages: Python, MATLAB
  - Libraries: librosa, Parselmouth (Praat in Python), openSMILE (eGeMAPS / ComParE), python_speech_features, spafe, torchaudio, SciPy signal, pyworld, CREPE / pYIN (pitch), Brouhaha (SNR/C50), Surfboard, DisVoice; MATLAB Audio Toolbox (mfcc, pitch, harmonicRatio), VOICEBOX, COVAREP
  - Software: Praat, openSMILE, Audacity
- **Speech quality assessment (perceptual and signal-based metrics)** — PhD (metrics study)
  - Languages: Python, MATLAB
  - Libraries: pesq (PESQ), pystoi (STOI), DNSMOS, NISQA, torchmetrics audio, speechmos, pysepm, WADA-SNR, ViSQOL; MATLAB Audio Toolbox
- **Score normalization (S-Norm / AS-Norm)** — PhD
  - Languages: Python
  - Libraries: NumPy, SpeechBrain, WeSpeaker
- **Speech enhancement evaluation (deep-learning models)** — PhD (BDCC paper)
  - Languages: Python
  - Libraries: SpeechBrain (MetricGAN+, SepFormer), Asteroid, DeepFilterNet, Demucs, noisereduce, PyTorch
- **Voice deepfake / synthetic speech detection** — PhD (CUWavestarGAN, MDTS paper, ELAD-SVDSR)
  - Languages: Python
  - Libraries: PyTorch, scikit-learn, librosa, AASIST / RawNet2 baselines, Coqui TTS / ElevenLabs / other voice-cloning tools (for synthetic data)
- **Emotion-robust speaker recognition: effect of emotional state on FAR / FRR, emotion-diverse annotated data, MFCC / prosodic features, CNN and LSTM models, multi-task learning (speaker + emotion)** — RAPS 2024 award-winning presentation
  - Libraries: PyTorch / TensorFlow-Keras, librosa, openSMILE; datasets: ESD, RAVDESS, CREMA-D, IEMOCAP
- **Emotional speech analysis** — PhD (ESD analysis)

- Languages: Python
- Libraries: pandas, Matplotlib, seaborn, SciPy stats

- **Age normalization of acoustic features (z-score, linear, polynomial growth models)** — PhD (children study)
  - Languages: Python, R, MATLAB
  - Libraries: NumPy, SciPy, statsmodels, scikit-learn; R lm
- **Audio recording, editing and inspection** — PhD (data collection)
  - Software: Audacity, Adobe Audition, Praat, Sonic Visualiser, iPhone call recording app, Inni Oasis R1 recorder

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Biometrics

- **Biometric quality assessment; ISO/IEC biometric sample quality framework (NFIQ 2, OFIQ as references)** — PhD
  - Languages: Python, C++
  - Software: NFIQ 2, OFIQ (reference implementations)
- **Face recognition and face image quality: face detection and alignment, deep face embeddings and cosine-similarity verification (AdaFace, ArcFace, LVFace, EdgeFace, FaceNet), ISO/IEC 29794-5 face quality components (OFIQ: lighting, sharpness, pose, framing, expression, occlusion, UQS), CR-FIQA, utility-based quality labels from matcher consensus, explainable quality models** — CITeR 25F-04W (FUQI); Beyond ROC-AUC (face matchers); PR24 (passport-photo capture)
  - Languages: Python, C++
  - Libraries: PyTorch, InsightFace, facenet-pytorch, ONNX Runtime, OpenCV, dlib, NumPy, pandas, scikit-learn, LightGBM, XGBoost, CatBoost, SciPy
  - Software: OFIQ v1.0.3 (ISO/IEC 29794-5 reference implementation), CR-FIQA, NVIDIA GPUs
  - Data: CelebA, XQLFW, CPLFW, CALFW, AgeDB, TinyFace, CFP, LFW
- **Biometric quality evaluation: ISO/IEC 29794-1 Error-versus-Discard Characteristic (EDC / ERC) and pAUC gain, DET by quality bins, Spearman rank agreement with utility, person-disjoint splits with held-out datasets and matchers** — CITeR 25F-04W (FUQI); PR1 (VQI)
  - Languages: Python, R
  - Libraries: NumPy, pandas, SciPy, scikit-learn, Matplotlib, pyeer, bob.measure
- **Biometric performance metrics: EER, FNMR/FMR, minDCF, d', AUC-ROC** — PhD
  - Languages: Python, MATLAB
  - Libraries: scikit-learn metrics (roc_curve, roc_auc_score), NumPy, SpeechBrain metrics, pyeer, bob.measure
- **DET curves, ranked DET, Error-vs-Reject Curves (ERC)** — PhD
  - Languages: Python, MATLAB
  - Libraries: Matplotlib, pyeer, bob.measure, NIST DETware
- **Multimodal biometric data capture in the field: fingerprint scanners, iris cameras, face capture, voice recording; portable capture stations; on-site quality checks and recapture; session metadata logging** — longitudinal child biometrics collection
  - Hardware / Software (typical): optical fingerprint readers (e.g. Crossmatch / HID, Integrated Biometrics), iris cameras (e.g. IriShield, Iritech), DSLR / webcam face capture, studio microphones, NFIQ 2 on-site quality checking
- **Longitudinal biometric analysis (template aging)** — PhD (children study)
  - Languages: Python, R
  - Libraries: pandas, statsmodels, R lme4

- **Multimodal biometrics knowledge (fingerprint, face, iris, gait)** — PhD, HI-POCT paper, FPGA review
  - Languages: Python, MATLAB, Verilog / VHDL
  - Libraries / Software: wearable IMU sensor software (gait paper), Xilinx Vivado / Vitis HLS / Intel Quartus (FPGA review)
- **Biometric systems (fingerprint, face, iris, voice, gait; enrollment, matching, liveness / presentation attack detection, template protection, fusion)** — EE 523 Intro to Biometrics, EE 622 Advanced Biometrics
  - Languages: Python, MATLAB, C++
  - Libraries: OpenCV, dlib, face_recognition, DeepFace, InsightFace, NBIS (NIST fingerprint tools), SourceAFIS, OSIRIS / USIT (iris), bob (Idiap biometrics toolkit)
  - Software: NIST NFIQ 2, OFIQ, Neurotechnology SDKs (VeriFinger, VeriLook, VeriEye, VeriSpeak)

Deep Learning (coursework + PhD)

- **Neural network fundamentals: loss functions (cross-entropy, mean squared error, focal, Dice, contrastive), backpropagation and automatic differentiation, gradient descent and optimizers (SGD with momentum, Adam / AdamW), regularization (dropout, weight decay, early stopping), learning-rate schedules, custom gradients** — CS 570 Deep Learning; Co-Instructor EE 402/502 (teaching); PhD (custom XGBoost / LightGBM gradient and Hessian, GAN training)
  - Languages: Python, MATLAB
  - Libraries: NumPy (from-scratch exercises), PyTorch (autograd, torch.optim), TensorFlow / Keras, JAX; MATLAB Deep Learning Toolbox
- **Deep neural networks: CNNs, RNNs / LSTMs / GRUs, autoencoders, attention and transformers, GANs, transfer learning, fine-tuning** — CS 570 Deep Learning, PhD
  - Languages: Python, MATLAB
  - Libraries: PyTorch, TensorFlow, Keras, PyTorch Lightning, Hugging Face Transformers, timm, torchvision, torchaudio; JAX / Flax (re-implemented small models from papers during literature reviews); MATLAB Deep Learning Toolbox
  - Software: Google Colab, Kaggle, Jupyter, TensorBoard, Weights & Biases, NVIDIA GPUs / CUDA / cuDNN
- **Training practice: optimizers (SGD, Adam), regularization (dropout, batch norm, weight decay), learning-rate schedules, early stopping, data loaders, mixed precision** — CS 570, PhD
  - Libraries: PyTorch, TensorFlow / Keras, torch.cuda.amp

High-Performance Inference & Model Efficiency (PhD, ImransLab, teaching)

- **LLM inference efficiency: paged KV cache and prefix caching, continuous batching, weight quantization (AWQ, GPTQ, FP8 / INT8), speculative decoding, FlashAttention kernels, dense vs mixture-of-experts trade-offs, quality checks after optimization** — PhD (LLM and TTS batch generation)
  - Libraries: vLLM, SGLang, TensorRT-LLM, Hugging Face Transformers, FlashAttention, bitsandbytes, AutoGPTQ, AWQ
- **Inference optimization: latency, throughput, real-time factor, batching, mixed precision (FP16 / INT8), quantization, KV caching, profiling and bottleneck analysis** — PhD (VQI labelling and app, speech enhancement, TTS generation); ImransLab (ONNX CPU inference); EE 513 TinyML booklet
  - Languages: Python, C++, CUDA C++
  - Libraries: PyTorch (TorchScript, torch.compile, AMP), ONNX Runtime, TensorRT, TensorRT-LLM, vLLM, SGLang, CuPy, Numba, pybind11
  - Software: NVIDIA Nsight Systems / Compute, PyTorch Profiler, py-spy, cProfile, nvidia-smi
- **GPU programming and multi-GPU parallelism: CUDA kernels, data and model parallelism, distributed inference** — PhD (labelling at scale, GAN / TTS training)

- Libraries: CUDA, CuPy, Numba CUDA, PyTorch DDP / FSDP, torch.distributed, DeepSpeed, NCCL
- Software: HPC GPU clusters (SLURM), NVIDIA GPUs

- **Low-latency ML services, concurrency, and real-time pipelines: microservices over REST and gRPC, asyncio and thread / process pools, request batching, Redis caching of embeddings and features, Kafka consumer groups with windowed aggregations, load testing for tail latency (p95 / p99), many models trained, served, and monitored side by side** — PhD (lab assistant and model services, LLM router, VQI models)
  - Libraries / Software: FastAPI, gRPC, asyncio, concurrent.futures, multiprocessing, Redis, Apache Kafka, Locust, Docker, Kubernetes (k3s), MLflow, SLURM job arrays, Prometheus / Grafana
- **Real-time and duplex streaming audio: streaming capture and playback, ring buffers, chunked / frame-based processing, overlap-add, streaming VAD, live scoring** — PhD (VQI app, streaming enhancement)
  - Libraries: PortAudio, sounddevice, PyAudio, webrtcvad / Silero VAD, torchaudio streaming
- **End-to-end audio pipeline optimization: decoding, resampling, feature extraction, model inference, and post-processing** — PhD (VQI, degradation study, enhancement)
  - Libraries: FFmpeg, torchaudio, librosa, NumPy / Numba, multiprocessing
- **Pulse-echo and active acoustic measurement: swept-sine and pulse excitation, deconvolution and matched filtering, echo detection and time-of-flight estimation, reverberation decay (RT60, C50), background-noise spectra (principles shared with sonar and radar)** — PhD (recording-room measurements, PR14 / VQI augmentation)
  - Languages / Libraries: Python (NumPy, SciPy signal, pyroomacoustics, librosa), MATLAB (Audio Toolbox impulse-response measurer); Software: Audacity, REW (Room EQ Wizard)
- **Safety-critical and high-reliability AI and hardware: life-critical clinical alerting and surgical-safety review tools (false-alarm vs miss trade-offs, thresholds set with clinicians, validation reports), production sign-off of phone antennas through EVT / DVT / PVT, reliability engineering and life testing** — ImransLab; BlackBerry; EE 603
- **Patent and prior-art searching as part of literature review; open-source licensing** — PhD (VQI, deepfake and denoising pipelines)
  - Software: Google Patents, Espacenet, Google Scholar
- **ML pipeline orchestration: scheduled data and training workflows with retries, logging, and versioned outputs** — PhD (lab data and training pipelines)
  - Tools: Apache Airflow, Prefect, Kubeflow Pipelines, MLflow, DVC, Docker, Kubernetes (k3s)

Large Language Models (LLMs) & Generative AI

- **AI security: red-teaming voice biometrics with deepfakes and voice conversion, presentation attack detection (ISO/IEC 30107), anti-spoofing models, threat modelling of biometric and LLM systems, prompt-injection and jailbreak testing of LLM agents, guardrails and least-privilege tool access, security scanning in CI/CD; content safety and moderation (harmful-content classifiers such as Llama Guard and Detoxify, LLM-judge policy checks, human review of borderline cases), abuse, spam, and bot-traffic detection on LLM APIs (rate and timing features, embedding near-duplicates, Isolation Forest), LLM-simulated attacker and user personas** — PhD (CITeR deepfake projects, CUWavestarGAN, lab LLM agents, VQI pipeline); McGill IT (network hardening)
  - Languages: Python
  - Tools: promptfoo, garak, PyRIT, Llama Guard, Detoxify, OpenAI moderation endpoint, Perspective API, NeMo Guardrails, sentence-transformers and scikit-learn (abuse and bot-traffic detection), RawNet2 / AASIST, ASVspoof, CodeQL, Bandit, pip-audit, Dependabot
- **Agent authorization and runtime security: intent verification of agent tool calls (embedding similarity, zero-shot classification, confidence scoring), policy-as-code (Open Policy Agent, AWS Cedar), inline low-latency guardrails in the API request path, prompt-injection and jailbreak defenses, false-block vs missed-**

attack evaluation; identity and authentication background from voice biometrics (speaker verification, presentation attack detection) — PhD (lab agents and assistant; voice biometrics research)
    - Tools: FastAPI, ONNX Runtime, sentence-transformers, Hugging Face zero-shot pipelines, Open Policy Agent, Cedar, LiteLLM, AWS Bedrock, promptfoo, garak
- **Natural language processing: text classification (BERT / DistilBERT, TF-IDF + linear models), named-entity recognition and information extraction (spaCy, LLMs), semantic search and embeddings (sentence-transformers, FAISS), topic modelling (BERTopic), clinical-note NLP (negation handling, concept extraction), ASR transcripts and text analysis, grapheme-to-phoneme conversion** — PhD (lab records, literature, speech corpora); ImransLab (sepsis clinical notes)
    - Libraries: spaCy, Hugging Face Transformers, sentence-transformers, BERTopic, gensim, NLTK, scikit-learn, phonemizer, jiwer
- **Product usage analytics: opt-in usage logging, feature and session analysis, funnel and error analysis, dashboards, using findings to prioritize improvements** — PhD (VQI software)
    - Tools: SQL, pandas, Plotly Dash, GitHub release and issue data
- **Agentic AI and post-training (small scale): multi-agent planning, tool use, grounding, memory, and personalization; supervised fine-tuning and DPO preference tuning of small open models; hallucination and faithfulness evaluation; GRPO with verifiable rule-based and judge-based rewards, compared with DPO and PPO; LoRA fine-tuning of open-weight vision-language models (Qwen2-VL)** — PhD (lab assistant and agents; script-generation model)
    - Languages: Python
    - Libraries: LangGraph, Google ADK, LangChain, RAGAS, Hugging Face TRL (SFTTrainer, DPOTrainer, PPOTrainer, GRPOTrainer, RewardTrainer), PEFT (LoRA / QLoRA), Transformers, Accelerate, DeepSpeed, bitsandbytes, vLLM (rollouts), OpenRLHF; Qwen2-VL, LLaVA, Llama, Qwen, Mistral open weights
- **Agent frameworks and self-hosted inference: LangGraph, Google ADK, LangChain, LlamaIndex agents with tool calls; self-hosted models with Ollama and vLLM; BERT-family text classifiers** — PhD (lab agents, RAG assistant, note classification); also CrewAI, AutoGen, and MCP tool servers
    - Libraries: LangGraph, google-adk, LangChain, LlamaIndex, Ollama, vLLM, Hugging Face Transformers
- **Streaming anomaly detection and decision support: change data capture (Debezium, Kafka), rolling z-score and seasonal baselines, Isolation Forest on event streams, LLM root-cause summaries with confidence scores and evidence links; Monte Carlo simulation of schedules; natural-language rules compiled into structured optimizer constraints (LLM + JSON schema)** — PhD (lab study database, data-collection scheduling)
    - Libraries / Software: Debezium, Apache Kafka, PostgreSQL, scikit-learn, statsmodels, NumPy (Monte Carlo), PuLP / OR-Tools, Pydantic, Claude / OpenAI APIs
- **Classical data science for operations: clustering (k-means, DBSCAN), anomaly detection (Isolation Forest), optimization (integer programming with PuLP / OR-Tools), time-series forecasting (ARIMA, Prophet)** — PhD (VQI data, data-collection planning)
    - Libraries: scikit-learn, PuLP, OR-Tools, statsmodels, Prophet
- **Recommendation and ranking: content-based recommendation from embeddings and user history, re-ranking with user feedback, ranking metrics (precision at k, NDCG); product matching and recommendation** — PhD (lab assistant paper and dataset recommendations); RA, NSU (fintech product recommendation, C9)
    - Libraries: sentence-transformers, FAISS, scikit-learn, LightGBM (LambdaRank), LightFM, pandas
    - Also: collaborative filtering from implicit feedback (matrix factorization), LLM-generated item summaries and tags for cold-start items, LLM re-rankers compared with cross-encoder and LambdaRank re-rankers — PhD (lab assistant recommendations)

- **Planning and supply-side analytics: demand forecasting (ARIMA, Prophet, gradient boosting), inventory and replenishment planning, scheduling optimization (integer programming), work with suppliers and contract manufacturers on tolerances, yield, and cost, procurement and vendor comparison** — PhD (lab operations); BlackBerry; McGill IT
  - Libraries / Software: statsmodels, Prophet, LightGBM, PuLP, OR-Tools, pandas, Excel
- **LLM and speech evaluation systems: automated judges (LLM-as-judge) validated against human review, regression suites, reference datasets, WER / DER for ASR and diarization, audio-quality measures (VQI, PESQ, STOI, SNR) to decide whether a recording can be trusted, weak labels from model consensus** — PhD (RAG evaluation harness, VQI, degradation study, CITeR deepfake project)
  - Libraries / Tools: RAGAS, LangSmith, jiwer, pyannote.audio, OpenAI Whisper, pesq, pystoi, pandas, SQL
- **LLM evaluation research: benchmark and test-set design, LLM judges (prompted and LoRA fine-tuned) validated against human review, LLM-based data synthesis with quality checks, evaluation efficiency (sampling, caching, cheaper judges), metric validity (checking that a metric tracks the capability it claims to measure)** — PhD (evaluation harness, VQI, Beyond ROC-AUC, SPEAD / ELAD script generation)
  - Libraries: Hugging Face Transformers / PEFT / TRL, RAGAS, LangSmith, OpenAI / Anthropic SDKs, vLLM, pandas, SciPy stats
- **Production agents and model routing: tool use, orchestration, guardrails, retries and fallbacks, trace logging; weighted routing and staged rollout of models and prompts; agent self-improvement loops (failed traces to regression cases and few-shot examples, automatic prompt optimization with DSPy, gated releases)** — PhD (lab RAG assistant and agents; local–cloud routing)
  - Languages: Python, TypeScript
  - Libraries / Software: LangGraph, LangChain, LlamaIndex, DSPy, LangSmith, Pydantic, LiteLLM, FAISS / Chroma, Claude API, Vertex AI, Google Cloud Run, Docker
- **Responsible AI and model risk: bias and fairness testing (per-group error rates, disparity measures), explainability (SHAP, LIME, Grad-CAM, per-feature explanations), model cards, documentation of intended use, limitations, and model risk, validation against standards and sponsor requirements** — PhD (VQI, speaker models, lab LLM tools); ImransLab (clinical models); EE 402/502 teaching
  - Languages: Python, R, SQL
  - Libraries: Fairlearn, AIF360, SHAP, LIME, Captum, pytorch-grad-cam, pandas and scikit-learn (per-group metrics; sepsis model fairness by sex, age group, and payer type, ImransLab)
- **Vendor and third-party AI assessment: benchmarking commercial SDKs and cloud LLM APIs against open-source and self-hosted models on fixed test suites; documenting capabilities, limitations, cost, latency, and governance implications; A/B testing of model and prompt versions** — PhD (VeriSpeak SDK, cloud vs local LLMs)
  - Tools: Neurotechnology VeriSpeak SDK, OpenAI / Anthropic / Gemini APIs, vLLM, Ollama, LangSmith, SciPy stats
- **Production feedback loops and telemetry: active feedback and passive behavioural signals, event streaming (Kafka), user-journey KPIs, dashboards; survey and questionnaire analytics (cleaning, open-ended response coding with NLP and LLMs)** — PhD (lab assistant, data-collection questionnaires, VQI usage analytics)
- **Customer-service AI: request intent classification and routing, agent-assist reply drafting with retrieval, human review before sending, service metrics (time to resolution, edit rate, satisfaction); help-desk ticket triage and escalation** — PhD (lab assistant); HAF General Hospital (Customer Support Consultant)
  - Tools: Hugging Face Transformers, LangChain, LlamaIndex, OpenAI / Anthropic SDKs, SQL, Plotly Dash
  - Tools: Apache Kafka, Amazon EKS, SageMaker, Athena, S3, Qualtrics, Google Forms, BERTopic, pandas, SQL, Plotly Dash
- **Deep-search agents, voice AI agents, and enterprise integration: iterative plan–search–reflect research agents with cited reports; streaming voice pipelines (ASR, LLM, TTS) over WebSockets / WebRTC with**

barge-in and per-stage latency; single sign-on (OAuth 2.0 / OIDC), permission-aware retrieval, webhooks, audit logs, encryption for data security — PhD (lab assistant)

- Libraries / Software: LangGraph, Semantic Scholar / arXiv / Crossref APIs, faster-whisper, Silero VAD, XTTS, FastAPI WebSockets, WebRTC (aiortc), OAuth 2.0 / OpenID Connect, HTTP / REST

- **Agent design trade-offs: context preload vs retrieval vs tool calling vs hybrid, latency / cost / quality tension, guardrails, confirm-first actions and failure containment, eval-driven development** — PhD (lab assistant and agents)
  - Frameworks: LangGraph, OpenAI Agents SDK, Google ADK, MCP, LangSmith; warehouses: PostgreSQL, BigQuery, Snowflake
- **AI evaluation methodology: metrics, test sets, acceptance thresholds and release gates, uncertainty (bootstrap confidence intervals, calibration), significance tests, drift monitoring in production** — PhD (LLM evaluation harness, VQI, Beyond ROC-AUC)
  - Also: probability calibration (temperature, Platt, isotonic scaling; reliability diagrams, expected calibration error) for classifiers and routers — PhD (note classifier, intent router, VQI)
  - Libraries: SciPy stats, statsmodels, scikit-learn, RAGAS, LangSmith, Evidently
- **Document AI: information extraction from unstructured documents (PDF parsing, OCR, layout-aware extraction, LLM structured outputs); financial documents (vendor quotes, invoices, receipts) with rule-based validation and invoice-to-purchase-request matching, human review of mismatches** — PhD (lab reports and protocols; lab procurement); McGill Financial Services (accounts payable domain knowledge)
  - Languages: Python, SQL
  - Libraries: PyMuPDF, Unstructured, Tesseract OCR, docTR, pdfplumber, Camelot (tables), LayoutLMv3 (Hugging Face), Pydantic, RapidFuzz, pandas, OpenAI / Anthropic SDKs
  - Software: Excel, university ERP / financial systems (Banner Finance), Azure AI Document Intelligence and AWS Textract (cloud alternatives)
- **Conversation intelligence: transcription (Whisper, faster-whisper), speaker diarization and attribution (pyannote plus speaker-recognition embeddings), structured extraction and source-backed synthesis from transcripts, stage-wise evaluation (WER, DER, extraction precision / recall, faithfulness), private on-premise processing** — PhD (ELAD recordings, lab meetings and sponsor calls)
  - Libraries: openai-whisper, faster-whisper, pyannote.audio, SpeechBrain (ECAPA-TDNN), jiwer, LangChain, Pydantic, vLLM
- **Information retrieval: hybrid search (BM25 plus dense embeddings), cross-encoder re-ranking, query rewriting, chunking, retrieval metrics (recall at k, MRR, NDCG); question answering over regulatory and contractual documents (IRB protocols, consent forms, data-use and licence agreements) with clause-level citations; approximate nearest-neighbour vector indexes (FAISS flat / IVF / HNSW); entity matching and de-duplication of records (blocking, fuzzy matching, embedding similarity, learned match classifier); managed cloud search (Azure AI Search: hybrid and vector search, semantic ranking)** — PhD (lab RAG assistant, paper and dataset catalogue)
  - Languages: Python, SQL
  - Libraries / Software: rank_bm25, Elasticsearch / OpenSearch, Azure AI Search, Amazon OpenSearch Service, Vespa, sentence-transformers (bi- and cross-encoders), FAISS, Chroma, LangChain, LlamaIndex, RapidFuzz, recordlinkage, scikit-learn, LightGBM
- **Human-in-the-loop ML: confidence-based review queues, label correction, active learning (uncertainty sampling), silver-to-gold labelling with adjudication and inter-annotator agreement, override-rate and review-time tracking** — PhD (lab note and request classifiers, LLM labelling)
  - Languages: Python, SQL
  - Libraries / Software: Label Studio, Prodigy, Argilla, CVAT, modAL, small-text, cleanlab, Snorkel, scikit-learn, Hugging Face Transformers, sentence-transformers, pandas, statsmodels (Cohen's kappa)

- **Knowledge graphs and GraphRAG: entity and relation extraction with LLMs into a graph of papers, datasets, collections, models, and experiments; graph queries combined with vector search for multi-hop questions** — PhD (lab RAG assistant)
  - Libraries / Software: Neo4j (Cypher), LangChain graph integrations, LlamaIndex PropertyGraphIndex, NetworkX, Pydantic
- **AI governance, security, and compliance standards: ISO/IEC 42001 (AI management system), NIST AI Risk Management Framework, OWASP Top 10 for LLM Applications, UK AI Cyber Security Code of Practice; privacy for human-subjects data (IRB, de-identification, least-privilege access)** — PhD (lab LLM assistant and agents; human-subjects data collections)
- **Multimodal learning and foundation models: multimodal fusion (structured records, clinical text, imaging), multimodal datasets (ECG / PPG / GSR physiology, multi-microphone audio, face / voice / iris / fingerprint biometrics), self-supervised foundation models (WavLM, HuBERT, Wav2Vec2, DINOv2), vision-language models for figure and table understanding and image auto-tagging (GPT-4o, Claude, Qwen2-VL; zero-shot CLIP / SigLIP labels), speech-text alignment with Whisper; LoRA fine-tuning of open-weight vision-language models on curated image-text pairs** — PhD; ImransLab (sepsis fusion); mentored fingerprint work
  - Languages: Python
  - Libraries: PyTorch, Hugging Face Transformers, timm, OpenCLIP (CLIP / SigLIP image-text embeddings), OpenAI / Anthropic SDKs, Whisper, pandas
- **Trustworthy AI: failure prediction and selective prediction (reject option), knowing when a model should not be trusted, robustness under distribution shift (noise, devices, codecs, ageing over years), synthetic-media and novelty detection, calibration, fairness, explainability** — PhD (VQI, children study, degradation study, deepfake projects); ImransLab
  - Libraries: scikit-learn, SciPy, statsmodels, SHAP, Fairlearn, PyTorch
- **LLM-assisted research software development: code generation, refactoring, debugging, unit tests, documentation** — PhD (VQI, children study, CUWavestarGAN, Beyond ROC-AUC, deepfake pipeline)
  - Languages: Python, R, Bash
  - Software: Claude Code, GitHub Copilot, ChatGPT (GPT-4o), Cursor, VS Code
- **Prompt engineering and LLM APIs: structured and few-shot prompting, output validation, batch generation** — PhD (SPEAD / ELAD text generation, VQI feedback messages)
  - Languages: Python
  - Libraries: OpenAI Python SDK, Anthropic Python SDK, Hugging Face Transformers, LangChain, LlamaIndex
  - Software: OpenAI API, Anthropic Claude API, Ollama (local models)
- **LLM-generated text data for speech corpora: phoneme-rich sentences, dialogues, tongue twisters, reading scripts, with coverage checks** — SPEAD, ELAD-SVDSR
- **Small language models (SLMs) on edge devices: running and compressing Llama 3.2 1B / 3B, Phi-3-mini, Qwen2.5, Gemma 2B locally; 4-bit / 8-bit quantization, LoRA / QLoRA, knowledge distillation; memory and latency budgets** — PhD (local script generation, prompt classification)
  - Languages: Python, C++
  - Libraries: llama.cpp (GGUF), Ollama, MLC LLM, ExecuTorch, ONNX Runtime GenAI, bitsandbytes, AutoGPTQ, AWQ, Hugging Face PEFT / Transformers
- **Local–cloud inference arbitration: routing requests between on-device models and cloud APIs by difficulty, latency, cost, and privacy** — PhD (text and speech generation)
- **Intent understanding and NLU: intent and style classification of text, keyword spotting, spoken-command recognition** — PhD (prompt classification); EE 402/502 and EE 513 booklets (keyword spotting)
  - Libraries: Hugging Face Transformers / PEFT, scikit-learn, spaCy, TensorFlow Lite Micro
- **Privacy-first on-device personalization: local enrollment and adaptation per user, no user data sent to the cloud** — PhD (VQI app)

- Libraries: OpenAI / Anthropic SDKs, g2p_en / CMUdict (phoneme coverage), NLTK, spaCy

- **Transformer models: self-supervised speech transformers (WavLM, HuBERT, Wav2Vec2), Whisper, transfer learning and fine-tuning** — PhD; EE 402/502 teaching
  - Libraries: Hugging Face Transformers, PEFT (LoRA / QLoRA), accelerate, datasets
- **LLM-assisted literature review, scientific writing, and reviewer responses** — PhD
  - Software: Claude, ChatGPT, Perplexity, Elicit, Consensus, Grammarly
- **LLM-assisted teaching content: course booklets, exercises, example code** — B1, B2 booklets
  - Software: ChatGPT, Claude

Research Engineering, MLOps & Reproducibility (PhD)

- **Experiment tracking, reproducible environments, CI, code review, and research data management** — PhD (all projects)
  - Languages: Python, Bash, SQL, YAML
  - Libraries: Hydra / OmegaConf (configs), MLflow, Weights & Biases, TensorBoard, DVC, pytest, Black / Ruff, pandas, PyArrow, h5py
  - Software: Git / GitHub (pull requests, Issues, Actions), Docker, conda / venv, SLURM, NVIDIA GPUs / CUDA, SQLite / PostgreSQL, Linux (Ubuntu)
- **Model packaging and deployment for end users: desktop app, command-line tool, web demos, model serialization** — PhD (VQI)
  - Libraries: joblib / pickle, ONNX, PyInstaller, FastAPI / Flask, Streamlit / Gradio
- **Cloud and big-data tools: cloud storage and GPU compute, distributed data processing, CI/CD, containerized deployment and orchestration, model monitoring, data lineage** — PhD (VQI data pipeline, research code, RAG assistant)
  - Languages: Python, SQL, Bash
  - Libraries: PySpark, Dask, boto3, Azure SDK for Python
  - Software: Azure (Blob Storage, Data Lake, ML), AWS (S3, EC2, SageMaker, Bedrock, Athena, Glue, Glue Data Catalog, Glue Data Quality, Step Functions, Lambda), Google Cloud / Colab, Databricks, Apache Spark, Docker, Kubernetes (k3s), GitHub Actions (CI/CD), MLflow, DVC, Prometheus / Grafana
- **AIOps and cloud-native architecture: drift detection (PSI, Kolmogorov–Smirnov), automated alerts, anomaly detection on service logs and metrics, runbooks and rollback; stateless containerized services with health checks and autoscaling; service-level objectives (latency, availability, error rate, answer quality) with alerting and incident review** — PhD (lab model services and RAG assistant)
  - Libraries / Software: Evidently, Prometheus, Grafana, Kubernetes (k3s, Amazon EKS), Docker, MLflow
- **LLM applications: retrieval-augmented generation (RAG), semantic search, summarization, information extraction, classification, agentic workflows, conversational assistants and multi-turn reasoning** — PhD (lab RAG assistant, metadata extraction)
  - Libraries: LangChain, LlamaIndex, sentence-transformers, FAISS, Chroma, OpenAI / Anthropic SDKs
  - Software: LangSmith (tracing), RAGAS (RAG evaluation), LLM-as-judge evaluation
- **Medical imaging AI product: COVID-19 detection from chest X-ray and CT, transfer-learning CNNs, lung segmentation (U-Net), Grad-CAM explainability, DICOM handling, clinician-facing GUI** — Machine Learning Engineer, ImransLab (2020–2021)
  - Libraries: TensorFlow / Keras, PyTorch, torchvision, timm, OpenCV, scikit-image, pydicom, SimpleITK, MONAI, pytorch-grad-cam
  - Software: Google Colab, Kaggle, Flask, PyQt / Tkinter, PyInstaller, Docker
- **Healthcare data: EHR-style patient records, admission / discharge / transfer (ADT) data, clinical notes, physiological signals, HL7 FHIR concepts, de-identification** — RA, NSU (Info Hospital, telemedicine); EE

402/502; CUADS; PhD
    - Libraries / Software: MySQL, Firebase, PhysioNet / MIMIC, NeuroKit2, WFDB, pandas
- **Medical image segmentation and detection: retained surgical item (gauze / sponge) detection in ultrasound, CT, and X-ray; 2D / 3D segmentation, object detection, annotation criteria and quality standards with radiologists, failure-mode analysis** — ML Engineer, ImransLab (2018–2019)
    - Languages: Python
    - Libraries: TensorFlow / Keras, PyTorch, segmentation-models-pytorch, Detectron / torchvision detection, YOLOv3 (Darknet), SimpleITK, pydicom, NiBabel, OpenCV, albumentations; newer equivalents: MONAI, nnU-Net, Ultralytics YOLO
    - Software: 3D Slicer, ITK-SNAP, LabelImg, CVAT
- **Self-supervised and representation learning: autoencoder and pretext-task pre-training on unlabelled scans (ImransLab, 2018–2019); self-supervised speech transformers WavLM / HuBERT / Wav2Vec2 (masked prediction, contrastive learning) and LoRA fine-tuning (PhD); knowledge distillation (ImransLab, VQI); DINOv2 features (mentored fingerprint work); Vision Transformers, contrastive learning (SimCLR / MoCo), and masked image modelling (coursework and self-study)** — ImransLab; PhD
    - Libraries: PyTorch, timm, Hugging Face Transformers, lightly
- **Clinical prediction from longitudinal patient data: sepsis early warning from vitals, labs, and clinical notes; irregular time series, imputation, gradient boosting, LSTM / 1D CNN, SHAP; multimodal fusion of structured data, clinical text features, and imaging features** — ML Engineer, ImransLab (2018–2019)
    - Languages: Python, SQL
    - Libraries: XGBoost, LightGBM, PyTorch, TensorFlow / Keras, scikit-learn, SHAP, gensim (word2vec / fastText), pandas; Data: MIMIC-III

Applied Machine Learning & Intelligent Systems (coursework)

- **Financial and insurance-related data: claims-style billing codes (ICD-9, DRG) and health-insurance payer fields (MIMIC-III; also used for per-payer fairness checks of the sepsis model), financial product portfolio data (fintech aggregator: accounts, cards, loans, rates, fees), accounting and reimbursement records (McGill)** — ML Engineer, ImransLab (sepsis); RA, NSU (fintech paper); McGill Accounting Clerk
    - Languages / Libraries: Python, SQL, pandas, scikit-learn, Fairlearn; MySQL; Excel
- **Applied ML pipelines: data cleaning, feature engineering, supervised / unsupervised learning, clustering (k-means, DBSCAN, GMM), model evaluation and tuning** — IA 651 Applied Machine Learning
    - Languages: Python, R
    - Libraries: scikit-learn, pandas, NumPy, SciPy, Matplotlib, seaborn, XGBoost; R caret / tidymodels
    - Software: Jupyter, Google Colab, Kaggle
- **Intelligent systems: fuzzy logic, expert systems, search and optimization, genetic algorithms, neural networks, intelligent agents, reinforcement learning basics** — EE 513 Intelligent System Design
    - Languages: Python, MATLAB
    - Libraries: scikit-fuzzy, DEAP, PyGAD, Gymnasium (OpenAI Gym), scikit-learn; MATLAB Fuzzy Logic Toolbox, Global Optimization Toolbox, Reinforcement Learning Toolbox
- **Reinforcement learning: contextual multi-armed bandits for model routing (epsilon-greedy, Thompson sampling), preference-based post-training (DPO; RLHF with a reward model and PPO), RL fundamentals (MDPs, value functions, Q-learning, policy gradients) with Gymnasium** — PhD (LLM router, script-generation model); EE 513 coursework
    - Libraries: Hugging Face TRL, Gymnasium, Stable-Baselines3, NumPy, PyTorch
- **Programming fundamentals: data types, control flow, functions, OOP, file I/O, debugging** — IA 503 Introduction to Programming
    - Languages: Python

- Software: VS Code, Jupyter, Git

Computer Vision & Image Processing (coursework)

- **Image understanding: image filtering, edge and feature detection (SIFT, SURF, ORB, HOG), segmentation, object detection and recognition, image classification with CNNs** — CS 572 Image Understanding
- **Vehicle detection and tracking from traffic video: YOLO / Faster R-CNN vehicle detection, multi-object tracking (SORT with Kalman filtering), vehicle counting and speed estimation from camera calibration** — CS 572 Image Understanding course project; estimated motion between frames with dense optical flow (Farneback, RAFT) and distance with monocular depth estimation (MiDaS); calibrated the camera (intrinsics with a checkerboard, extrinsics and camera pose from known road points with OpenCV solvePnP) to map image positions to road distances
  - Libraries: OpenCV, PyTorch, Ultralytics YOLO, filterpy
- **Multispectral remote sensing: land-cover classification of Sentinel-2 multispectral imagery (EuroSAT, 13 bands) with CNNs, spectral indices (NDVI), band selection, and a comparison of RGB-only and all-band inputs** — CS 572 Image Understanding course assignment
  - Libraries: PyTorch, torchvision, rasterio, NumPy, scikit-learn
  - Languages: Python, MATLAB, C++
  - Libraries: OpenCV, scikit-image, Pillow, torchvision, YOLO (Ultralytics), Detectron2, MediaPipe; MATLAB Image Processing Toolbox, Computer Vision Toolbox

Embedded Systems, IoT & Instrumentation (coursework)

- **Instrumentation and IoT: sensors, data acquisition, signal conditioning, microcontrollers, wireless communication (Wi-Fi, BLE, MQTT), cloud dashboards** — EE 597 Instrumentation & IoT for Engineers; HI-POCT wearable sensor paper
  - Languages: C / C++ (Arduino), Python (Raspberry Pi), MicroPython, LabVIEW (G)
  - Libraries: Arduino libraries, RPi.GPIO, paho-mqtt, PySerial
  - Software / Hardware: Arduino, Raspberry Pi, ESP32, NI LabVIEW, NI DAQ, oscilloscopes, multimeters, function generators, ThingSpeak / Node-RED / AWS IoT
- **Computer architecture: processor design, pipelining, memory hierarchy and caches, instruction set architectures (MIPS / RISC-V / ARM), performance analysis** — EE 566 Computer Architecture; FPGA review
  - Languages: Verilog, VHDL, C, Assembly (MIPS / RISC-V / ARM)
  - Software: Xilinx Vivado, Intel Quartus, ModelSim, gem5, MARS / Venus simulators
- **Computer networks: OSI / TCP-IP models, routing, switching, subnetting, transport protocols, network security basics, socket programming** — EE 507 Computer Networks
  - Languages: Python, C
  - Libraries: socket, Scapy
  - Software: Wireshark, Cisco Packet Tracer, GNS3, Mininet, ns-3
- **Digital control systems: z-transforms, discrete-time system modelling, stability analysis, digital controller design (PID, state-space), sampling** — EE 551 Digital Control
  - Languages: MATLAB, Python
  - Libraries: MATLAB Control System Toolbox, Simulink; python-control, SciPy signal

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Statistics & Research Methods

- **Statistical software and enterprise analytics: SAS, SAS Viya, R, Python statistics stack; data visualization and BI dashboards** — EE 603 coursework; McGill data mining; PhD reporting
  - Languages: SAS, R, Python, SQL

    - Software: SAS / SAS Viya (SAS OnDemand for Academics), SAS Enterprise Guide, Power BI, Tableau, Plotly Dash, Excel
- **Reliability engineering and life-data analysis: failure distributions (Weibull, exponential, lognormal), survival analysis, accelerated life testing, reliability prediction** — EE 603 Statistical Methods: Reliability and Life Testing
    - Languages: R, Python, MATLAB
    - Libraries: R survival / WeibullR; Python lifelines, reliability, SciPy stats; MATLAB Statistics and Machine Learning Toolbox
- **Effect sizes and non-parametric tests: Cliff's delta, Mann-Whitney U (one-sided), Spearman rank correlation; least-squares decomposition of paired scores** — CITeR 25F-04W (FUQI attribute checks and utility labels); PhD
    - Languages: Python, R
    - Libraries: SciPy stats, NumPy, pandas, statsmodels, pingouin; R effsize, coin
- **Causal inference on observational data: confounder handling, inverse-probability weighting, double machine learning, heterogeneous treatment effects (causal forests), causal feature ablation** — PhD (children longitudinal study, VQI feature ablation)
    - Libraries: DoubleML, EconML, statsmodels, scikit-learn, LightGBM / XGBoost
    - Software: Minitab, JMP, Excel
- **Mathematical optimization and operations research: integer and linear programming (scheduling under room, device, and staff constraints), custom objective functions with gradient-based optimization (custom gradient and Hessian in XGBoost / LightGBM), Bayesian hyperparameter optimization (Optuna TPE), threshold and operating-point optimization; probability and decision-making under uncertainty (bandits, calibration, expected-cost trade-offs)** — PhD (lab scheduling, VQI, LLM router, Beyond ROC-AUC); ImransLab (alert thresholds)
    - Libraries: PuLP, OR-Tools, SciPy optimize, CVXPY, Optuna, XGBoost / LightGBM custom objectives, NumPy
- **Offline policy evaluation and campaign analytics: replay of historical logs, off-policy evaluation (inverse propensity scoring, doubly robust), A/B tests, channel-level response and conversion rates, cost per acquisition and return on investment** — PhD (LLM router; participant-recruitment campaigns)
    - Libraries: pandas, NumPy, SciPy stats, statsmodels
- **Customer and participant modelling: response and show-up prediction from campaign history (calibrated classifiers), CRM-style contact and campaign records into model-ready tables, LLM-simulated users (synthetic personas from logged sessions, PyRIT attacker personas) checked for faithfulness against real sessions, Monte Carlo simulation, multi-tenant data isolation (per-project storage, access, and training runs)** — PhD (data collections, participant recruitment, lab assistant)
    - Languages: Python, SQL, R
    - Libraries: pandas, NumPy, scikit-learn, LightGBM, XGBoost, statsmodels, SciPy, Prophet, lifelines, SimPy, SDV (synthetic tabular data), OpenAI / Anthropic SDKs; R tidymodels, forecast
    - Software: PostgreSQL, BigQuery, Snowflake, Qualtrics, Google Forms, Excel
- **Linear mixed-effects regression (LMER)** — PhD (children study)
    - Languages: R, Python, MATLAB
    - Libraries: R lme4, lmerTest, emmeans; Python statsmodels MixedLM, pymer4; MATLAB fitlme
    - Software: RStudio
- **Causal ablation analysis** — PhD (children study)
    - Languages: Python
    - Libraries: NumPy, pandas

- **Correlation analysis (Spearman), variance explained (R^2), significance testing** — PhD
  - Languages: Python, R, MATLAB
  - Libraries: SciPy stats, statsmodels, pingouin, pandas
  - Software: SPSS, Excel
- **Experimental design; cross-system and cross-dataset evaluation** — PhD
- **Human-subject research: recording protocols, informed consent and child assent (pediatric research), IRB approval; CITI-certified in Biomedical Research (Human Subjects), valid to Feb 2027** — PhD (VPQAD, VSEA-DC, ELAD, CUADS, children dataset)
  - Software: IRB submission system, CITI human-subjects training, Qualtrics / Google Forms (surveys, demographics), scheduling tools
- **Dataset curation, metadata, documentation, and public release** — PhD
  - Software: IEEE DataPort, CSV / JSON metadata, Zenodo, Hugging Face Datasets

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Software Development

- **Desktop GUI application (record/upload, dual gauges, waveform/spectrogram, feedback text, report export)** — PhD (VQI app)
  - Languages: Python
  - Libraries: PyQt5 / PyQt6 / PySide6 / Tkinter / CustomTkinter, Matplotlib (embedded plots), pyqtgraph, sounddevice / PyAudio (recording), ReportLab (PDF reports)
  - Software: PyInstaller / cx_Freeze (packaging), Qt Designer
- **Command-line tool** — PhD (VQI CLI)
  - Languages: Python
  - Libraries: argparse / click / typer
- **Web demos / dashboards** — PhD
  - Libraries: Streamlit, Gradio, Flask / FastAPI
- **Version control and open-source release** — PhD
  - Software: Git, GitHub, GitHub Actions
- **Development environments and reproducibility** — PhD
  - Software: Anaconda / conda, venv / pip, Jupyter Notebook, VS Code, PyCharm, Spyder, Docker, Google Colab
- **Testing and code quality** — PhD
  - Libraries: pytest, unittest, Black / Ruff
- **Agile development cycle (Scrum-style sprints, backlog and user stories, sprint planning, stand-ups, reviews, retrospectives; definition of done with code review, tests, and documentation)** — Graduate Research Assistant, AVHBAC Lab (PhD); RA, NSU (Jira / Trello)
  - Software: GitHub Projects, GitHub Issues, Jira, Trello, Confluence
- **Operating systems** — PhD
  - Software: Linux (Ubuntu), Windows, macOS
- **AI coding assistants (LLM-assisted development: code generation, refactoring, debugging, tests, documentation)** — PhD (VQI, children study, CUWavestarGAN, Beyond ROC-AUC, deepfake pipeline)
  - Software: Claude Code, GitHub Copilot, ChatGPT (GPT-4o), Cursor

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Teaching & Educational Technology

- **Course support: lab instruction, tutorials / recitations, grading, exam and quiz proctoring, office hours, assignment and rubric design** — Graduate Teaching Assistant (Jan 2024 – Jun 2026)
  - Software: Moodle, Canvas, Blackboard, Gradescope, Microsoft Teams, Zoom, Google Classroom, Piazza, Microsoft Excel (grade books), Kahoot / Mentimeter / Poll Everywhere (active learning)
- **Course site and gradebook management; academic integrity checks; FERPA-compliant handling of student records** — all TA courses
  - Software: Moodle, Gradescope, Turnitin, MOSS (code plagiarism detection), Excel / Google Sheets
- **Lab kit and equipment setup and maintenance: development boards, instruments, lab computers** — EE 260, ES 250
  - Software / Hardware: Arduino IDE, Code Composer Studio, driver and toolchain installation, Windows lab images
- **Biomedical measurements lab instruction: sensors, transducers, bridges, signal conditioning, DAQ, biosignal recording (ECG, EMG, PPG, blood pressure, respiration), calibration and uncertainty** — Lab Instructor, BME 338L (SIUC)
  - Languages: MATLAB, LabVIEW, Python
  - Software / Hardware: BIOPAC MP36 / Student Lab (BSL), AcqKnowledge, NI LabVIEW, NI myDAQ / USB DAQ, Excel, oscilloscopes, multimeters, function generators, strain gauges, thermistors, thermocouples, pressure transducers, photodiodes
- **Engineering lab instruction: circuits, electronics, signals, microcontrollers, measurement equipment** — Graduate Teaching Assistant
  - Languages: MATLAB, Python, C / C++ (Arduino), LabVIEW
  - Software / Hardware: MATLAB / Simulink, LTspice / Multisim / PSpice, Arduino IDE, oscilloscopes, multimeters, function generators, power supplies, breadboards
- **Creating instructional content: slides, lab handouts, solution keys, tutorial videos** — Graduate Teaching Assistant
  - Software: PowerPoint, LaTeX / Beamer, Overleaf, Microsoft Word, OBS Studio / Camtasia, Jupyter notebooks
- **Course design and delivery as co-instructor: lectures, syllabus and schedule, assignments, projects, exams, grading policy** — Co-Instructor, EE 402/502 (Fall 2025, Fall 2026)
  - Software: Moodle, Gradescope, PowerPoint, LaTeX / Beamer, Jupyter notebooks, Google Colab, GitHub Classroom, Zoom / Teams, Panopto / Echo360 (lecture capture)

Electrical Science & Circuits (teaching: ES 250)

- **Circuit analysis: Ohm's law, KVL / KCL, nodal and mesh analysis, Thevenin / Norton equivalents, superposition, RC / RL / RLC transients, AC phasor analysis, power, operational amplifiers, diodes and transistor basics** — Graduate TA, ES 250 Electrical Science (4 semesters)
  - Languages: MATLAB, Python
  - Software: LTspice, Multisim, PSpice, Falstad circuit simulator, MATLAB / Simulink, Tinkercad Circuits
  - Hardware: breadboards, digital multimeters, oscilloscopes, function generators, DC power supplies, resistors / capacitors / inductors, op-amps, NI myDAQ / Analog Discovery

Embedded Systems (teaching: EE 260)

- **Microcontroller programming: GPIO, interrupts, timers, PWM, ADC / DAC, serial communication (UART, SPI, I2C), polling vs. interrupt-driven design, real-time constraints, sensor and actuator interfacing, debugging** — Graduate TA, EE 260 Embedded Systems; EE 597 Instrumentation & IoT
  - Languages: C, C++, Assembly (ARM / AVR / MSP430), MicroPython, Python

- Libraries: Arduino core, AVR-libc, STM32 HAL, TI DriverLib, FreeRTOS
- Software: Arduino IDE, Code Composer Studio (TI), STM32CubeIDE, Keil µVision, MPLAB X, PlatformIO, Tinkercad, logic analyzers
- Hardware: Arduino Uno / Mega, TI MSP430 / Tiva C LaunchPad, STM32 Nucleo, Raspberry Pi, ESP32, sensors, LEDs, motors, LCDs, breadboards

Biomedical Signal Processing & ML (teaching: EE 402/502)

- **Wearable and metabolic health data: sleep staging (heart rate, HRV, accelerometry), stress detection (EDA / GSR, HRV; WESAD), continuous glucose monitor (CGM) trend forecasting (OhioT1DM), per-person baselines and subject-wise evaluation** — Co-Instructor, EE 402/502 (course projects); CUADS (stress and emotion physiology)
  - Libraries: NeuroKit2, pandas, scikit-learn, statsmodels, PyTorch
- **Biomedical signals: ECG, EEG, EMG, PPG, EDA / GSR, respiration, accelerometer / IMU; filtering, denoising, artifact removal, R-peak and QRS detection, heart rate variability, time / frequency / time-frequency features (FFT, wavelets, spectrograms)** — Co-Instructor and TA, EE 402/502; HI-POCT gait paper; CUADS affective dataset
  - Languages: Python, MATLAB
  - Libraries: SciPy signal, NumPy, NeuroKit2, BioSPPy, MNE-Python, WFDB, HeartPy, PyWavelets, pyHRV; MATLAB Signal Processing Toolbox, Wavelet Toolbox, EEGLAB
  - Data: PhysioNet (MIT-BIH Arrhythmia, PTB-XL), UCI HAR, DEAP / WESAD (affective computing)
- **EMG and IMU gesture and intent detection: surface EMG (envelope, RMS, onset detection), jaw-clench and hand-gesture detection, head-motion and gesture recognition from IMU, sensor fusion of EMG + IMU, sliding-window streaming classification** — TA / Co-Instructor EE 402/502 (EMG / IMU lab); EE 513 booklet (IMU gesture recognition); AIDrawPen (mentored)
  - Languages: Python, C / C++, MATLAB
  - Libraries: SciPy signal, NeuroKit2, BioSPPy, PyTorch, TensorFlow / Keras, TensorFlow Lite Micro, CMSIS-NN; Data: NinaPro, CapgMyo, UCI HAR
- **Temporal deep learning for biosignals: 1D CNNs, temporal convolutional networks (TCN, dilated causal convolutions), CNN-LSTM hybrids, lightweight models for edge inference** — CUADS / UEMCON benchmark; EE 402/502
  - Libraries: PyTorch, TensorFlow / Keras, pytorch-tcn, keras-tcn
- **Biosignal evaluation: subject-wise and leave-one-subject-out splits, cross-session drift, electrode contact and fit effects, leakage checks between training and test sets** — CUADS benchmark; children longitudinal study (drift over years); EE 402/502
- **Multi-device data synchronization: Lab Streaming Layer (LSL), BLE data transport, timestamp alignment across sensors and recorders** — CUADS (Shimmer3 + app), ELAD-SVDSR (5 synchronized microphones), VSEA-DC (2 devices), AIDrawPen (BLE)
  - Libraries / Tools: pylsl, LabRecorder, pyxdf, bleak (BLE), Shimmer SDK
- **ML on biomedical signals: feature extraction, classification (arrhythmia, emotion, activity, stress), deep learning (1D CNNs, LSTMs), cross-validation by subject, class imbalance, clinical evaluation metrics (sensitivity, specificity)** — Co-Instructor and TA, EE 402/502
  - Languages: Python, MATLAB
  - Libraries: scikit-learn, PyTorch, TensorFlow / Keras, XGBoost, imbalanced-learn, tsfresh, sktime
  - Software: Jupyter, Google Colab

Voice & Speech Processing for Health and ML (teaching: EE 402/502 lectures)

- **Speech analysis fundamentals: source–filter model, acoustic phonetics, sampling / quantization, energy, zero-crossing rate, autocorrelation, STFT, windowing, spectrograms, mel scale, cepstrum, LPC, MFCC, filter banks, delta features** — Co-Instructor EE 402/502; PhD
  - Languages: Python, MATLAB
  - Libraries: librosa, torchaudio, SciPy signal, NumPy, python_speech_features, Parselmouth, spafe; MATLAB Audio Toolbox, Signal Processing Toolbox, VOICEBOX
  - Software: Praat, Audacity, Sonic Visualiser, Wavesurfer
- **Voice pathology and health analytics: jitter, shimmer, HNR, CPP, voice disorder detection, Parkinson's voice markers, cough and respiratory sound analysis, speech-based emotion, stress, and depression detection** — Co-Instructor EE 402/502
  - Languages: Python, MATLAB
  - Libraries: Parselmouth / Praat, openSMILE (eGeMAPS), DisVoice, Surfboard, librosa, scikit-learn
  - Data: Saarbrücken Voice Database, UCI Parkinson's Speech datasets, Coswara, COUGHVID, RAVDESS, CREMA-D, IEMOCAP, ESD, DAIC-WOZ
- **Classical and statistical speech models: GMM, GMM-UBM, HMM, k-NN, SVM, Random Forest** — Co-Instructor EE 402/502; PhD
  - Libraries: scikit-learn, hmmlearn, pomegranate; MATLAB Statistics and Machine Learning Toolbox
- **Deep learning for speech: spectrogram CNNs, RNNs / LSTMs, transformers, self-supervised models (wav2vec 2.0, HuBERT, WavLM), transfer learning** — Co-Instructor EE 402/502; PhD
  - Libraries: PyTorch, TensorFlow / Keras, Hugging Face Transformers, SpeechBrain, torchaudio, ESPnet
- **Speech applications: speaker recognition, speech emotion recognition, automatic speech recognition, spoofing / deepfake detection** — Co-Instructor EE 402/502; PhD
  - Libraries / Software: SpeechBrain, OpenAI Whisper, Vosk, Kaldi, ESPnet, NVIDIA NeMo, Hugging Face pipelines, ASVspoof baselines
- **Applied audio ML for teaching: keyword spotting (Random Forest), speaker classification (SVM with ANOVA / chi-squared / k-fold validation), speech emotion recognition, voice-based gender identification (EDA, dimensionality analysis, model compression and deployment), MFCC-based deepfake detection; ML under embedded-system constraints** — EE 502 course booklet, GitHub teaching repos
  - Libraries: librosa, NumPy, pandas, scikit-learn, Matplotlib, seaborn, joblib, SciPy stats
- **Responsible AI for voice data: privacy, consent, bias and fairness across speakers, demographics, and languages** — Co-Instructor EE 402/502; PhD
- **Teaching materials for speech processing: Jupyter notebook labs, starter code, datasets, auto-graded assignments, project rubrics** — Co-Instructor EE 402/502
  - Software: Jupyter, Google Colab, GitHub / GitHub Classroom, Moodle, Gradescope, LaTeX / Beamer, PowerPoint
- **Programming instruction and online teaching: Python, Java, data structures and algorithms; live online sessions; code review for learners** — Instructor, ImransLab (Jan 2018 – Nov 2023)
  - Languages: Python, Java, C / C++
  - Software: VS Code, Jupyter, Google Colab, GitHub, Replit, Zoom / Google Meet, Google Classroom, LMS platforms
- **Secondary-level STEM tutoring: O Level / A Level physics and mathematics (mechanics, electricity, waves, calculus, algebra, statistics); Cambridge and Edexcel syllabi, past-paper and mark-scheme practice** — Independent Tutor
  - Software: GeoGebra, Desmos, PhET simulations, Google Docs / Sheets, Zoom

Research Administration (Graduate Research Assistant)

- **Sponsored research: grant-funded project work, progress reports, sponsor deliverables, research data management** — Graduate Research Assistant (Jun 2026 – present)
  - Software: Microsoft Office / Google Workspace, Overleaf, Git / GitHub, shared lab storage / OneDrive / Google Drive, Slack / Teams
- **Lab operations: equipment and computing resource management, dataset storage and backup, onboarding new lab members** — Graduate Research Assistant, AVHBAC Lab
  - Software: Linux server administration, HPC / SLURM, NAS / cloud storage, GitHub organization management

Biomedical Engineering (M.Eng. coursework)

- **Neuroengineering: neural signals and neurophysiology, EEG / ECoG / spike recording, brain–computer interfaces, neural decoding, neurostimulation (DBS, TMS, FES), neuroprosthetics** — BME 417 Neuroengineering
  - Languages: MATLAB, Python
  - Libraries: EEGLAB, FieldTrip, BCILAB, MNE-Python, BrainFlow, PyTorch; MATLAB Signal Processing Toolbox
  - Software / Hardware: OpenBCI, g.tec / BioSemi EEG systems, NEURON simulator
- **Medical instrumentation: biopotential amplifiers, electrodes, ECG / EMG / EEG front ends, blood pressure and pulse oximetry, filtering and noise reduction, electrical safety, and medical device standards (IEC 60601, FDA / Health Canada device regulation)** — BME 438 Medical Instrumentation
  - Languages: MATLAB, C / C++ (Arduino), LabVIEW
  - Software / Hardware: LTspice, Multisim, NI LabVIEW, NI DAQ, BIOPAC, AD8232 ECG front end, MAX30102 pulse oximeter, oscilloscopes
- **Biomedical measurements and lab: transducers and sensors (strain, temperature, pressure, optical), Wheatstone bridges, signal conditioning, amplification, filtering, analog-to-digital conversion, calibration, uncertainty and error analysis, lab reports** — BME 338 / 338L Biomedical Measurements and Lab
  - Languages: MATLAB, LabVIEW, Python
  - Software / Hardware: NI LabVIEW, NI myDAQ / DAQ, BIOPAC Student Lab, Arduino, breadboards, multimeters, oscilloscopes, Excel
- **Biomedical imaging: X-ray, CT, MRI, ultrasound, PET / SPECT, optical imaging; image formation and reconstruction (Fourier / Radon transform, filtered back-projection), image quality (resolution, contrast, SNR), image processing and segmentation** — BME 532 Biomedical Imaging; CS 572 Image Understanding
  - Languages: MATLAB, Python
  - Libraries: MATLAB Image Processing Toolbox, scikit-image, SimpleITK, ITK, pydicom, NiBabel, MONAI, OpenCV
  - Course project (BME 532): segmented the left and right ventricles and myocardium on short-axis cardiac cine MRI (ACDC dataset) with a U-Net in PyTorch / MONAI, derived end-diastolic and end-systolic ventricular volumes and ejection fraction from the masks, and compared them with the reference values
  - Keeps up with medical imaging AI research (MICCAI, Medical Image Analysis, Radiology: Artificial Intelligence) for new segmentation and validation methods
  - Software: 3D Slicer, ImageJ / Fiji, ITK-SNAP, OsiriX / Horos

Cellular Neuroscience & Wet-Lab Research (M.Eng. research, Frontiers paper)

- **Patch-clamp electrophysiology: cell-attached and outside-out recordings, mechanosensitive channel currents, high-speed pressure clamp, I–V and pressure–response (Boltzmann) curve fitting, P50 and slope sensitivity, single-channel conductance** — M.Eng. research (Piezo1)
  - Software: pClamp 10 (Clampex, Clampfit), Axon MultiClamp Commander, Origin 2020b, GraphPad Prism 9, MATLAB, Python (pyABF, NumPy, SciPy)

- Hardware: Axon MultiClamp 700B amplifier, Digidata digitizer, ALA Scientific HSPC-2 high-speed pressure clamp, micromanipulators, pipette puller, inverted microscope, Faraday cage, anti-vibration table

- **Cell culture and cell biology: astrocyte cell lines (C8-S), aseptic technique, media preparation, passaging, seeding, LPS-induced neuroinflammation models, poly-D-lysine coating** — M.Eng. research
  - Equipment: biosafety cabinet, CO_2 incubator, centrifuge, hemocytometer / cell counter, BSL-2 practices
- **Molecular biology: siRNA gene silencing and transfection (Dharmacon, DharmaFECT), knockdown validation (qPCR / Western blot)** — M.Eng. research
- **Pharmacology: channel agonists and antagonists (Yoda1, GsMTx4), dose preparation, reversibility tests** — M.Eng. research
- **Fluorescence imaging: calcium imaging (Cal-520 AM), confocal laser scanning microscopy (Olympus FV3000), time-lapse imaging, ROI-based $\Delta F/F_0$ analysis** — M.Eng. research
  - Software: ImageJ / Fiji, Olympus FluoView / cellSens, MATLAB, Python (scikit-image, tifffile)
- **Spectral imaging and optical measurement data: lambda-scan (x, y, λ) image cubes, excitation / emission spectra, linear spectral unmixing, blind unmixing (NMF, PCA), ratiometric calibration to chemical concentration (Fura-2, Grynkiewicz equation), imaging-sensor calibration (dark-frame and flat-field correction, background and photobleaching correction, reference slides and beads), per-image quality metrics, acquisition metadata for traceability** — M.Eng. research (Olympus FV3000 confocal)
  - Libraries / Software: Python (NumPy, scikit-learn NMF / PCA, scikit-image, tifffile), Olympus FluoView (spectral unmixing), ImageJ / Fiji (LUMoS spectral unmixing, BaSiC flat-field correction), MATLAB
- **Light and fluorescence microscopy: brightfield, phase contrast, differential interference contrast (DIC), epifluorescence / widefield fluorescence, confocal laser scanning, spinning-disk confocal, multiphoton / two-photon, total internal reflection fluorescence (TIRF), super-resolution basics (STED, SIM, STORM / PALM), live-cell and time-lapse imaging, Z-stacks and 3D reconstruction, multichannel imaging, sample prep for microscopy** — M.Eng. research (Dr. Bae's lab)
  - Microscope systems: Olympus (FV3000, IX-series inverted), Nikon (Eclipse, A1R), Zeiss (LSM, Axio Observer), Leica (SP8, DMi8)
  - Cameras / detectors: sCMOS, EMCCD, CCD, PMT / GaAsP detectors
  - Acquisition software: Olympus FluoView / cellSens, Nikon NIS-Elements, Zeiss ZEN, Leica LAS X, Micro-Manager, MetaMorph
  - Analysis software: ImageJ / Fiji, Imaris, CellProfiler, Icy, napari, QuPath, MATLAB Image Processing Toolbox, Python (scikit-image, OpenCV, tifffile, aicsimageio)
- **Fluorescent labelling and staining: immunocytochemistry / immunofluorescence, fluorescent dyes (Cal-520, Fluo-4, Fura-2, DAPI / Hoechst, phalloidin), GFP-tagged constructs, fixation and permeabilization, mounting** — M.Eng. research
- **Image quantification: cell counting, segmentation, intensity and $\Delta F/F_0$ measurement, colocalization, morphology, migration tracking (TrackMate, MTrackJ)** — M.Eng. research
  - Software: ImageJ / Fiji (TrackMate, Coloc 2), CellProfiler, Imaris, MATLAB, Python
- **Cell migration assays: wound-healing / scratch assays (micropipette, Ibidi inserts), migration velocity quantification** — M.Eng. research
  - Software: ImageJ / Fiji (Wound Healing Size Tool, MRI plugin)
- **Biostatistics: Mann-Whitney tests, t-tests, one-way ANOVA with post-hoc tests, mean ± SEM reporting, significance testing** — M.Eng. research
  - Software: GraphPad Prism, Origin, R, Python SciPy / statsmodels, Excel
- **Lab safety and compliance: chemical and biological safety, lab notebooks, SOPs, animal-derived cell line handling** — M.Eng. research

Core Electrical & Electronic Engineering (B.Sc. EEE foundation)

- **Circuits and electronics: DC / AC circuit analysis, analog electronics (diodes, BJTs, MOSFETs, op-amps, amplifiers, oscillators), digital logic design (combinational / sequential circuits, FSMs), PCB design** — B.Sc. EEE
  - Languages: Verilog, VHDL, MATLAB
  - Software: LTspice, PSpice / OrCAD, Multisim, Proteus, Logisim, Quartus / Vivado, KiCad, Altium Designer, Eagle
- **Signals, systems and communications: continuous and discrete signals, Fourier / Laplace / z-transforms, DSP, filter design, analog and digital communications (AM / FM, PSK / QAM), information theory, wireless and mobile communications** — B.Sc. EEE
  - Languages: MATLAB, Python
  - Software: MATLAB / Simulink (Communications, DSP System Toolbox), GNU Radio, LabVIEW
- **Electromagnetics, microwave and antennas: Maxwell's equations, transmission lines, Smith chart, waveguides, antenna theory** — B.Sc. EEE
  - Software: CST Studio Suite, Ansys HFSS, ADS, MATLAB RF / Antenna Toolbox
- **Power and energy: electrical machines (transformers, DC / induction / synchronous machines), power systems analysis, power electronics (rectifiers, inverters, converters), renewable energy** — B.Sc. EEE
  - Software: MATLAB / Simulink (Simscape Electrical), PSIM, PowerWorld, ETAP, PSCAD
- **Control systems: transfer functions, root locus, Bode / Nyquist, PID control, state-space** — B.Sc. EEE
  - Software: MATLAB Control System Toolbox, Simulink
- **Microprocessors and microcontrollers: 8086 / ARM assembly, interfacing, Arduino / AVR / PIC** — B.Sc. EEE
  - Languages: Assembly, C
  - Software: emu8086, Keil, MPLAB, Atmel Studio, Arduino IDE, Proteus
- **Programming foundations: structured and object-oriented programming, data structures and algorithms** — B.Sc. EEE
  - Languages: C, C++, Java, Python

Mathematics & Engineering Foundations (B.Sc. coursework)

- **Engineering mathematics: calculus I–III, differential equations, linear algebra, complex variables, Fourier / Laplace transforms, numerical methods, probability and statistics** — MAT 120 / 130 / 250 / 350, MAT 125, MAT 361
  - Languages: MATLAB, Python, R
  - Libraries / Software: NumPy, SciPy, SymPy, MATLAB Symbolic Math Toolbox, Wolfram Mathematica, Maple, Excel
- **Physics, chemistry and biology foundations: mechanics, electricity and magnetism, optics, chemistry, cell biology** — PHY 107 / 108, CHE 101, BIO 103 (supports later biomedical and neuroscience work)
- **Engineering drawing and CAD: technical drawing, orthographic projection, electrical schematics and layouts** — CEE 110 / EEE 154 Engineering Drawing
  - Software: AutoCAD, AutoCAD Electrical, SolidWorks, Fusion 360, Visio, draw.io
- **Engineering economics: time value of money, cost–benefit analysis, NPV / IRR, depreciation, project evaluation, feasibility studies** — EEE 452 Engineering Economics
  - Software: Excel (financial functions), MS Project
- **Programming foundations: C programming, algorithms, debugging** — CSE 115 Programming Language I (+ lab)
  - Languages: C
  - Software: Code::Blocks, Dev-C++, GCC, VS Code

- **Engineering design projects: requirements, design, prototyping, testing, documentation, team presentations** — EEE 299 Junior Design, EEE 499 / 499B Senior Design
  - Software: Proteus, Fritzing, KiCad / EasyEDA, Arduino IDE, MS Project / Trello, PowerPoint

High Voltage & Power Engineering (B.Sc. concentration)

- **High voltage engineering: generating and measuring HV AC / DC / impulse voltages, insulation and dielectric materials, breakdown in gases, liquids and solids, partial discharge, corona, lightning and switching overvoltages, insulation coordination, HV testing standards (IEC 60060, IEEE 4)** — B.Sc. EEE (HV concentration)
  - Software: ATP-EMTP / ATPDraw, PSCAD / EMTDC, COMSOL Multiphysics (electrostatics / field simulation), Ansys Maxwell, MATLAB / Simulink
  - Equipment: impulse voltage generators, HV AC / DC test sets, hipot testers, insulation resistance testers (Megger), partial discharge detectors, sphere gaps, HV dividers and probes
- **Power system analysis and protection: load flow, fault / short-circuit analysis, stability, protective relaying (overcurrent, distance, differential), circuit breakers and switchgear, grounding / earthing, substation design** — B.Sc. EEE; ECE 589 Power Distribution
  - Software: ETAP, PowerWorld, PSS/E, DIgSILENT PowerFactory, PSCAD, MATLAB Simscape Electrical, OpenDSS
- **Transmission and distribution: overhead lines and underground cables, line parameters, corona loss, sag and tension, HVDC and FACTS basics** — B.Sc. EEE
  - Software: MATLAB, PSCAD, PowerWorld
- **Electrical machines and power electronics: transformers, induction / synchronous / DC machines, rectifiers, inverters, DC-DC converters, motor drives** — B.Sc. EEE
  - Software: MATLAB / Simulink, PSIM, PLECS, LTspice
- **Power stations and generation: thermal, hydro, nuclear, gas-turbine, and combined-cycle plants; plant economics; load curves; tariffs** — B.Sc. EEE (EEE 362 Power Systems, power concentration)
  - Software: MATLAB, Excel, ETAP
- **Power system operations and reliability: economic dispatch, unit commitment, load forecasting, reliability indices (LOLP, SAIFI, SAIDI), contingency analysis** — EEE 461 Power System Operations & Reliability
  - Software: MATLAB (MATPOWER), PowerWorld, PSS/E, Python (pandapower, PyPSA)
- **Motor drives and energy-efficient control: three-phase induction motor modelling (d–q / equivalent circuit), V/f control, field-oriented (vector) control, direct torque control, loss-minimization / flux-optimization control, variable-frequency drives, PWM and space-vector PWM inverters, efficiency and power-factor analysis** — B.Sc. directed research (Dr. Riasat Khan)
  - Languages: MATLAB, C (embedded DSP)
  - Software: MATLAB / Simulink, Simscape Electrical (SimPowerSystems), Motor Control Blockset, PSIM, PLECS
  - Hardware (typical platforms): TI C2000 DSP, dSPACE, VFDs, induction motor test benches, power analyzers
- **Fuzzy logic control: fuzzy inference systems (Mamdani / Sugeno), membership function design, rule-base design, defuzzification, fuzzy-PI / adaptive fuzzy controllers, comparison with PI / PID** — B.Sc. directed research; EEE 513 Intelligent System Design
  - Languages: MATLAB, Python
  - Libraries / Software: MATLAB Fuzzy Logic Toolbox (Fuzzy Logic Designer), Simulink, scikit-fuzzy, simpful
- **Renewable energy and smart grid: solar PV, wind, energy storage, grid integration, smart metering** — B.Sc. EEE
  - Software: HOMER Pro, PVsyst, MATLAB / Simulink, SAM (NREL)
- **Electrical safety and codes: electrical installation design, earthing, NEC / IEC / BNBC wiring standards** — B.Sc. EEE

- Software: AutoCAD Electrical, DIALux (lighting design), EPLAN

Embedded Systems Thesis (B.Sc.)

- **Wearable IoT medical device prototyping: sensor selection, analog front-end signal conditioning, Arduino firmware, Bluetooth data link, smartphone app, hardware integration, validation against reference devices** — B.Sc. thesis
  - Languages: C / C++ (Arduino), Java / Kotlin (Android), MIT App Inventor blocks, Python
  - Hardware: Arduino Uno / Nano / Mega, HC-05 / HC-06 Bluetooth, MAX30100 / MAX30102 (pulse oximetry), pulse sensor, AD8232 (ECG), LM35 / DS18B20 / MLX90614 (temperature), BP sensor modules, breadboards, perfboards / PCBs, Li-ion batteries and regulators
  - Software: Arduino IDE, Android Studio, MIT App Inventor, Proteus, Fritzing, Tinkercad Circuits, EasyEDA / KiCad (PCB), Serial Plotter, Excel / MATLAB (validation analysis)

Antennas, RF & Wireless (BSc-era conference papers)

- **Wireless body-area networks (WBAN) and 5G mm-wave for healthcare: on-body channel modelling, path loss, wearable radio links, UWB** — B.Sc. research with Dr. M. M. Khan
  - Software: CST Studio Suite (voxel body models), Ansys HFSS, MATLAB, SEMCAD X / Sim4Life
- **Mobile-device antenna engineering (industry): LTE / 3G, Wi-Fi, Bluetooth, GNSS, NFC antennas in device housings; impedance matching (Smith chart, lumped elements); prototyping and tuning; VNA S-parameter measurement; anechoic-chamber radiation pattern, gain, and efficiency; OTA TRP / TIS; SAR and HAC; regulatory pre-compliance (FCC, ISED, PTCRB); hand / head phantom effects; antenna isolation and coexistence** — Antenna Engineering Intern, BlackBerry (RIM), 2016–2017
  - Software / Hardware: CST Studio Suite, Ansys HFSS, Keysight ADS, FEKO, MATLAB, Python; Keysight / R&S VNAs, spectrum analyzers, Satimo / ETS-Lindgren OTA chambers, SAR systems
- **Antenna design and analysis: compact wideband mm-wave antennas (V-band, W-band, 60 GHz), transparent antennas, body-centric / wearable communication, S-parameters, return loss, VSWR, gain, radiation patterns, bandwidth, specific absorption rate (SAR)** — IEEE CCWC 2022, ICAEEE 2022 papers
  - Software: CST Studio Suite, Ansys HFSS, FEKO, ADS (Keysight), MATLAB Antenna Toolbox, COMSOL RF Module
  - Hardware: vector network analyzer (VNA), anechoic chamber

IoT, Telemedicine & Healthcare Systems (BSc-era papers)

- **IoT health monitoring: wearable vital-sign sensors (heart rate, SpO_2, temperature, ECG, blood pressure), real-time data transmission, cloud dashboards, alerts** — B.Sc. thesis; Smart Telemedicine System (CCWC 2022)
  - Languages: C / C++ (Arduino), Python, JavaScript
  - Hardware: Arduino, ESP8266 / ESP32 / NodeMCU, Raspberry Pi, MAX30100 / MAX30102, DS18B20, AD8232, LM35, GSM (SIM800 / SIM900), GPS (NEO-6M), Bluetooth HC-05
  - Software / Cloud: Arduino IDE, Blynk, ThingSpeak, Firebase, MQTT, Node-RED, Fritzing, Proteus
- **Safety and embedded sensing systems: accident detection (accelerometer / gyroscope), rear-view camera, GPS / GSM emergency alerts** — Smart Helmet paper (CCWC 2022), RA, NSU
  - Hardware: Arduino, MPU6050 accelerometer / gyroscope, camera modules, GPS / GSM modules, vibration / tilt sensors, buzzers
  - Software: Arduino IDE, Proteus, Fritzing

Web & Mobile Application Development (BSc-era papers)

- **Web and mobile apps: healthcare system (Info Hospital), mental well-being app (COVID-19), financial aggregator service; front end, back end, databases, authentication, APIs, UI / UX** — ICCMC 2022 papers
  - Languages: Java, Kotlin, Dart, JavaScript, TypeScript, PHP, Python, HTML, CSS, SQL

- Frameworks / Libraries: Android SDK, Flutter, React, React Native, Node.js / Express, Laravel, Django / Flask, Bootstrap, jQuery
- Databases / Cloud: MySQL, PostgreSQL, SQLite, Firebase (Auth, Firestore, Realtime Database), MongoDB, REST APIs
- Software: Android Studio, VS Code, XAMPP, Postman, Figma, Git / GitHub

- **App features for healthcare and fintech: user authentication, role-based access (patient / doctor / admin), appointment booking, video consultation, e-prescriptions, health records, push notifications, chat, payment gateways, third-party financial APIs, dashboards, responsive design** — RA, NSU (telemedicine, Info Hospital, mental well-being, financial aggregator)
  - Libraries / Services: WebRTC, Agora / Twilio / Jitsi (video), Firebase Cloud Messaging, Firebase Auth, SSLCommerz / Stripe / PayPal APIs, Chart.js, Google Maps API, REST / JSON
  - Software: Figma / Adobe XD (UI / UX), Postman, Android Studio, VS Code, Git / GitHub, cPanel / Heroku / Firebase Hosting
- **Software engineering practice: requirements gathering, UML / use-case diagrams, database design (ER diagrams), testing, deployment, documentation** — RA, NSU
  - Software: draw.io, Lucidchart, StarUML, MySQL Workbench, Jira / Trello

Assistive Technology & HCI (preprint)

- **Face capture and face-image quality: real-time face detection and 68-point landmarks (Dlib, OpenCV), pose and framing checks, ISO/IEC 19794-5 photo rules, automatic pan-tilt-zoom camera control on an NVIDIA Jetson, child face capture** — PR24 face-capture system (lab contributor); DC1 child biometrics collection
  - Hardware / Software: NVIDIA Jetson Nano, IMX477 PTZ camera, Dlib, OpenCV
- **Robotics and embodied AI: vision-guided grasping and manipulation, multi-sensor fusion (camera, force sensors, IMUs, time-of-flight distance), gesture-based control, servo actuation, pick-and-place testing, real-time object detection on an FPGA (YOLOv7-tiny, Vitis AI, quantization-aware training)** — PR23 robotic prosthetic arm (lab contributor); AIDrawPen; EE 260 / EE 513 embedded work
  - Hardware / Software: Xilinx Kria KV260 (Vitis AI, DPU), MPU-6500 IMUs, VL6180X ToF sensor, force sensors, SG90 servos, PyTorch, OpenCV
- **AI-powered gesture-based drawing for pediatric rehabilitation: hand tracking, gesture recognition, real-time interaction** — AIDrawPen (TechRxiv 2025); guided the child-facing drawing game (Pygame), with real-time IMU-driven pen strokes and gesture-triggered on-screen animations, and a Unity prototype of the game scene
  - Languages: Python
  - Libraries: MediaPipe, OpenCV, TensorFlow / PyTorch, NumPy, Tkinter / PyQt, Pygame
- **Human motion modelling and animation: IMU motion capture (APDM Opal), gait and balance analysis, gesture recognition, hand and pose tracking (MediaPipe), skeleton animation from orientation data (Blender, Unity), generative models for motion sequences (conditional VAE, diffusion), real-time interactive prototypes (Pygame, Unity)** — PR11 gait study; AIDrawPen (mentored); EE 402/502 EMG / IMU lab
  - Libraries / Software: PyTorch, NumPy, SciPy, MediaPipe, Blender (Python API), Unity, Pygame

Biometric Evaluation Methodology (preprint)

- **Operating-point performance reporting beyond ROC-AUC: FMR / FNMR at fixed thresholds, EER, DET analysis, confidence intervals, bootstrap, ISO/IEC 19795 reporting practice** — "Beyond ROC-AUC" (arXiv 2026)
  - Libraries: NumPy, SciPy, scikit-learn, Matplotlib, pyeer

Affective Computing (CUADS, ECG emotion preprint, HI-POCT)

- **Emotion recognition from physiological signals (ECG, EDA, PPG) and gait / balance sensors; deep learning model benchmarking; affective dataset collection** — CUADS (2025), ECG emotion recognition (arXiv 2026), anger detection (HI-POCT 2024)
  - Languages: Python, MATLAB
  - Libraries: NeuroKit2, BioSPPy, HeartPy, scikit-learn, PyTorch, TensorFlow / Keras
  - Hardware: Shimmer / Empatica E4 wearables, APDM Opal / IMU mobility sensors, Polar chest straps

Hardware Product Development & NPI

- **Hardware build stages and production release: EVT / DVT / PVT builds, issue tracking, sign-off for production, production test limits and fixtures, failure and rework analysis** — Antenna Engineering Intern, BlackBerry (RIM)
- **Design for manufacturing and assembly (DFM / DFA): part tolerances, assembly variation, yield, supplier and contract-manufacturer coordination, component cost reduction** — BlackBerry (RIM); smart helmet and wearable BOMs (NSU)
- **Mechanical CAD and prototyping: SolidWorks enclosures and mounts, 3D printing, importing CAD into EM simulation** — NSU smart helmet and wearable thesis; BlackBerry (RIM)
  - Software: SolidWorks, Creo, Fusion 360, CST Studio Suite, Ansys HFSS
- **PCB and electronics prototyping: custom PCBs, component selection, soldering, bring-up and debugging with oscilloscopes and logic analyzers** — AIDrawPen (mentored); B.Sc. thesis; EE 260 labs
  - Software: KiCad, Altium (layout review)
- **Vision and sensor systems in physical products: cameras, IMUs, heart-rate and vital-sign sensors, GPS / GSM, BLE; field testing on roads and in vehicles** — smart helmet (NSU); VSEA-DC (in-vehicle recordings); wearable thesis
- **Supplier and vendor management: requirements, vendor and quote comparison, purchasing, asset tracking, SDK licensing** — McGill IT; PhD lab procurement; BlackBerry suppliers
- **Quality and standards: ISO / IEC standards work, test protocols, root-cause analysis, regulatory pre-compliance (FCC, ISED)** — PhD (ISO/IEC biometric standards); BlackBerry (RIM)

Edge Computing & Embedded AI (M.Eng. coursework)

- **Edge computing: edge / fog / cloud architectures, on-device inference, latency and bandwidth trade-offs, IoT protocols (MQTT, CoAP, HTTP / REST), containerization at the edge, offloading and scheduling** — ECE 517 Edge Computing
  - Languages: Python, C / C++
  - Libraries: TensorFlow Lite, PyTorch Mobile / ExecuTorch, ONNX Runtime, OpenVINO, TensorRT, Edge Impulse, paho-mqtt, Flask / FastAPI
  - Software / Hardware: NVIDIA Jetson Nano / Xavier, Raspberry Pi, Google Coral TPU, ESP32, Arduino Nano 33 BLE Sense, Docker, K3s / Kubernetes, AWS IoT Greengrass, Azure IoT Edge, Node-RED
- **On-device accelerators and local–cloud arbitration: mobile and embedded GPUs, NPUs / TPUs, DSPs; delegating inference to accelerators; deciding in real time whether to run a model locally or offload it to the cloud (latency, bandwidth, power, privacy)** — ECE 517 Edge Computing; EE 513 booklet (MVP accelerator); PhD (local SLM vs. cloud LLM routing)
  - Languages: Python, C / C++
  - Libraries: TensorFlow Lite GPU / NNAPI delegates, Core ML, Qualcomm AI Engine / QNN, ExecuTorch, ONNX Runtime execution providers, CMSIS-NN / CMSIS-DSP, TensorRT
  - Hardware: NVIDIA Jetson GPUs, Google Coral Edge TPU, smartphone NPUs, Silicon Labs MVP accelerator, Arm Cortex-M DSP extensions

- **TinyML and model compression: quantization, pruning, knowledge distillation, efficient architectures (MobileNet)** — ECE 517 Edge Computing; PhD (VQI inference-time optimization)
  - Libraries: TensorFlow Lite Micro, PyTorch quantization, ONNX, Edge Impulse
- **Embedded systems design: microcontroller architecture, real-time operating systems, interrupts, timers, peripherals, low-power design, hardware / software co-design** — ECE 536 Embedded Systems; TA EE 260
  - Languages: C, C++, Assembly, Python
  - Libraries: FreeRTOS, Zephyr, STM32 HAL, Arduino core
  - Software / Hardware: STM32CubeIDE, Keil µVision, Code Composer Studio, PlatformIO, ARM Cortex-M boards, TI LaunchPads, logic analyzers

TinyML on Silicon Labs Microcontrollers (EE 513 booklet)

- **End-to-end TinyML: data collection on device, model design in Google Colab, TensorFlow Lite / TFLite Micro conversion, post-training quantization (int8), conversion to C arrays, deployment and profiling of memory, power, and latency on a BLE microcontroller** — EE 513 *Embedded Machine Learning* booklet
  - Languages: C, C++, Python
  - Libraries: TensorFlow, Keras, TensorFlow Lite / TFLite Micro, TensorFlow Model Optimization Toolkit, NumPy, xxd
  - Software: Silicon Labs Simplicity Studio, Gecko SDK, Edge Impulse Studio (impulse design, spectral features, anomaly detection), Google Colab
  - Hardware: Silicon Labs EFR32MG24 / EFR32XG24 dev kits (BLE), IMUs, thermal infrared sensor, LEDs, LCD, buttons, PWM
- **Embedded ML applications: keyword spotting (mel-spectrogram CNN), handwriting digit recognition, IMU gesture recognition with sensor fusion, real-time posture detection, gesture and anomaly detection, thermal-IR flame detection** — EE 513 booklet
- **Responsible and ethical ML: fairness and bias, transparency, safety, governance frameworks** — EE 513 booklet (chapter 2)
- **On-device audio and IMU AI on the Silicon Labs EFR32 board: microphone audio capture, framing and windowing, pre-emphasis and band-pass filtering, MFCC / log-mel features with CMSIS-DSP, keyword spotting and speaker recognition (verification) with small CNNs; IMU gesture recognition with filtering, windowing, and sensor fusion; INT8 quantization, TFLite Micro, CMSIS-NN, and the MVP accelerator; profiling of memory, latency, and power** — EE 513 booklet; PhD (Ajan, Sep 30, 2026)
  - Languages: C, C++, Python
  - Libraries: TensorFlow / TFLite Micro, CMSIS-DSP, CMSIS-NN, librosa (feature design), Edge Impulse, NumPy
  - Software / Hardware: Simplicity Studio, Gecko SDK; EFR32MG24 / EFR32XG24 dev kits (BLE) with on-board microphone and IMU

Microelectronics, VLSI & Hardware Reliability (M.Eng. coursework)

- **VLSI circuit testing: fault models (stuck-at, bridging, delay), fault simulation, automatic test pattern generation (ATPG), design for testability (DFT), scan chains, built-in self-test (BIST), boundary scan / JTAG (IEEE 1149.1), fault coverage** — ECE 522 VLSI Circuit Testing
  - Languages: Verilog, VHDL, SystemVerilog, Tcl
  - Software: Cadence (Virtuoso, Genus, Modus), Synopsys (Design Compiler, TetraMAX / TestMAX), Siemens / Mentor Tessent, ModelSim / QuestaSim, Xilinx Vivado
- **MEMS and micro-engineering: microfabrication (photolithography, deposition, etching, bulk and surface micromachining), MEMS sensors and actuators (accelerometers, gyroscopes, pressure sensors, microphones), scaling laws, microfluidics** — ECE 459 MEMS and Micro-Engineering
  - Software: COMSOL Multiphysics, CoventorWare, ANSYS, L-Edit / KLayout (layout), SolidWorks, MATLAB

- **Computer systems reliability: fault tolerance, redundancy (TMR, N-version), error-detecting and error-correcting codes, reliability block diagrams, Markov models, MTBF / MTTR, availability, failure-rate modelling** — ECE 419 Computer Systems Reliability; EE 603 Reliability and Life Testing
  - Languages: Python, MATLAB, R
  - Libraries / Software: Python reliability, lifelines, SciPy stats; MATLAB; ReliaSoft / Isograph Reliability Workbench, Minitab

Power Systems (M.Eng. coursework)

- **Electric power distribution: distribution system design, radial and network feeders, transformers, load flow / power flow, voltage regulation, power factor correction, short-circuit and fault analysis, protection (fuses, relays, reclosers), distributed generation and renewables integration** — ECE 589 Electric Power Distribution
  - Languages: MATLAB, Python
  - Libraries: MATLAB Simscape Electrical / Simulink, pandapower, PyPSA, OpenDSSDirect.py
  - Software: ETAP, PowerWorld Simulator, OpenDSS, PSS/E, PSCAD, CYME, GridLAB-D

Machine Learning (M.Eng. coursework)

- **Machine learning foundations: linear / logistic regression, decision trees, SVM, k-NN, Naive Bayes, ensembles, clustering, PCA, neural networks, bias–variance, regularization, model evaluation** — ECE 568 Machine Learning; IA 651; PhD
  - Languages: Python, MATLAB, R
  - Libraries: scikit-learn, NumPy, pandas, Matplotlib, PyTorch, TensorFlow / Keras; MATLAB Statistics and Machine Learning Toolbox

Independent Research (M.Eng.)

- **Independent research project: literature review, problem formulation, experimentation, technical reporting** — ECE 592 Special Investigations (Piezo1 / neuroinflammation research)

Classical Speech Enhancement & Early PhD Pipeline (Sep 2024 seminar)

- **Classical denoising and filtering: Kalman filtering for speech enhancement (state estimation from noisy measurements), high-pass Butterworth IIR filtering (4th order, 150 Hz cutoff), energy-threshold VAD segmentation, MFCC feature extraction** — PhD early work, T4 seminar
  - Languages: MATLAB, Python
  - Libraries / Software: MATLAB Signal Processing Toolbox (butter, filtfilt), SciPy signal, pykalman / FilterPy, librosa, python_speech_features
- **Deep-learning source separation for speech denoising: Wave-U-Net (1D CNN, multiscale raw-waveform processing) trained on music source-separation data** — PhD early work, T4 seminar; BDCC paper
  - Datasets: MUSDB18, ccMixter
  - Libraries: TensorFlow / PyTorch Wave-U-Net implementations
- **Speech-quality metrics: MOS, PESQ, segmental / weighted / frequency-weighted SNR, Speech Transmission Index (STI) and ASR-based intelligibility** — PhD early work, T4 seminar
  - Libraries: pesq, pystoi, pysepm, speech-to-text engines (e.g. Whisper / Google Speech API) for intelligibility
- **Additional speech corpora used:** SpEAR, CABank English CallHome, ESD, MUSDB18, ccMixter, MS-SNSD (Microsoft Scalable Noisy Speech Dataset) — T4 seminar, MCCNNY 2024 poster
- **CNN-based speech denoising and before / after quality benchmarking (SNR, MOS, PESQ) against VeriSpeak recognition accuracy** — MCCNNY 2024 poster
  - Libraries: PyTorch / TensorFlow, pesq, MS-SNSD scripts, NumPy / SciPy

- **Neural-network speaker recognition baseline: train / validation / test splits, true-acceptance-rate evaluation** — T4 seminar
  - Libraries: TensorFlow / Keras or PyTorch

Deepfake Audio Generation & Voice Cloning (CITeR 23F-03C)

- **Text-to-speech and voice cloning: zero-shot / few-shot cloning, hyperparameter tuning of TTS models, evaluating deepfakes against speaker-recognition systems** — CITeR 23F-03C, ELAD-SVDSR
  - Models / Tools: Tortoise TTS, ElevenLabs, YourTTS, XTTS (Coqui), MeloTTS, MetaVoice, Tacotron 2, WaveNet, MelGAN, Real-Time Voice Cloning (SV2TTS)
  - Libraries: Coqui TTS, PyTorch, Hugging Face
- **Speech denoising for synthesis: spectral gating, VAD-based adaptive denoising, spectral-statistics and multi-band noise profiling, deep-learning Denoiser (Meta / Facebook Research, Demucs-based)** — CITeR 23F-03C
  - Libraries: noisereduce, facebookresearch/denoiser, webrtcvad / Silero VAD, SciPy
- **Speaker diarization and audio annotation** — CITeR 23F-03C
  - Tools: pyannote.audio, Audacity (manual annotation and cleanup)
- **Speaker-embedding scoring:** WeSpeaker (cosine similarity), VeriSpeak thresholds calibrated to FAR — CITeR 23F-03C

Emotional Voice Conversion & Anti-Spoofing (CUWavestarGAN)

- **Emotion-conditioned many-to-many voice conversion: StarGAN-VC-style multi-discriminator GANs, cycle-consistency, hierarchical prosody disentanglement, hybrid losses (adversarial, speaker-identity cosine, emotion-transfer, differentiable PESQ-proxy), loss-weight ablations** — CUWavestarGAN
  - Libraries / Models: PyTorch, torchaudio, HiFi-GAN vocoder, Tacotron-style seq2seq with Bahdanau dual attention, speaker-embedding conditioning
- **Voice anti-spoofing / deepfake detection models: RawNet2, AASIST, LFCC-LCNN; cross-emotional, cross-synthesizer, and cross-dataset evaluation; paired t-tests** — CUWavestarGAN
  - Datasets: ASVspoof 2019 LA, ESD, EmoFake-style emotional spoofs
- **Audio conditioning for synthesis: FFmpeg resampling (24 kHz mono), multi-band spectral gating, dynamic-range compression (acompressor), fixed-length cropping and padding** — CUWavestarGAN

Quality–Performance Analysis for Law-Enforcement Audio (VSEA, FBI)

- **Controlled degradation experiments: 5-level denoising protocol, enrollment–probe verification design, correlation and variance-explained analysis across 7,645 comparisons, scatter-matrix and distribution visualization** — VSEA
  - Speaker-recognition systems: Neurotechnology VeriSpeak, SpeechBrain ECAPA-TDNN, Resemblyzer (GE2E)
  - Metrics: SNR, PESQ, STOI, jitter, shimmer; Pearson / Spearman correlation
- **Sponsor technical reporting: final technical reports with operational recommendations for a federal (FBI) sponsor, delivered under NDA** — VSEA

Noisy-Environment Speaker Recognition (CITeR 23S-07C)

- **Extended Kalman filter (EKF) denoising, frame-based processing (1.5–46.5 ms frames, 50% overlap), frame-wise FFT power-spectrum SNR estimation** — CITeR 23S-07C
  - Languages / Libraries: MATLAB, Python (NumPy FFT, SciPy, FilterPy)
- **Deep speaker recognition (ResNet) for text-dependent and text-independent verification** — CITeR 23S-07C

- Libraries: PyTorch, TensorFlow / Keras

- **Meta-learning for adaptive denoising: model-agnostic meta-learning (MAML) for fast adaptation to new noise conditions (proposed)** — CITeR 23S-07C
  - Libraries: PyTorch, learn2learn, higher

Datasets, Models & Tools Used in Publications

- **Speech / speaker datasets:** VoxCeleb1 / 2, VCTK, CN-Celeb, VOiCES, LibriSpeech, Common Voice, TIMIT, CHiME-6, SpEAR, VoiceBank-DEMAND (Valentini), Fake-or-Real (FoR), ESD, MUSAN, RIR corpora; own datasets VPQAD, VSEA-DC, ELAD-SVDSR, SPEAD — VQI, Signals, BDCC, MDTS, ELAD papers
- **Speech enhancement models:** Wave-U-Net, CMGAN (Conformer-based Metric GAN), U-Net — BDCC paper
  - Hardware: NVIDIA RTX 4080 GPU
- **Speech synthesis / deepfake generation:** Tortoise TTS; synthetic voice cloning from long-form recordings — ELAD-SVDSR, SPEAD
- **Audio codec and channel simulation:** telephone band-limiting (300–3400 Hz, 8 kHz), MP3 and AAC encode–decode roundtrips, RT60 reverberation — Signals paper
  - Software: FFmpeg, SoX, pesq, pystoi
- **Multi-modal biometric matchers and datasets:** face (ArcFace, FaceNet, MTCNN alignment; LFW), voice (ECAPA-TDNN, x-vector; VoxCeleb1), iris (open-iris by Worldcoin, log-Gabor; CASIA-Iris-Thousand), fingerprint (SourceAFIS; FVC2002) — Beyond ROC-AUC preprint
  - Libraries: facenet-pytorch, InsightFace, SpeechBrain, open-iris, SourceAFIS, scikit-learn, NumPy, SciPy
- **Statistical evaluation:** bootstrap confidence intervals, paired bootstrap tests, McNemar's test, ANOVA / ANCOVA, Spearman correlation, multiple regression, mixed-effects models; ISO/IEC 19795-1 and ISO/IEC 29794-1 reporting — Beyond ROC-AUC, MDTS, HI-POCT, Signals, VQI
- **Classical classifiers for audio forensics:** Logistic Regression, LDA, QDA, Gaussian Naive Bayes, linear / RBF SVM, GMM — MDTS paper
- **Physiological sensing and affective computing:** Shimmer3 ECG / PPG / GSR sensors, APDM Opal mobility sensors (IMU), mCTSIB balance test, POMS-SF mood survey, Big Five personality inventory; ARDT / ARRC toolkits; CUADS, ASCERTAIN, DREAMER datasets; temporal CNN, PETSFCNN, CNN-LSTM models — CUADS, UEMCON, HI-POCT papers
- **Wearable / embedded hardware:** Silicon Labs EFR32MG BLE microcontroller, ICM-20689 IMU, custom PCB, coin-cell power; Raspberry Pi camera, heart-rate sensor, GPS / GSM modules — AIDrawPen, Smart Helmet
  - Software: Simplicity Studio (Silicon Labs), KiCad / Altium, Raspberry Pi OS, Python
- **Recording hardware for data collection:** Audio-Technica AT2020, Shure SM58, TOZO A1 earbuds, Inni Oasis R1 recorder, spy-pen recorder, iPhone 16 Pro — VPQAD, ELAD-SVDSR, VSEA-DC
- **mm-wave antenna engineering:** patch antennas on glass (transparent) substrates, skin-equivalent phantoms, 60 / 94 / 110 GHz simulation — antenna papers
- **Real-time communication in apps:** Jitsi Meet API (video consultation), in-app chat, payment integration — mental well-being and telemedicine apps
- **FPGA biometric acceleration (review):** HLS, quantization / fixed-point, streaming dataflow, CPU–FPGA partitioning, vendor DL IP (Xilinx Vitis AI / DPU, Intel OpenVINO FPGA) — FPGA review
- **Synthetic biometric data generation and evaluation: age-conditioned generative models for fingerprints, candidate filtering, real-to-synthetic similarity and synthetic-to-synthetic diversity analysis** — UEMCON 2026 fingerprint paper
  - Software / Libraries: NIST NFIQ 2, NIST NBIS (MINDTCT, BOZORTH3), DINOv2 (Meta, texture embeddings), PyTorch, diffusion / GAN fingerprint generators, OpenCV
- **Generalized linear mixed models (GLMM) and nested LMER model comparison; bootstrapped confidence intervals for FNMR at fixed FMR** — TBIOM children study

- Libraries: R lme4 (glmer), lmerTest, Python statsmodels, NumPy / SciPy bootstrap

- **Grant proposal development: NSF proposals (Engineering Research Initiation and ERI CIVIC: 2-page summaries, 10-page project descriptions, broader impacts, data management, and budget justification), FBI proposals (1-page initial and 2-page detailed proposals), human-subjects IRB protocols, industry-consortium (NSF IUCRC / CITeR) proposals, progress reports, and review presentations** — NSF ERI #2501916, NSF ERI (SIUC), NSF ERI CIVIC, FBI projects, CITeR projects
  - Software / Systems: NSF Research.gov, SciENcv (biosketches), LaTeX / Word proposal templates, budget spreadsheets

Research Mentoring & Collaboration Tools

- **Supervising student research: shared codebases, experiment tracking, code review, shared writing** — mentoring 5 junior researchers
  - Presentation coaching: PowerPoint, Google Slides, LaTeX Beamer, Canva, Zoom / Teams (recorded mock defenses)
  - Software: Git / GitHub (pull requests, issues, code review), GitHub Projects / Trello, Overleaf (shared papers), Weights & Biases / TensorBoard (shared experiment tracking), Google Drive / OneDrive, Slack / Teams / Zoom, Zotero group libraries

LaTeX & Technical Typesetting (taught as an invited seminar)

- **LaTeX document engineering: article, report, thesis, resume / CV, and Beamer presentation and research-poster templates (beamerposter, tikzposter, baposter); BibTeX / BibLaTeX citation management; figures, subfigures, tables, equations, algorithms; cross-referencing; IEEE / ACM / Elsevier / MDPI journal templates; collaborative editing and version history** — invited ECE seminars (2025, 2026); papers and dissertation
  - Languages: LaTeX, BibTeX / BibLaTeX, TikZ / PGFPlots
  - Packages: amsmath, graphicx, booktabs, siunitx, hyperref, cleveref, algorithm2e, subcaption, IEEEtran, moderncv / awesome-cv
  - Software: Overleaf, TeX Live, MiKTeX, TeXstudio, VS Code (LaTeX Workshop), Zotero / JabRef (reference export), GitHub (template sharing)

IT, Systems & Network Administration (Work Experience)

- **Network design and administration: LAN / Wi-Fi, TCP/IP, DHCP, DNS, VLANs, switches and routers, VPN, firewall rules, secure multi-user lab networks (65 users)** — Network and IT Administrator, McGill University
  - Hardware / Software: managed switches, routers, wireless access points, pfSense / hardware firewalls, Cisco / Netgear / Ubiquiti equipment

- **Server and workstation administration: Windows Server, Active Directory, Group Policy, Linux servers, user accounts and permissions, patching, imaging and deployment, remote administration (RDP / SSH)** — McGill University
  - Tools: Active Directory Users and Computers, WSUS, Clonezilla / WDS / MDT, PowerShell, Bash

- **Backup, storage, and disaster recovery: multi-terabyte, multi-site backups; NAS / SAN; RAID; SMB / NFS shares; restore testing** — McGill University, Dogma Hospital
  - Tools: Veeam, rsync, robocopy, Windows Server Backup, cloud storage

- **Monitoring and security: server uptime monitoring, log review, antivirus / endpoint protection, least-privilege access control** — McGill University
  - Tools: Nagios / Zabbix / PRTG, Windows Event Viewer, Windows Defender / endpoint security

- **IT procurement and asset management: requirements gathering, vendor quotes, purchasing, inventory** — McGill University

- **Help-desk and end-user support: troubleshooting hardware, software, VPN, telephony, and wireless; ticket triage and escalation; supporting clinical staff** — HAF Hospital, Dogma Hospital, McGill University
  - Tools: help-desk ticketing systems, VPN clients, IP / PBX telephony, Microsoft Office
- **Specialized equipment support: eye-tracking systems and research / diagnostic computers** — Dogma Diagnosis and Hospital

Data Extraction & Bibliometrics (Work Experience)

- **Web data extraction and bibliometric analysis: Web of Science / Web of Knowledge exports, HTML parsing, structured CSV datasets, data cleaning and de-duplication, publication statistics** — Publication Data Mining, McGill University
  - Languages / Libraries: Java (jsoup, regular expressions, file I/O), CSV
  - Software: Web of Science, Microsoft Excel (pivot tables, charts)

Finance & Accounting Operations (Work Experience)

- **Accounting operations: accounts payable / receivable, invoice coding and processing, account and ledger reconciliation, expense reimbursements, journal entries, chart-of-accounts coding, month-end close support, audit documentation, internal controls** — Accounting Clerk, McGill University Financial Services
  - Software: university ERP / Banner Finance, Microsoft Excel (reconciliations, pivot tables, VLOOKUP / XLOOKUP), Microsoft Office

HR & Business Administration (Work Experience)

- **Employee relations and HR administration: performance evaluation, workplace investigations, HR records management, onboarding, policy application** — Employee Relations, Volt Workforce Solutions
  - Software: HRIS / applicant-tracking systems, Microsoft Office (Excel, Word, Outlook)

Scientific Writing, Publishing & Productivity Tools

- **Journal and conference papers, dataset descriptors, dissertation, defense slides** — PhD
  - Languages: LaTeX
  - Software: Overleaf, Beamer (slides), Microsoft Word, PowerPoint, Google Docs, Zotero / Mendeley / EndNote, draw.io / Inkscape (figures), Grammarly, Claude, ChatGPT
- **Project and lab organization**
  - Software: Microsoft Office (Excel, Teams), Google Workspace, Notion / Trello, Slack

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Tools & Platforms

- Neurotechnology VeriSpeak SDK (commercial speaker recognition) — PhD
- GitHub (open-source software release) — PhD
- IEEE DataPort (public dataset publishing) — PhD

Hardware & Recording Equipment

- Studio microphones (Audio-Technica AT2020, Shure SM58), audio recorder (Inni Oasis R1), iPhone 16 Pro call recording — PhD (data collection)
- GPU computing: NVIDIA GPUs, CUDA — PhD (deep models)
- HPC cluster — PhD (large-scale processing)
- Wearable IMU / mobility sensors — HI-POCT paper (gait and balance)
- FPGA development boards — FPGA review

Spoken Languages

- English: fluent
- Bengali: native

Soft Skills

Leadership & Initiative

- **Research leadership:** defined research questions and led a multi-part dissertation end to end, from data collection through modelling, software release, and publication — PhD
- **Independence and self-direction:** identified a gap in ISO/IEC standards (no voice quality metric) and built a solution — PhD (VQI)
- **Initiative and ownership:** proposed, built, and released open-source software beyond publication requirements — PhD (VQI app)
- **Vision and strategic thinking:** positioned VQI as a future ISO/IEC voice-quality reference alongside NFIQ 2 and OFIQ — PhD
- **Lead-author responsibility:** first author on 14 papers and abstracts (13 published / preprint, 1 under review), including VQI (IEEE Access), Signals, and VPQAD — PhD
- **Research mentoring and supervision:** mentored 5 junior researchers (graduate students Faheem Ahmad, Yash Sukhdeve, and Batuhan Uzunoglu; undergraduate John Parker; and master's student Leonel Akanji) from problem definition to publication, producing 2 accepted conference papers, 1 conference poster, 2 published datasets / descriptors, and 1 manuscript under review — PhD
- **Mentoring across hardware and software:** guided a mentee's embedded AI hardware project (BLE pen, IMU, custom PCB, deep-learning gesture model) to an IEEE MDTS 2025 poster — mentoring (Leonel Akanji)
- **Presentation coaching:** taught graduate students to structure talks, design clear slides, tell a story with data, and handle questions — mentoring (Batuhan Uzunoglu)
- **Proposal and defense preparation:** prepared students for research proposal defenses through mock defenses, feedback on proposal writing, and practice with likely committee questions — mentoring (Batuhan Uzunoglu)
- **Mentoring undergraduates in research:** brought an undergraduate researcher to co-authorship on a published IEEE dataset — mentoring (John Parker)
- **Developing others:** taught juniors the research workflow: literature review, experiment design, coding, evaluation, and scientific writing — PhD
- **Delegation and coordination:** split multi-person projects (datasets, deepfake detection) into tasks and tracked progress — PhD
- **Constructive code and paper review:** reviewed mentees' code, results, figures, and drafts with actionable feedback — PhD
- **Team coordination:** coordinated multi-author, multi-institution projects — PhD
- **Leading a multi-university project:** led the Clarkson–WVU CITeR face image quality project from setup to final report and program review presentation — CITeR 25F-04W
- **Decision-making:** chose deployed models by weighing accuracy, error reduction, and inference speed — PhD (VQI)

Project & Time Management

- **Project management:** delivered several parallel projects (VQI, children study, deepfake detection, 4+ datasets) to publication — PhD
- **Planning and scheduling:** planned multi-environment data collections (6 environments, 2 devices, 100 participants) — PhD (VSEA-DC)

- **Time management and prioritization:** balanced dissertation research, co-authored papers, and reviewer revisions to finish the PhD in under 3 years (January 2024 to June 2026 defense) — PhD
- **Meeting deadlines:** met journal submission, revision, and conference deadlines — PhD
- **Handling large, complex workloads:** managed 11 datasets, ~2.6 M utterances, and ~29 M verification pairs — PhD (VQI)
- **Resource management:** managed compute, storage, and recording equipment — PhD
- **Accountability to sponsors:** delivered grant-funded research results and progress reports — Graduate Research Assistant
- **Multitasking:** balanced teaching duties, research, publications, and coursework at the same time — Graduate TA + PhD
- **Agile teamwork:** planned work in two-week sprints with a shared backlog, stand-ups, reviews, and retrospectives with lab members and partner developers — Graduate Research Assistant (PhD)

Communication

- **Presenting to internal and external stakeholders:** presented research findings to lab and department audiences and to external sponsors (CITeR members, FBI) and international workshops, explaining complex technical concepts clearly — PhD
- **Scientific writing:** 24 publications (9 journal, 12 conference / poster, 1 abstract, 1 under review, 1 preprint) and 4 public datasets, including 3 dataset descriptors; 144 citations, h-index 6 — BSc, M.Eng., PhD
- **Invited speaking:** 6 invited talks and seminars (2024–2026), including an invited talk at the International Martigny Biometrics Workshop (Switzerland), an invited CITeR webinar, and 4 invited ECE department seminars; invited 2 years in a row to teach LaTeX / Overleaf — PhD
- **Research talks to academic audiences:** presented dissertation research (VQI) at the ECE graduate seminar, making a technical biometrics topic accessible to students from other ECE specialties — ECE seminar (March 2026)
- **Explaining a field to newcomers:** built a talk that starts from biometrics fundamentals and works up to open research problems, making it accessible to non-specialists — T4 seminar (Sep 2024)
- **Participant recruitment through outreach:** used seminar talks to recruit volunteers for voice data collection — T4 seminar
- **Iterative problem solving:** recognized the limits of a classical method (Kalman filter inconsistency across subjects) and moved to deep learning (Wave-U-Net) — PhD early work
- **Supporting peers' professional development:** shared reusable LaTeX templates for papers, reports, resumes, and slides — ECE seminar, GitHub
- **Poster and oral presentations:** poster at IEEE MDTS 2025 (AIDrawPen); oral at IEEE MDTS 2026 (deepfake baselines); talks at IEEE CCWC, ICCMC, HI-POCT — PhD, RA NSU
- **Securing research funding:** co-wrote 12 funded proposals totaling over $1.1M: 3 NSF ERI awards ($200K each), 3 FBI awards ($120K, $108K, $50K), and 6 CITeR projects ($45K each, one funded by DHS); also co-wrote proposals that were not funded — M.Eng., PhD
- **Writing for federal sponsors:** contributed to proposals funded by the National Science Foundation, the Department of Homeland Security (via CITeR), and the FBI Criminal Justice Information Services (CJIS) Division, including multi-institution (Clarkson–WVU) subawards — PhD
- **Grant writing and sponsor communication:** contributed to a multi-university CITeR research proposal (Clarkson + West Virginia University) and presented proposals and progress posters to industry and government sponsors at CITeR program reviews — PhD
- **International research engagement:** gave an invited talk at the International Martigny Biometrics Workshop 2025 (European Association for Biometrics, CITeR, Idiap) in Switzerland, engaging with the European and US biometrics communities — PhD
- **Evidence-based pivoting:** benchmarked classical against deep-learning denoising, recognized the Kalman filter's inconsistency, and changed the project's direction based on data — CITeR 23S-07C

- **Leading a sponsored project as a student:** lead student researcher on CITeR project 23S-07C, reporting directly to industry and government sponsors — PhD
- **Reporting lessons learned and changing direction:** communicated what did not work (TTS models failing on real noise, inconsistent Kalman filtering) and adjusted research plans — CITeR 23F-03C, 23S-07C
- **Presenting to industry stakeholders:** communicated research to CITeR member organizations (industry and government) under an industry-consortium model — PhD
- **Explaining to senior and governance audiences:** explained methods, results, uncertainty, and trade-offs to sponsor review boards, IRB reviews, and department leadership — PhD
- **Curiosity about the data under the model:** opened the raw signals to find out why a method worked or failed (quality metrics that did not predict recognition, Kalman filtering that was inconsistent across speakers, look-alike structures in scans) instead of treating models as black boxes — PhD; ImransLab
- **Technical judgment on new AI tools:** compared new frameworks and models against a fixed regression suite before adopting them (LangGraph vs CrewAI / AutoGen; prompted LLMs vs fine-tuned BERT; local vs cloud models), separating lasting value from hype — PhD
- **Patience with detail:** spent many hours reviewing data and model outputs by hand and wrote guidelines so others could review consistently — PhD; ImransLab (annotation adjudication)
- **Handling peer review (journals):** addressed first-round reviewer comments and submitted revisions for IEEE Access (VQI, accepted) and IEEE TBIOM (children study) — PhD
- **Corresponding-author responsibility:** managed submission, reviewer correspondence, and revisions as corresponding author — VQI (IEEE Access), Signals
- **Open science and reproducibility:** released 4 public datasets on IEEE DataPort and open-source code (VQI app, data scripts) — PhD
- **Leading data collection teams:** led and supervised human-subject data collections (VPQAD 50, ELAD 36, CUADS 38, VSEA-DC 100 participants) and worked on the school-based longitudinal child biometrics collection (379 children, 9 years) — PhD
- **Working with children, schools, and families:** ran on-site biometric collections with children aged 4–18 on ~7 school visits per semester across elementary, middle, and high schools, coordinated with school staff, and followed consent / assent protocols — longitudinal child collection
- **Field operations and logistics:** planned and ran off-site collection days with portable equipment under time limits — longitudinal child collection, VSEA-DC
- **Critical evaluation of research practice:** challenged common but misleading metric reporting (ROC-AUC) and argued for standards-based operating-point reporting — Beyond ROC-AUC
- **Research versatility:** published across neuroscience, biometrics, speech processing, affective computing, antennas, IoT healthcare, and software systems — BSc, M.Eng., PhD
- **Wet-lab collaboration:** worked in an interdisciplinary cellular neuroscience team; contributed to experimental design and data analysis — M.Eng. (Frontiers in Cellular Neuroscience)
- **Patience and precision:** mastered delicate, time-intensive bench techniques (patch clamp, confocal imaging) where small errors ruin experiments — M.Eng. research
- **Fast hands-on learning:** trained in several microscopy methods and electrophysiology within a Master's program — M.Eng. research
- **Lab rigor and record-keeping:** followed SOPs, aseptic technique, and careful lab notebooks — M.Eng. research
- **Entrepreneurial and product thinking:** built user-facing health, well-being, and fintech apps — BSc-era papers
- **Oral presentation:** defended dissertation before a 5-member committee; presented at IEEE conferences — PhD
- **Communicating to non-experts:** translated technical features into plain language (e.g. "how echoey the room sounds") — PhD (VQI)
- **User-focused communication:** wrote in-app explanations telling users what each score means and how to improve it — PhD (VQI app)
- **Visual communication:** designed figures, diagrams, charts, and slide decks to explain complex results — PhD
- **Technical documentation:** wrote dataset documentation, software READMEs, and appendices — PhD

- **Handling peer review:** answered reviewer critiques and addressed all comments in journal revisions — PhD (VQI, IEEE Access)
- **Active listening and taking feedback:** incorporated committee and advisor feedback — PhD

Teaching, Mentoring & Student Support

- **Teaching effectiveness:** average student evaluation of 4.95 out of 5 across all teaching assistant roles — Graduate TA (ES 250, EE 260, EE 402/502, EE 622)
- **Sustained teaching commitment:** taught programming and computer science for about six years (January 2018 – November 2023) alongside undergraduate and graduate studies, in person and online — Instructor, ImransLab
- **Teaching and instruction:** supported 4 courses across 9 offerings, from first-year Electrical Science to graduate Advanced Biometrics — Graduate TA (Jan 2024 – Jun 2026)
- **Lecturing and course leadership:** delivered 12 lectures to 42 senior undergraduate and graduate students as co-instructor of EE 402/502 Machine Learning on Biomedical Signals (Fall 2025, Fall 2026) — Co-Instructor
- **Public speaking and classroom presence:** presented technical material to a 42-student class across 12 lectures — Co-Instructor
- **Assessment design:** designed assignments and course projects that test both theory and hands-on skills — Co-Instructor
- **Project mentoring:** guided students through open-ended ML projects from idea to results — Co-Instructor
- **Turning research into teaching:** built lectures from dissertation expertise in voice and speech processing — Co-Instructor
- **Student engagement and accessibility:** held office hours for a 42-student course, supporting students with different backgrounds — Co-Instructor
- **Earned trust and promotion:** moved from TA (Fall 2024) to co-instructor of the same course — EE 402/502
- **Instructional writing and textbook authoring:** wrote the entire ~120-page, 9-chapter course booklet (*Machine Learning on Audio Signals*, EE 502) and the text of the ~150-page, 11-chapter *Embedded Machine Learning* booklet (EE 513), pairing concise theory with runnable code, hardware labs, and exercises: ~270 pages of course material in total
- **Curriculum and course development:** assignments, projects, and exams for a cross-listed undergraduate/graduate course — Co-Instructor
- **Teaching across levels:** first-year, upper-level undergraduate, and graduate students — ES 250, EE 260, EE 402/502, EE 622
- **Interdisciplinary teaching:** taught engineering and biomedical-signal ML to students from electrical, computer, and biomedical engineering backgrounds — EE 402/502
- **Inclusive teaching and equity, diversity, and inclusion:** taught diverse classes (international and domestic students, many majors, first-generation learners), used clear rubrics, multiple ways to engage, and flexible office hours so every student could succeed — ES 250, EE 402/502, ImransLab
- **Community outreach:** presented lab research at public showcases and seminars, recruited volunteers for data collection, and ran coding camps for school and university students — PhD; ImransLab
- **Knowledge of course learning outcomes:** wrote course learning outcomes and mapped them to lectures, labs, and assessments — EE 402/502
- **Active and project-based learning:** used live coding, Jupyter notebook labs, in-class activities, and open-ended team projects on real biomedical and voice data — EE 402/502
- **Hands-on lab teaching:** guided students through circuit building, instrument use, and microcontroller programming — ES 250, EE 260
- **Consistency and reliability:** asked back to TA ES 250 for 4 semesters — ES 250
- **Teaching across learner types:** university students, online / ed-tech learners (ImransLab), and high-school O / A Level students — all teaching
- **Personalized instruction:** diagnosed individual learning gaps and built tailored study plans — O / A Level tutoring

- **Parent and stakeholder communication:** reported student progress and goals to parents — O / A Level tutoring
- **Entrepreneurial self-management:** ran an independent tutoring practice (scheduling, student recruitment, fees) — O / A Level tutoring
- **Real-time classroom support:** as in-lecture TA, answered questions on the spot and helped students during live instruction — EE 622
- **Independent lab instruction:** ran lab sessions as the instructor in charge, from briefings to supervision and grading — Lab Instructor, BME 338L (SIUC)
- **Teaching at two universities:** taught at SIUC and Clarkson, adapting to different curricula and student groups — BME 338L, Clarkson courses
- **Human-subject lab safety:** supervised safe biosignal recording on student volunteers — BME 338L
- **Balancing research and teaching from early on:** held an RA post while also instructing a lab during the M.Eng. — SIUC
- **Managing large classes:** supported ~120-student sections while keeping grading and feedback consistent — ES 250
- **Scaling teaching across class sizes:** from 12-student graduate courses to 120-student core courses; ~665 students in total across 2 universities — all courses
- **Teaching non-majors:** helped mechanical, civil, chemical, and other engineering students build confidence with electrical concepts — ES 250
- **Lab safety and responsibility:** enforced safe handling of electrical and embedded lab equipment; basic first aid training (2012) — ES 250, EE 260, BME 338L
- **Emergency preparedness:** basic first aid training (Bangladesh, 2012), able to respond to injuries and medical emergencies in labs, classrooms, and data-collection sessions
- **Hands-on troubleshooting coaching:** taught students to debug code and hardware systematically rather than handing them answers — EE 260, EE 402/502
- **Team project mentoring:** guided student teams through scoping, execution, and reporting of embedded and ML projects — EE 260, EE 402/502
- **Teamwork with fellow TAs:** coordinated grading consistency across sections and TAs — ES 250
- **Academic integrity:** managed submissions, extensions, and integrity checks fairly — all TA courses
- **Written communication with students:** answered forum and email questions clearly and promptly — all TA courses
- **Peer-level teaching:** helped classmates while taking the same graduate course, balancing student and TA roles — EE 622
- **Explaining complex ideas simply:** adapted explanations to students with different backgrounds and learning styles — Graduate TA
- **Giving constructive feedback:** graded assignments, labs, and exams fairly and consistently, with actionable written feedback — Graduate TA
- **Student mentoring and support:** held office hours and helped students troubleshoot code, circuits, and concepts — Graduate TA
- **Patience and empathy:** supported struggling students and students from diverse backgrounds — Graduate TA
- **Classroom and lab management:** kept lab sessions safe, organized, and on schedule — Graduate TA
- **Fairness and confidentiality:** handled grades and student records under FERPA (FERPA-trained, Clarkson), with a signed confidentiality agreement for student data systems — Graduate TA
- **Collaboration with faculty:** coordinated with instructors on course logistics, grading standards, and exam preparation — Graduate TA
- **Balancing teaching and research:** held a teaching assistantship while publishing 11 papers and completing the dissertation — PhD + Graduate TA
- **Onboarding and mentoring lab members:** guided newer students on tools, datasets, and research methods — Graduate Research Assistant

Collaboration & Interpersonal

- **Cross-functional collaboration:** worked with senior researchers, engineers, product managers, and industry and government sponsors on shared research goals — PhD (CITeR, FBI projects); ML Engineer, ImransLab
- **Brainstorming and idea generation:** contributed and developed new research ideas in lab brainstorming sessions and sponsor planning meetings, several of which became funded proposals — PhD
- **Collaboration:** co-authored with 30+ researchers across Clarkson University, UNC Charlotte, Southern Illinois University Carbondale, and North South University — BSc, M.Eng., PhD
- **Interdisciplinary breadth:** M.Eng. coursework spanning biomedical engineering (neuroengineering, instrumentation, imaging), computing (edge, embedded, ML), microelectronics (VLSI, MEMS), and power systems — M.Eng.
- **Lab discipline and technical reporting:** wrote structured lab reports with calibration and error analysis — BME 338L
- **Cross-disciplinary teamwork:** worked across engineering, computer science, business, and health research (gait and anger detection) — PhD
- **Working with external experts:** collaborated with a leading biometrics researcher (Dr. Stephanie Schuckers) — PhD
- **Participant relations:** recruited, briefed, and recorded 150 adult participants with professionalism and empathy — PhD (VPQAD, VSEA-DC)
- **Working with children and families:** longitudinal dataset of 292 children — PhD (children study)
- **Cultural adaptability:** international researcher working across the US and Canada — PhD
- **Running many projects at once:** led 7 projects at the same time across hardware, RF, and software as a research assistant, all through to publication — RA, NSU (2021–22)
- **Switching between domains:** moved between antenna simulation, embedded hardware, and full-stack app development on different projects — RA, NSU
- **User-centred design:** designed apps around the needs of patients, doctors, people seeking mental-health support, and consumers comparing financial products — RA, NSU
- **Conference presentation:** presented papers at international IEEE conferences (CCWC, ICCMC, ICAEEE) — RA, NSU
- **Research initiative as an undergraduate:** took on faculty-supervised directed research in energy-efficient motor control alongside the thesis — B.Sc. (Dr. Riasat Khan)
- **Sustainability mindset:** worked on reducing energy waste in industrial motors, which use a large share of industrial electricity — B.Sc. directed research
- **Working with multiple supervisors:** delivered projects under different faculty (Dr. M. M. Khan, Dr. Riasat Khan) with different research styles — B.Sc.
- **Formal public speaking training:** ENG 111 Public Speaking; applied later in 12 lectures, conference talks, and the dissertation defense — B.Sc. to PhD
- **Professional and engineering ethics:** PHI 104 Introduction to Ethics; applied in human-subject research and responsible AI — B.Sc. to PhD
- **Written communication foundations:** ENG 102 / 103 Composition — B.Sc.
- **Business and economic awareness:** engineering economics, cost–benefit and feasibility analysis — EEE 452
- **Well-rounded education:** coursework in economics, political science, sociology, and history alongside engineering — B.Sc. university core
- **Team leadership (undergraduate):** led a 4-member thesis team through design, prototyping, testing, and the final defense — B.Sc. thesis
- **Social impact focus:** designed affordable healthcare technology for patients with limited access to care in developing countries — B.Sc. thesis, telemedicine and Info Hospital papers
- **Productive long-term mentor relationship:** worked with Dr. M. M. Khan across the thesis and 7 papers, covering antennas, IoT, telemedicine, and software — B.Sc.
- **Early research productivity:** published 7 conference papers within a year of finishing the B.Sc. — B.Sc.

- **Breadth across engineering:** combined a high-voltage power concentration with an embedded-systems thesis, then moved into biomedical engineering and AI — B.Sc. to PhD
- **Canadian work experience:** 16-month antenna engineering internship at BlackBerry (RIM), Waterloo, plus IT administration, data mining, and employee relations roles in Montreal (McGill University, Volt Workforce Solutions) from 2014 — Work Experience
- **Cross-functional engineering teamwork:** worked with mechanical, PCB, RF, and test teams through hardware build cycles — BlackBerry (RIM)
- **Service orientation and user support:** supported doctors, hospital staff, students, and employees as the main point of contact for technical and workplace issues — Work Experience
- **Confidentiality and discretion:** handled confidential employee investigations, HR records, and university financial data — Volt Workforce Solutions, McGill Financial Services
- **Numerical accuracy and attention to detail:** reconciled accounts and processed payments under internal controls — Accounting Clerk, McGill
- **Juggling concurrent roles:** worked as an accounting clerk and a network / IT administrator at McGill at the same time, alongside an employee-relations role — Montreal, 2015–2016
- **Judgment and fairness in people decisions:** recommended hiring, promotion, and disciplinary actions based on performance evidence — Volt Workforce Solutions
- **Conflict resolution and interviewing:** investigated employee claims through structured interviews and worked with HR on fair outcomes — Volt Workforce Solutions
- **Reliability under operational pressure:** kept servers and hospital IT running, minimizing downtime for critical users — McGill University, hospital IT roles
- **Stakeholder liaison:** acted as the link between IT departments and hospital clients, escalating issues effectively — HAF Hospital
- **International education and adaptability:** studied and worked in Bangladesh and the US, and now lives in Canada — B.Sc., M.Eng., PhD
- **Multilingual communication:** fluent English, native Bengali; comfortable working in diverse, multicultural teams
- **Relationship building:** built a professional network of faculty, co-authors, and collaborators — PhD

Analytical & Problem Solving

- **Algorithm design:** designed new algorithms for hard speech problems: a dual-score voice quality index, age normalization of acoustic features, and emotion-conditioned GAN voice conversion — PhD
- **Critical thinking:** spotted the circularity risk of using recognition-model embeddings and designed a system-independent alternative — PhD (VQI)
- **Problem solving:** diagnosed why children's verification degrades and developed age-normalization fixes — PhD (children study)
- **Analytical reasoning:** used mixed-effects models and causal ablation to separate the drivers of performance changes — PhD
- **Data-driven decision-making:** chose methods based on systematic comparison of 8 model families and 7 dimensionality-reduction settings — PhD (VQI)
- **Transferring methods across domains:** carried the utility-based quality approach from voice (VQI) to face (FUQI) and adapted it to face matchers, datasets, and the ISO/IEC 29794-5 components — PhD and CITeR 25F-04W
- **Reporting where a competitor wins:** stated plainly where CR-FIQA outperformed FUQI and why, instead of reporting only favourable results — CITeR 25F-04W
- **Attention to detail:** kept protocols consistent across 6 recording environments and 2 devices — PhD (VSEA-DC)
- **Creativity and innovation:** created a new dual-score quality index and an emotion-aware GAN for deepfake detection — PhD
- **Research design:** designed experiments with held-out datasets and systems to test generalization — PhD
- **Adopting new technology:** built LLM tools into research workflows (coding, text-data generation, writing) while verifying every output — PhD

- **Critical evaluation of AI output:** checked LLM-generated code, text, and phoneme coverage before use — PhD, SPEAD
- **Client communication:** gathered requirements, set scope and timelines, and reported progress to a paying client — ML Engineer, ImransLab
- **Team leadership:** led a student team on a client project — ML Engineer, ImransLab
- **User-centred design with clinicians:** collected doctors' feedback and changed the product to fit their workflow — ML Engineer, ImransLab
- **Clinical collaboration:** worked with surgeons, radiologists, and hospital clinicians to frame problems, set annotation criteria, review failure cases and false alarms, and present validation results — ML Engineer, ImransLab (surgical-item detection, sepsis)
- **Self-direction and ownership:** ran research projects end to end with little supervision, from question to release — PhD
- **Organization:** kept code, data, experiments, and documentation organized across several parallel projects — PhD

Professionalism & Personal Qualities

- **Scientific integrity:** documented limitations openly and evaluated honestly on held-out data — PhD
- **Research ethics:** handled human-subject research, informed consent, and participant privacy; CITI Program certified (Biomedical Research, Human Subjects, 2024–2027) — PhD
- **Award-winning presentation skills:** won 1st Prize in the 3-Minute Thesis competition and Best Graduate Oral Presentation (CS & Computational Modeling) at Clarkson's RAPS, Spring 2024, in the first PhD semester — PhD
- **Poster presentation:** Best Presenter Award (Poster) at the Mathematics Conference and Competition of Northern New York (MCCNNY) 2024, a regional multi-institution conference — PhD
- **Explaining research to non-specialists:** 3MT winner, distilling doctoral research into 3 minutes and one slide for a general audience — RAPS 2024
- **Sustained academic excellence since school:** Outstanding Pearson Learner Award (A Level) and 3 Daily Star Awards: 10 As at O Level, 2 A* and 3 As at A Level, and Distinction in O Level Physics (highest marks in the world, 2010) — secondary education
- **Recognized merit:** awarded a $6,750 ECE Department merit scholarship (50% tuition) on entering the Clarkson PhD (Spring 2024) — PhD
- **Academic excellence:** A grades in Biometrics, Advanced Biometrics, Deep Learning, and Intelligent System Design — PhD
- **Commitment to professional development:** attended the ECE graduate seminar for 6 semesters — PhD
- **Perseverance and resilience:** carried long projects through multiple rounds of peer review — PhD
- **Adaptability and fast learning:** learned new toolkits, models, and domains (speech, biometrics, statistics, software engineering, FPGA) — PhD
- **Accountability:** responsible for end-to-end delivery of dissertation work and released software — PhD
- **Commitment to open science:** publicly released datasets and open-source software — PhD
- **Curiosity and lifelong learning:** extended research into emotion, children's voices, and deepfakes — PhD
- **Working under pressure:** delivered revisions and the defense on schedule — PhD

Certifications & Training

- **CITI Program: Biomedical Research, Basic Course** (Human Subjects Research: Biomedical Research, Basic / Refresher), CITI Program (Collaborative Institutional Training Initiative), through Clarkson University. Completed February 19, 2024; valid until February 19, 2027. Record ID 61360613. Verify: www.citiprogram.org/verify/?wdfdde743-f115-4e92-b697-8d721955dacc-61360613
- **FERPA Compliance Training ("FERPA Bootcamp for Students")**, Clarkson University (Moodle). Family Educational Rights and Privacy Act training required for teaching assistants handling student records: FERPA

principles and quiz, privacy regulations for student workers, and a signed confidentiality agreement for student data systems. Completed Summer 2024.
- **Basic First Aid Training**, Dhaka, Bangladesh, 2012. Not a current Canadian certification; consider renewing with Standard First Aid + CPR Level C / AED (Canadian Red Cross or St. John Ambulance) if a job requires it.

Awards, Honors & Scholarships

- **1st Prize, 3-Minute Thesis (3MT) Competition**, Research and Project Showcase (RAPS), Clarkson University, Spring 2024 (April 2024). Won the 3MT competition with a 3-minute, single-slide talk for a non-specialist audience on the same research as the Best Graduate Oral Presentation: how speech emotionality affects speaker recognition accuracy.
- **Best Graduate Oral Presentation, Computer Science and Computational Modeling category**, Research and Project Showcase (RAPS), Clarkson University, April 19, 2024.
  - **Presentation:** "The Influence of Speech Emotionality on ASR Metrics and Recognition Accuracy" (single author). Examined how emotion changes pitch, tone, speed, and volume and so degrades automatic speaker recognition; proposed a three-part approach (emotion-diverse annotated data collection, preprocessing with MFCC / pitch / tone / speed features, and CNN / LSTM models trained with cross-entropy loss). Reported lower FAR and FRR across 8 emotions (e.g. anger FRR 0.20 to 0.17, happiness FRR 0.05 to 0.02), and outlined future work on transformers and multi-task learning of speaker and emotion.
  - **Highlights:** Won Best Graduate Oral Presentation at Clarkson's Research and Project Showcase for research on how speech emotion affects speaker recognition, showing lower false-acceptance and false-rejection rates across 8 emotional states with CNN / LSTM models.
- **Best Presenter Award (Poster)**, Mathematics Conference and Competition of Northern New York (MCCNNY), April 2024.
  - **Poster: A. Ahmed**, M. Imtiaz, and S. Schuckers, "Quantifying the Impact of Deep Learning-Enhanced Speech Preprocessing on ASR Accuracy." Applied CNN-based denoising to noisy speech (MS-SNSD dataset), measured speech quality before and after denoising (SNR, MOS, PESQ), measured speaker recognition performance with Neurotechnology VeriSpeak, and correlated the quality metrics with recognition accuracy to find which metrics predict it; this line of work led to the Signals paper and VQI.
  - **Highlights:** Won Best Presenter (Poster) at MCCNNY 2024 for research correlating deep-learning speech enhancement and speech-quality metrics (SNR, MOS, PESQ) with commercial speaker-recognition accuracy.
- **Merit Scholarship ($6,750; 50% tuition)**, Department of Electrical and Computer Engineering, Clarkson University, Spring 2024.
- **Outstanding Pearson Learner Award (OPLA), International A Level**, Pearson Edexcel, 2012. Awarded for outstanding overall results: the number of top grades achieved (2 A* and 3 A)
- **The Daily Star Award for Outstanding Results, A Levels** (2 A* and 3 A grades), The Daily Star (Bangladesh's leading English-language daily), 2012
- **The Daily Star Award for Outstanding Results, O Levels** (10 A grades), The Daily Star, 2010
- **The Daily Star Award for Distinction in Physics, O Level: Highest Marks in the World**, The Daily Star, 2010. Achieved the highest marks in the world in O Level Physics

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Personal Interests

- **Video games:** plays a lot of EA SPORTS FC (user, Oct 7, 2026); interest in how game AI, player animation, and motion feel to players. Use on game-industry applications (e.g. EA).

Professional Memberships

- **IEEE Graduate Student Member**, Spring 2024 – present